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**VISUAL COMPUTING AND AI TECHNOLOGIES
TO ANALYSE MOVEMENT DISORDERS
ASSOCIATED WITH PARKINSON'S DISEASE FOR
DIAGNOSTIC PURPOSES**

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**Visual Computing and AI Technologies to Analyse
Movement Disorders Associated with Parkinson's
Disease for Diagnostic Purposes**

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Abstract

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Title: Development of Visual Computing and AI Technologies to Analyse Movement Disorders Associated with Parkinson's Disease for Diagnostic Purposes

Keywords: Computer vision, object detection, Parkinson's disease, movement disorders, feature extraction, symptom severity, time series signals, machine learning, classification

Abstract: Clinical assessments of Parkinson's disease (PD) centre on clinicians' inherently subjective visual interpretations of characteristic motor signs like bradykinesia. This research investigates comprehensive video and signal processing techniques and their capacity to yield clinically meaningful representations of PD movement disorders. PD and healthy control participants were video recorded performing three upper limb motor examinations, including finger tapping, opening-closing and pronation-supination hand movements. Clinicians reviewed movement performance in each video and assigned corresponding MDS-UPDRS scores, providing motor symptom severity class labels for classification and correlation analysis. A real-time computer vision method used a custom-trained YOLO model to evaluate finger tapping videos, analysing computer features and their association with clinical ratings using Spearman coefficients. An automated framework employing MediaPipe Hands evaluated all three motor tasks, transforming extracted motion features into principal components for classification. Results indicate promising classification accuracy, effectively discriminating between motor symptom severity levels in both binary (mild/ severe) and multiclass (mild/ moderate/ severe) classifications.

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Research Achievements

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List of Abbreviations

PD - Parkinson's Disease

HC – Healthy Control

OC – Older Control

YC – Younger Control

VC – Visual Computing

MDS-UPDRS - Movement Disorder Society-Unified Parkinson's Disease Rating Scale

MBRS – Modified Bradykinesia Rating Scale

UPDRS-ME – Unified Parkinson's Disease Rating Scale-Motor Examination

MDS – Movement Disorder Society

YOLO - You Only Look Once

IMU - Inertial Measurement Unit

IoU - Intersection over Union

ANOVA - Analysis of Variance

CV – Coefficient of Variation

ICC - Interclass Correlation Coefficient

PCA - Principal Component Analysis

Chapter 1: Introduction

1.1 Thesis Background

Neurological disorders such as Parkinson's disease (PD) can impose a substantial burden on numerous aspects of an individual's quality of life. PD is a chronic progressive neurodegenerative disorder and is the second most prevalent neurodegenerative condition worldwide, with its incidence having doubled between 1990 and 2016 (Naghavi, 2019). The disease is known to primarily affect older adults, though it can also impact younger individuals (Adam et al., 2023). PD encompasses a wide spectrum of clinically significant motor and non-motor symptoms that affect functional mobility, mental health and the ability to lead an independent life. The definitive pathological changes that occur in the brain associated with PD can only be confirmed postmortem. As such, these complex underlying changes in the brain are often well-advanced by the time symptoms emerge. Critically, there is no treatment that will halt or stop the progression of the disease (Sveinbjornsdottir, 2016). Therefore, treatment with dopaminergic drugs aim to correct motor impairment, one of the most disabling and characteristic features of the disease. These motor impairments include bradykinesia, tremor, muscular rigidity, and postural and gait disturbances (Kalia & Lang, 2015). The progressively debilitating nature of PD demands early intervention together with routine clinical assessments of motor abnormalities to preserve quality of life. In clinical practise, the visual judgement of visible motor symptoms and gauging the effectiveness of treatment response remain central to guiding treatment decisions (Meara, 1999). Consequently, this research focuses on the development of visual computing (VC) techniques to enhance the clinical assessment of movement disorders associated with PD. More

specifically, VC was employed as a measurement method to quantify movements captured in video recordings of distinct parkinsonian movement tests, establishing an objective assessment framework for the characterisation and symptom severity classification of PD motor symptoms.

Parkinson's disease was first described primarily through observable signs by James Parkinson, with hallmark movement disorder symptoms forming the foundation of the disease's clinical characterisation. In clinical assessments, visual examination of these characteristic motor symptoms has long served as the foundation for diagnoses and assessing the severity of the disease. Clinicians remain heavily reliant on these visible examination signs that prevail over the course of the disease to guide long-term treatment decisions and address the specific needs of patients. However, even to the trained/ experienced eye, the era of precision medicine has brought into focus the variability in observer interpretation. A further difficulty complicating the judgment of PD is the variation in perception of symptom severity by both patients and clinicians (Politis et al., 2010; Jankovic, 2008). This underscores the critical role of clinical rating scales for motor examinations in consistent characterisation and grading of PD symptoms, aligning definitive diagnostic markers with disease severity and appropriate treatment management. The challenges and limitations associated with the inherently subjective visual judgement of movement disorders in achieving precise and consistent PD evaluations has motivated significant efforts to develop both assistive and alternative assessment techniques that quantify, characterise and objectively grade PD symptom severity. Notably, this thesis presents the acquisition of a substantial parkinsonian movement video dataset. The assessment methods applied to the video data demonstrate the strong

potential of the proposed VC and AI techniques in providing a contactless, accurate, objective, automated and clinically meaningful evaluation of PD movement disorder.

1.2 Parkinson's Disease

1.2.1 Aetiology and Pathology

The underlying cause of PD has been investigated extensively, highlighting a multifactorial aetiology attributed to age, environmental and genetic influences. Warner & Schapira (2003) highlighted the increasing evidence supporting a genetic role in the disease and the identification of PD-related genes. Lees et al. (2009) also recognised genetic predisposition as a causal factor with susceptibility genes including α -synuclein, leucine rich repeat kinase 2 (LRRK-2), and glucocerebrosidase (GBA). In recent years, there has been increasing evidence to support that in most cases there is a complex interplay between genetic and environmental influences in the causation of PD (Cherian & Divya, 2020; Warner & Schapira, 2003; Wang et al., 2021). The majority of PD cases occur sporadically, Warner & Schapira (2003) outlined that the earlier the age of PD onset, the more likely it is that genetic factors play a dominant role. Despite the involvement of inheritable factors in the prevalence of PD, it has been recognised that environmental factors may contribute to the onset and, importantly, the progression of the disease. The discovery that environmental factors can induce parkinsonism and trigger neurodegeneration has been linked to exposure to neurotoxins such as pesticides and metals (Di Monte et al., 2002; Ball et al., 2019). A small percentage (approximately 5% of all cases) of PD patients present with symptoms before the age of 60 years, with the condition continuing to be identified as one predominantly associated with ageing. Reeve

et al. (2014) maintained that advancing age remains the biggest risk, with the interaction of multiple ageing-related processes weakening substantia nigra neurons, making them more susceptible to degeneration.

In Parkinson's disease, the degeneration of dopaminergic neurons in the substantia nigra pars compacta (SNc) is the initial defining pathological feature (Vázquez-Vélez & Zoghbi, 2021). This progressive loss of neurons disrupts the nigrostriatal pathway, leading to dopamine depletion in the striatum, which is a key characteristic of the disorder. Critically, motor symptoms in patients typically emerge after 50-80% of dopaminergic neurons in the substantia nigra have degenerated (Cheng et al., 2010). Another cardinal pathological hallmark of PD is the presence of fibrillar aggregates in neurons termed Lewy bodies (LBs), which have been detected both within and outside the substantia nigra pars compacta (Wakabayashi et al., 2012; Shults, 2006). Lewy bodies contain aggregates of the protein alpha-synuclein and their formation has been considered to be a marker for neuronal degeneration. Importantly, it was recognised that Lewy bodies indicate an early, presymptomatic stage of PD, particularly as their prevalence increases with age (Gibb & Lees, 1988).

1.2.2 Motor Symptoms

The long-established cardinal motor symptoms that prevail over the course of PD are central to its diagnosis and tailored symptomatic treatment. Therefore, the understanding and characterisation of key motor symptoms are necessary to guide the development of a clinically relevant assessment framework in this research. Angel et al. (1970) explained that movement poverty can stem from any level of the motor system, "from the highest level down to the muscles themselves". Furthermore, the terminology used to characterise disturbances to

parkinsonian movement have been recognised to lack unity, heightening the challenge in documenting these motor abnormalities (Schilder et al., 2017). Therefore, it is critical to establish detailed motor aspects, be it through descriptive terms, to account for movement deficits at all levels. This chapter details the defining motor deficits of PD, highlighting their associated symptom presentations. Previous studies have reported important links between PD motor symptoms, suggesting overlapping neural areas that underly different motor abnormalities (Moustafa et al., 2016). As such, the broad spectrum of PD motor symptoms are explored, together with their correlations and disturbance to normal movement, with a focus on establishing these commonalities among patients.

The distinction between bradykinesia, hypokinesia and akinesia is critical to this research. Bradykinesia and hypokinesia are inextricably linked to PD (Schilder et al., 2017). Bradykinesia, derived from 'brady-' meaning slow and '-kinesis' meaning movement, is considered the hallmark motor symptom of PD. It manifests as a slowness in voluntary movement and further encompasses low amplitude movements known as hypokinesia. Furthermore, these movements often suffer from the sequence effect, described as a gradual reduction in amplitude during repetitive movements. These various features of bradykinesia often respond differently to clinical interventions and treatment, with L-dopa improving movement amplitude and speed but not the sequence effect (Bologna et al., 2016). Previous studies have established crucial symptom presentations related to these movement abnormalities, using them to distinguish PD from other disorders. In a study by Ling, H. et al. (2012), the presence of hypokinesia without decrement in finger tapping was demonstrated to be characteristic of individuals

with progressive supranuclear palsy (PSP), helping to set them apart from PD subjects.

Akinesia is a confounding motor symptom of PD and is frequently used to address the underlying condition. Spay et al. (2019) described this component of motor deficiency as a clinical concept with blurred boundaries, reporting a lack of studies that account for this dysfunctional brain process. Akinesia is associated with difficulties or failures in executing voluntary movements. Individuals who present with akinesia struggle to initiate and execute the normal impulses of movement. In the study by Angel et al. (1970), the movement disorders that can arise from dysfunctions at varying levels with potential to impede movement quality were addressed. It was highlighted that poverty in movement can be attributed to peripheral factors such as muscular weakness and rigidity, thus, prolonged reaction times and delayed movement initiations may not directly point towards a deficit at the brain's highest level. Importantly, it was suggested that prolonged reaction time may offer limited insight into brain function and that a more meaningful measure related to the highest level i.e. the brain, would be the time required to halt a voluntary movement that an individual perceives as inaccurate. The study concluded that if it could be shown that patients with akinesia struggle to stop their own 'false' movements, a delay in the transmission of "stop orders" from the brain could be assumed.

Tremor is another characteristic, noticeable and potentially highly disabling symptom of PD, with a reported presence of up to 80% of patients with autopsy-confirmed cases of the disease. Tremor has previously been defined as a rhythmic and involuntary movement of any body part (Elias & Shah, 2014). It typically occurs in the upper limbs with its repetitive high frequency nature often

being compared with pill rolling (Klockgether, 2004). Parkinsonian tremor typically occurs at rest, with a frequency of 3-7 Hz and can fluctuate, often diminishing or intensifying in response to motor and cognitive tasks (Dovzhenok & Rubchinsky, 2012). PD patients can experience different types of tremor, but the most common is an asymmetric resting tremor which can manifest along with kinetic tremor and secondary postural tremor. Kinetic tremor becomes evident during distal limb movement tasks such as finger tapping (Zalyalova et al., 2024). Essential tremor is an isolated action tremor affecting both upper limbs and is often confused with PD. Distinguishing between essential tremor and PD has been a longstanding debated topic and previous research has highlighted inconsistencies in diagnostic criteria as being a critical hindrance (Tarakad & Jankovic, 2019). As such, the multifaceted motor symptoms of PD serve to confirm the presence of PD. Importantly, it has further been suggested that in the end stages of PD, the disorder transitions from being primarily characterised by tremor to one where akinesia becomes more pronounced (Carr, 2002).

Rigidity is another predominant symptom of PD and is characterised by increased resistance to passive (without the patient actively moving) movement of an extremity (Baradaran et al., 2013). Notably, limb rigidity is an enhanced muscular response to being stretched and can occur in all movement directions remaining constant throughout the full range of movement. This muscular rigidity is commonly described as one that resembles the resistance of a lead-pipe, and in PD patients, the combination of rigidity and resting tremor presents a unique type of resistance named cogwheel-like rigidity (Klockgether, 2004). Given that rigidity is not a visually observable examination sign, clinicians assess this symptom

based on the evoked stiffness of limbs. In clinical practice, this is done by manipulating patients' limbs passively and scoring the level of joint resistance felt.

1.3 Challenges in the Clinical Assessment of Parkinson's Disease

Visual judgement of movement performance is fundamental to the diagnosis and ongoing evaluations of PD, but it is fraught with challenges, including observer subjectivity, variability in clinician agreement, and the complexities of accurately characterising and quantifying symptom presentation and severity, particularly in the context of comorbidities. These factors carry significant implications that span the entire diagnostic and treatment process. Recent developments over the past 10 years include the validation of clinical diagnostic criteria for PD outlined by the 2015 Movement Disorder Society (MDS). This chapter delves into the challenges and implications of determining PD motor symptom characteristics and severity, through clinical visual assessment, for both clinical practise and patient outcomes.

1.3.1 Clinical Rating Scale

The MDS clinical diagnostic criteria for PD are intended for use in clinical research and to guide clinicians in diagnosis by providing a systematic approach that ensures consistency and reproducibility. The criteria outlines that the presence of parkinsonism is fundamental to the diagnosis of PD. Parkinsonism is defined by a core movement abnormality termed bradykinesia, combined with either resting tremor or rigidity (Postuma et al., 2015). Bradykinesia refers to slowing of movement and the simplification of complex motor tasks and more specifically, the precise definition used by the MDS describes bradykinesia as "slowness of movement AND decrement in amplitude or speed (or progressive hesitations/halts) as movements are continued" making it a core criterion of

parkinsonism (Postuma et al., 2015). Tremor is defined as a rhythmic and involuntary movement of any body part (Elias & Shah, 2014). The tremor in PD is distinct from bradykinesia and rigidity and the classic tremor associated with the disease is typically observed at rest but also during movements after a brief pause (Hallett, 2012).

The Movement Disorder Society sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS) offers the comprehensive diagnostic framework used to evaluate PD and serves as the current standard for both clinical practise and research. This criteria expands on the original UPDRS by incorporating both motor and nonmotor aspects of PD, capturing the full spectrum of the disease. Importantly, Part III retains the original assessment of parkinsonism, but all tests now have explicit instructions for consistent data collection. With its four-part structure being: Part I "nonmotor experiences of daily living," Part II "motor experiences of daily living," Part III "motor examinations," and Part IV "motor complications." Goetz et al. (2008) noted that the total rater involvement time for administering the full MDS-UPDRS can be achieved in approximately 30 minutes. The study emphasised that despite the inclusion of non-motor aspects in Part I, Part III (motor examination) remains central to clinical assessments of PD. These assessments include the unilateral upper limb tests: finger tapping, opening-closing hand movements and pronation-supination hand movements.

In practise, the 30-minute goal set to administer the entire MDS-UPDRS is challenging to achieve in fast-paced clinical settings. Therefore, clinicians often visually examine a select number of movement tests that target limb bradykinesia for a more streamlined assessment. These typically include the finger tapping,

opening-closing, and pronation-supination hand movement tests, corresponding to items 3.4, 3.5 and 3.6 in the motor examination section. In the finger tapping test, patients are instructed to tap their index finger against thumb 10 times as quickly and as big as possible, with the corresponding scoring criteria outlined in **Table 1**. In the opening-closing hand movement test, MDS-UPDRS scores are clinician-assigned by observing patients open the hand 10 times as fully and as quickly as possible (see **Table 2**). For pronation-supination movements of the hand, patients turn the palm up and down alternately 10 times as fast and as fully as possible (see **Table 3**). In clinical assessments that grade these motor examinations using the MDS-UPDRS criteria, the amplitude, speed and rhythm components of limb movements are incorporated into a single score to depict symptom severity. Scores are assigned on a five-point scale, progressing from 0 (normal) to 4 (severe), with each score linked to explicit performance metrics – see **Tables 1-3**. Conversely, the Modified Bradykinesia Rating Scale (MBRS) is a clinical rating scale that considers the amplitude, speed and rhythm components of movement independently. Subscores are assigned for each movement component resulting in a more specific representation of movement severity, see **Table 4**. Thus, motor examinations of PD involve clinician visual observation of critical movement tests and assigning clinical rating scores to reflect the judgment of movement performance.

Table 1: MDS-UPDRS Finger Tapping Scoring Criteria

Finger Tapping (Item 3.4)	
Instructions to examiner: Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as big as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.	
0: Normal:	No problems.
1: Slight:	Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the tapping movement; b) slight slowing; c) the amplitude decrements near the end of the 10 taps.
2: Mild:	Any of the following: a) 3 to 5 interruptions during tapping; b) mild slowing; c) the amplitude decrements midway in the 10-tap sequence.
3: Moderate:	Any of the following: a) more than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st tap.
4: Severe:	Cannot or can only barely perform the task because of slowing, interruptions, or decrements.

Table 2: MDS-UPDRS Opening-Closing Hand Movements Scoring Criteria

Hand Movements (Item 3.5)	
Instructions to examiner: Test each hand separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to make a tight fist with the arm bent at the elbow so that the palm faces the examiner. Have the patient open the hand 10 times as fully AND as quickly as possible. If the patient fails to make a tight fist or to open the hand fully, remind him/ her to do so. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.	
0: Normal:	No problems.
1: Slight:	Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the movement; b) slight slowing; c) the amplitude decrements near the end of the task.
2: Mild:	Any of the following: a) 3 to 5 interruptions during the movements; b) mild slowing; c) the amplitude decrements midway in the task.
3: Moderate:	Any of the following: a) more than 5 interruptions during the movement or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st open-and-close sequence.
4: Severe:	Cannot or can only barely perform the task because of slowing, interruptions, or decrements.

Table 3: MDS-UPDRS Pronation-Supination Hand Movements Scoring Criteria

Pronation-Supination Movements of Hands (Item 3.6)	
Instructions to examiner: Test each hand separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to extend the arm out in front of his/her body with the palms down, and then to turn the palm up and down alternately 10 times as fast and as fully as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.	
0: Normal:	No problems.
1: Slight:	Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the movement; b) slight slowing; c) the amplitude decrements near the end of the sequence.
2: Mild:	Any of the following: a) 3 to 5 interruptions during the movements; b) mild slowing; c) the amplitude decrements midway in the sequence.
3: Moderate:	Any of the following: a) more than 5 interruptions during the movement or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st supination-pronation sequence.
4: Severe:	Cannot or can only barely perform the task because of slowing, interruptions, or decrements.

Table 4: Modified Bradykinesia Rating Scale (MBRS)

Score	Speed	Amplitude	Rhythm
0	Normal	Normal	Regular, no arrests or pauses in ongoing movement
1	Mild slowing	Mild reduction in amplitude in later performance, most movements close to normal	Mild impairment, up to two brief arrests in the 10 seconds, none lasting > 1 second
2	Moderate slowing	Moderate, reduction in amplitude visible early in performance but continues to maintain 50% amplitude through most of the tasks	Moderate, 3 to 4 arrests in 10 seconds; OR 1 or 2 lasting > 1 second
3	Severe slowing	Severe, less than 50% amplitude through most of the task	Severe, 5 or more arrests/10 seconds; OR more than 2 lasting > 1 second
4	Can barely perform the task	Can barely perform the task	Can barely perform

1.3.2 Variability in Clinician Ratings

Although the Unified Parkinson's Disease Rating Scale-Motor Examination (UPDRS-ME) section provides a standardised assessment criteria to guide clinicians in conducting consistent assessments of PD, scoring symptom severity through visual observation of movement tests can lead to inconsistencies both between different raters (inter-rater variability) and within the same rater over time (intra-rater variability). These inconsistencies are particularly evident in cases where patients exhibit subtle movement abnormalities in the early stages of the disease. Therefore, the leading factors influencing the visual judgement of PD symptom severity are believed to emanate from observer experience and symptom presentation. In light of this, studies have investigated the impact of inter-rater variability among neurologists. A study that extended to comparing the agreement between different healthcare professionals, specifically neurologists and advanced practise nurses established only moderate inter-rater agreement (Kappa = 0.53) when evaluating whether all items in the UPDRS-ME were rated as normal (Palmer et al., 2010). Post et al. (2005) called attention to the significance of clinician experience in their study, which evaluated the inter-rater variability in the UPDRS-ME using video recordings of 50 PD patients. The study compared the ratings of nurse practitioners, residents in neurology, and a movement disorder specialist to those of a senior movement disorder specialist. Both nurses, residents, and even the movement disorder specialist consistently assigned higher scores than the senior movement disorder specialist. Mean differences ranged between 1.7 and 5.4 (all differences $P < 0.05$), with wide 95% limits of agreement, underscoring the importance of experience and specialisation in the visual judgement of PD. Siderowf et al. (2002) set out to evaluate intra-rater variability, administering the UPDRS-ME using a 'test-retest'

approach where 400 participants with early-stage PD were evaluated on two separate occasions with a mean interval of 14.6 days apart. The UPDRS-ME scores demonstrated consistent intra-rater variability between the screening and baseline assessments of all participants, with an interclass correlation coefficient (ICC) of 0.90. Notably, reliability was slightly improved in patients retested within 14 days (ICC = 0.92) compared to those retested after 14 days (ICC = 0.87). This suggests that while the same clinician may provide consistent assessments of motor symptoms over short periods, the long-term reality of advancing symptom presentations combined with the involvement of multiple clinicians may introduce significant variability that impacts assessment outcomes.

The importance of quantifying inter-rater variability before starting longitudinal studies of disease progression or clinical trials was emphasised by Post et al. (2005), though it may be the case that intra-rater variability exerts less influence on UPDRS-ME outcomes, it is important to recognise that both high rater agreement and low variability between assessments conducted by the same clinician do not necessarily indicate the correctness or accuracy of symptom severity assessments. Furthermore, even in cases of high intra-rater agreement, inclusion of only early-stage PD patients raises concerns about whether high consistency can be maintained with broader symptom profiles (Siderowf et al., 2002). This holds particular significance for PD given that ageing (time) is a critical factor in symptom advancement (Reeve et al., 2014). The terms used to define and characterise PD have seen considerable changes over time, Bologna et al. (2019) highlighted that in many cases the term bradykinesia is used interchangeably to depict a range of movement abnormalities associated with the disease. Additionally, the diagnostic criteria for essential tremor have also

experienced significant revisions, contributing to ongoing ambiguity among clinicians (Louis, 2018). Complicating matters further, studies have indicated that items such as resting tremor, which hinge on visual observation, are insufficiently standardised in the MDS-UPDRS (Goetz et al., 2008; Ebersbach et al., 2006). With the growing prevalence of PD, the diagnosis and management of the disease is likely to become increasingly frequent in general practise (Waller et al., 2021). Therefore, there is compelling rationale for understanding the factors that influence clinicians' visual judgement and their lasting implications on patient outcomes.

1.3.3 Factors Shaping Visual Judgement

Previously, the variability in clinician-assigned symptom severity scores, arising from the inherently subjective visual judgement of PD movement tests, was highlighted by the presence of inter- and intra-rater variability. While this provides insights into the inconsistency of PD motor examinations, it does not directly address the factors that shape symptom severity scoring through observational evaluation. Even as the benchmark tool in clinical practise and research, the application of the MDS-UPDRS by eye is not immune to the challenge of observer interpretation. Shulman et al. (2010) explored the degree of change in MDS-UPDRS scores that is considered clinically meaningful for PD patients, substantiating the impact symptom severity scores have in managing the disease. Given the current reliance on clinician judgment of movement disorders by eye, specifically within the framework of the UPDRS-ME, it is essential to understand the specific factors that influence visual interpretation through the lens of a clinician.

There is no doubt that among numerous factors, clinician experience and specialisation are key determinants in disease diagnosis and management (Post et al., 2005). In recognition of this, the International Parkinson and Movement Disorder Society introduced an MDS-UPDRS certificate program to ensure that clinicians have the proper training to apply the MDS-UPDRS. This program includes both verbal and visual instructions to guide clinicians on the correct application and scoring of the MDS-UPDRS, with a particular emphasis on motor examinations. Despite the training designed to ensure uniformity and clear understanding of the performance measures outlined in the MDS-UPDRS, there remains an inherent subjectivity in the visual judgement of PD symptom severity. Whilst visual observation remains central to the assignment of MDS-UPDRS scores, even the expert eye cannot accurately discern all movement disorders a PD patient may have, let alone quantify the subtle and abnormal movements associated with early-stage PD. More specifically, accurately detecting and quantifying mild bradykinesia or tremor poses a critical difficulty for clinicians. This may be especially true for PD where tremor and other motor symptoms can overlap with those of other neurological disorders. The nuanced nature of tremor may further contribute to misclassifications of symptom severity given that PD harbours several subtypes of tremor that are variable in frequency, amplitude, distribution, consistency, contexts in which they occur, and provoking situations (Zach et al., 2015). Intensifying the problem, environmental stressors could exacerbate tremor or bradykinesia, leading to overestimation of symptom severity (Goetz et al., 2005).

Unsurprisingly, the expert eye alone is inevitably constrained, not only in its ability to observe a complex interplay of movement abnormalities but also in its capacity

to quantify subtle movements accurately. This limitation becomes even more pronounced when combined with the inherently subjective interpretation of observable measures. Beyond that, clinicians are not immune to information bias and a patient's demographic information, such as age, sex and ethnicity can affect the perceived severity of symptoms (FitzGerald & Hurst, 2017; Saposnik et al., 2016). They can also give undue weight to initial impressions or recent experiences, which can further result in diagnostic inaccuracies (Saposnik et al., 2016). Despite the refinement of the terminology and criteria used to evaluate key items of the UPDRS-ME, the limited accuracy of visual assessments makes it increasingly inadequate. While determining symptom severity has long been a fundamental aspect of PD assessments, the integration of objective methods could provide the level of precision needed to quantify complex movement disorder symptom presentations and establish objective benchmarks to monitor disease progression.

1.4 Implications of Symptom Presentations for Clinical Practice

The variability in PD symptom presentation makes both assessing symptom severity and pinpointing symptoms that contribute the most to movement disturbance challenging. Despite the detailed UPDRS-ME, which is designed to enhance the diagnostic accuracy of PD, the complex and inconsistent interplay between PD-related motor symptoms make clinical assessments of the disease particularly challenging. The clinical concepts surrounding classic hallmark symptoms of PD are continually evolving in an effort to improve the characterisation and identification of the disease. This is supported by the comparative and largely descriptive terms such as 'cogwheel-like' rigidity, the so-called 'pill rolling' in hand tremor and the 'sequence effect' in movement amplitude

decrement used to describe key motor symptoms. Evidenced by existing literature, a major difficulty and one that continuous to cause controversy is distinguishing PD from other disorders like essential tremor (Tarakad & Jankovic, 2019). Another clinically relevant problem lies in establishing the underlying cause of movement deteriorations and whether they stem from brain level or peripheral abnormalities (Angel et al., 1970). This is perhaps particularly challenging in PD which is known to be a disease of ageing (Reeve et al. (2014). Although previous studies have suggested that PD is strikingly heterogeneous in terms of the age onset and response to treatments (Tolosa et al., 2021), traditional assessment tools like the MDS-UPDRS rely on a one-time clinical evaluation that may not capture the full spectrum of a patient's symptoms. In the study by Tolosa et al. (2021), it was reported that while diagnosing PD can be simple in cases where patients exhibit typical motor examination signs like asymmetric movement quality and present with a classic history of the disease, atypical motor deficits complicate diagnosis, and 10% of individuals diagnosed with PD have alternative pathologies. Majority misdiagnoses of PD have been connected to difficulties in distinguishing it from essential tremor (Tarakad & Jankovic, 2019) and Alzheimer's disease (Meara, 1999). Provided that definitive pathological changes in the brain can only be confirmed postmortem, the diagnosis, characterisation and symptom severity determination of PD remain contingent on key motor examination signs. Unsurprisingly, in an attempt to address these challenges, there is a growing interest in the development of automated assessment techniques that objectively quantify and evaluate the numerous motor subtypes of PD discussed. As such, the review of existing methods that capture these complex motor presentations and how they incorporate clinically significant concepts is a crucial part of this research.

1.5 Motivation for Visual Computing in Parkinson's Movement Disorders

In the previous chapters, the limitations of clinician visual judgement in accurately scoring items of the UPDRS-ME were addressed. It is evident that visual observation of the wide spectrum of PD motor examination signs is constrained by the limited accuracy in detecting detailed and concurrent movement abnormalities by eye, while the interpretation of these motor symptoms is inherently subjective and requires experienced clinicians, whose availability is limited. Given these limitations, VC and AI techniques are well-positioned to overcome the challenges associated with achieving accurate and routine PD motor examinations.

There has been a growing body of research aimed at establishing objective, quantitative movement disorder measures for the commonly observed upper limb tests including finger tapping, opening-closing and pronation-supination hand movements. Initially, the majority of research focused on objectively assessing parkinsonian movement primarily centred around wearable technologies. The application of sensor-based technology for PD hand movement tests has expanded significantly over the years and can generally be divided into two main categories. The first includes dedicated measurement devices such as accelerometers, gyroscopes and Inertial Measurement Units (IMUs) to capture motion data. The second category being sensors integrated in ubiquitous devices such as smartphones, which were used for clinical movement analysis. Widely used devices including smartphones and keyboards have also been repurposed to act as contact sensors, specifically to quantify finger tapping (Giancardo et al., 2016). Additionally, marker-based methods have been utilised to monitor hand movements. Wearable devices and in particular marker-based motion capture

technologies often require specialised software, professional setup and regular calibration to ensure accurate data collection, all of which increase the overall cost and complexity of deployment.

Visual computing (VC) is a sub-field within AI and has been described as a domain that studies how computers can ‘see’ by processing images captured with a camera (Matsuzaka & Yashiro, 2023). Similar to how movement is perceived through observation by eye, which involves identifying an object and then monitoring its spatial location, VC techniques offer a contactless method in which objects can be detected and then tracked across successive frames, enabling the computational observation and quantification of movement. Object detection is a transformative VC task that is centred around providing one of the most fundamental insights required for analysing digital images: what objects are present and where are they located? As such, the two most critical metrics for object detection are accuracy (including classification and localisation accuracy) and inference speed (Zou et al., 2023). Given that movement data captured using wearable technologies is typically obtained in an intrusive manner such as placing markers or a wearable on the body (Gavrila, 1999), object detection has increasingly permeated human-centred applications by treating body parts as ‘objects’ of interest in detection tasks such as biometric identification, surveillance systems and sports analytics.

Particularly in the context of human movement analysis, object detection approaches can be classified in various ways, with one primary distinction being the distinction between 2D and 3D object detection (Gavrila, 1999). Polana & Nelson (1996) referred to “how to get your man without finding his body parts” developing a real-time system that can classify repetitive motion in normal gray-

scale image sequences. The study emphasised that any repetitive motion is such a strong cue that it does not necessitate identification of a specific body part. Instead, the moving actor can be segmented, normalised spatially and temporally, and recognised by a spatiotemporal template of its motion features. This simplified approach, described as ‘bottom up processing’, bypasses the pose reconstruction process altogether, demonstrating the strong potential of 2D features extraction for motion analysis. Despite the multifaceted movement abnormalities associated with PD, there is a substantial body of research (Lones et al., 2014) supporting the role of pattern discovery in PD symptom severity classification, suggesting there are underlying patterns even within these motor abnormalities. The knowledge that such movement abnormalities, though irregular, are not entirely chaotic and still exhibit detectable patterns provides a strong basis for applying VC techniques. While 3D object detection involves greater model complexity, the addition of a depth parameter in the spatial location of objects carries obvious advantages, particularly for more complex movements that involve moving closer to or further away from a camera.

Pulli et al. (2012) emphasised that for VC tasks to meet the demands of many real-world scenarios, video frames must be processed within 30-40 milliseconds. This indicates that object detection models should have a minimum inference speed of 25 FPS to qualify as real-time. Temporal resolution and inference speed fundamentally dictate the capacity of VC systems, particularly in tasks that demand either: 1) high sensitivity to subtle movements; or 2) accurate tracking of rapid movements. This is especially valid for parkinsonism, where detecting the subtle movements associated with limb bradykinesia and high-frequency tremor-induced movements necessitate both highly sensitive and accurate limb tracking.

In turn, this PhD explores two main methods for feature extraction from videos of PD hand movement tests in the UPDRS-ME section: You Only Look Once (YOLO) and MediaPipe Hands.

1.5.1 Hand Landmark Detection and Tracking with YOLO and MediaPipe

At its core, Visual computing (VC) transforms pixels into higher-level information, allowing machines to extract and interpret symbolic information from images. VC has previously been described as the inverse of computer graphics: while computer graphics transform an object description into pixels, VC transforms pixels and maps them back to meaningful data (Pulli et al., 2012). Hence, the foundation of extracting human movement data from videos using YOLO and MediaPipe Hands lies in analysing pixel coordinates across consecutive frames, with the transition from locating hand landmarks to tracking their movement achieved by computing changes in these coordinates over time (video frames). This section outlines specific spatial location extraction techniques used to analyse videos of the finger tapping test. The technique used to describe an object's location within a frame using pixel coordinates must consider three critical aspects: (1) the known or annotated "ground truth" coordinates of a target reference point's location within a frame; (2) how the chosen coordinate representation method will capture dynamic movements (e.g., bounding boxes or keypoint tracking); and (3) the capacity of the coordinate representation method to characterise complex hand movements and translate them into meaningful numerical movement data.

Deep learning-based object detection can be generally divided into two categories: two stage and single-stage object detectors. Two-stage models first generate preliminary region proposals, which are then refined in a subsequent

stage for more accurate detection. Conversely, single-stage models like YOLO predict object location and class labels in one evaluation, resulting in faster and more efficient object identification and localisation (Wang et al., 2023). A significant optimisation of the MediaPipe Hands model is that the initial palm detector, used to localise the hand, is only activated when the hand tracker confidence score falls below a set threshold. This infrequent re-engagement of the palm detector allows the system to rely on hand landmarks in the current video frame to predict its location in subsequent frames, facilitating both fast detection and reduced computation time (Google Research, 2019). With inference speeds of 45 FPS for base YOLO and 30 FPS for MediaPipe hands, both models demonstrate strong potential for tracking parkinsonian hand movements. Additionally, the flexibility to optimise model parameters allows for data extraction techniques, evaluation metrics and detection overlay to be tailored to assessing video recordings of the finger tapping test.

Particularly for tracking human movement, techniques must consider instances of occlusion, including self-occlusion where parts of the body cover each other, as well as cases where body parts may move partially or completely out of frame, or become obscured by external objects. YOLO detects an object by outputting a bounding box, defined as the minimal area rectangle that encompasses the entire object region (Murrugarra-Llerena et al., 2022) and captures the object's features using anchor points that define its position and size. Yao et al. (2023) stressed that on top of complex backgrounds, effective bounding box representations are complicated by the dense arrangement and arbitrary orientations of objects. In hand tracking, bounding box contours are simultaneously affected by the hand's positional shifts and its perceived shape

changes across frames, caused by inter-joint articulations. **Figure 1.1** illustrates two bounding box representations that can be implemented using YOLO to extract hand movement data from finger tapping videos: one approach uses a bounding box that encompasses the entire hand, while the other captures the index finger and thumb separately. In both methods, 2D (x, y) pixel coordinates can be extracted from either the centre or vertices of a bounding box. For the entire hand, pixel coordinates of the vertices can be used to derive the rectangle area. While tracking the overall change in this area could provide useful insight of hand movements, a more refined approach commonly adopted for the finger tapping test is to monitor the relative movements of the index finger and thumb.

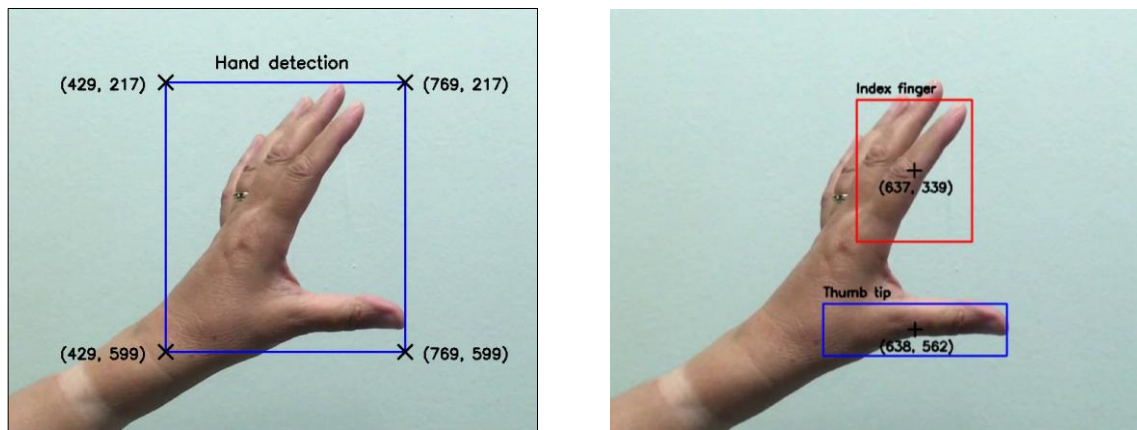


Figure 1.1. Two bounding box representations used for tracking finger tapping movement: full hand detection with the coordinates of bounding box vertices used to reflect gross contour changes of the hand (left); binary class detection using the index finger and thumb bounding box centre coordinates to track relative movements of the finger and thumb (right).

MediaPipe was developed by Google Research and is a versatile and open-source framework for building pipelines that enable inference on a wide range of sensory data of different modalities, including video and audio data, with particular capabilities for detecting human movements through its advanced pose estimation models. MediaPipe offers a range of inference models, allowing

researchers to deploy specialised models for VC tasks such as hand and face detection, full body pose estimation, and object tracking (Lugaresi et al., 2019). The flexible framework for building ‘perception pipelines’ leverages CNN for tasks involving object detection from visual data. Each pipeline in MediaPipe is structured as a direct graph, where nodes represent computational units known as “calculators”. The MediaPipe graph for hand tracking consists of two subgraphs, one for hand detection and another for hand landmarks. The model offers high-fidelity hand and finger tracking by employing machine learning (ML), supported by the manual annotation of approximately 30,000 real-world images with 21 3D coordinates for each hand, **Figure 1.2**.

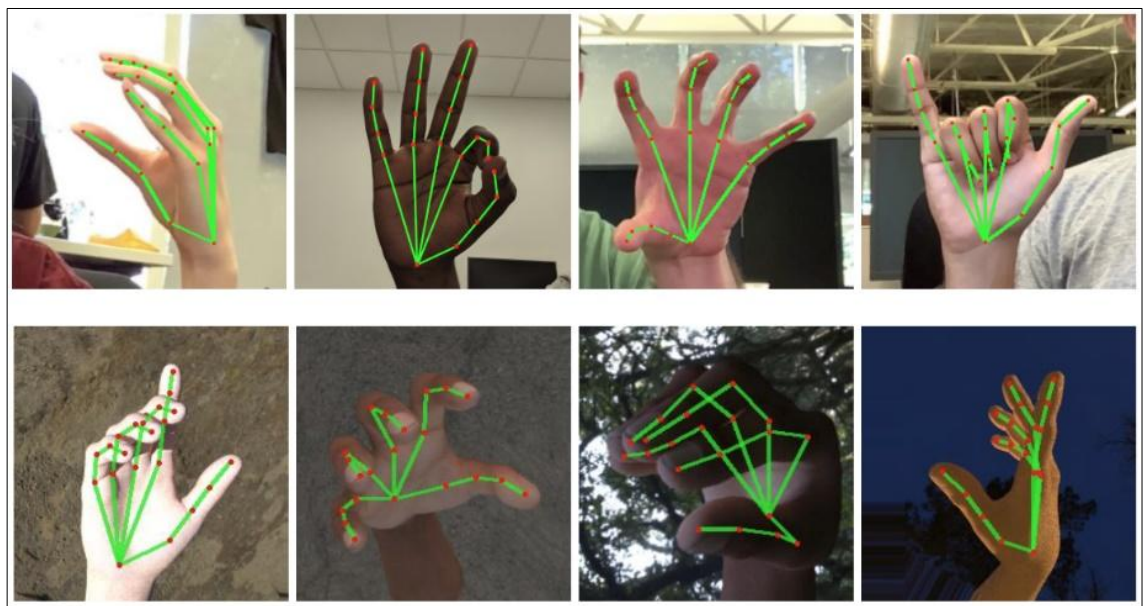


Figure 1.2. MediaPipe Hands landmark model: manual annotation of approximately 30,000 real-world images with 21 3D coordinates (Google Research, 2019).

The landmarks extracted using MediaPipe Hands have anatomical relevance with their reference descriptions provided in **Figure 1.3**. The model outputs 3D (x, y, z) pixel coordinates of 21 hand landmarks, with a significant reference point being the wrist, often termed ‘origin’, which can serve as a key anchor for tracking

the baseline movement of the hand during finger tapping. **Figure 1.3** shows an example of model inference where 3D pixel coordinates of the index fingertip and thumb tip are extracted. The estimation of rigid objects like palms and fists is significantly more straightforward than detecting the entire hand during complex movements which involve multi-joint movements, self-occluded instances and articulation of the fingers. As such, the model detects initial hand locations using the palm, a single-shot detector model called BlazePalm is employed to establish this region of interest. Following palm detection, the hand landmark model is used to infer 3D landmarks for each hand from a single frame, in real-time (Google Research, 2019). The 'z' coordinate represents the depth information relative to the camera, enabling the representation of hand movements in 3D space.

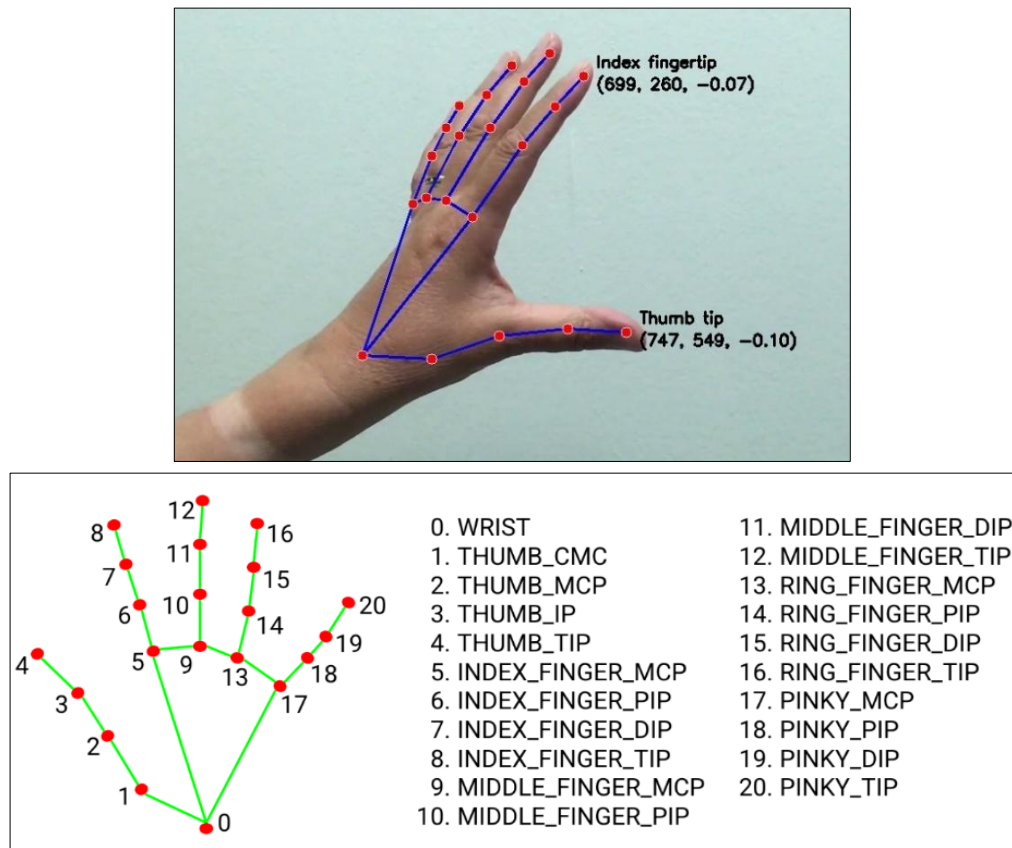


Figure 1.3. MediaPipe Hands inference example: extracting the index fingertip and thumb tip landmarks from the 21 hand landmarks detected (Google Research, 2019).

Pixel coordinates are effective for tracking hand landmarks in videos, providing detailed representation of movement across frames. It must be noted that, pixel-based measures are arbitrary units that are relative to video resolution and camera setup. Therefore, converting these measures into units such as centimetres can provide a more meaningful interpretation particularly in finger tapping movement analysis. This can be achieved by including a calibration object of known size in videos allowing pixel coordinates to be scaled to more interpretable units. In a wider context, this process allows movement data to be directly compared across datasets and against real-world performance measures. **Figure 1.4** illustrates the use of a checkerboard as a calibration object, where its known height is manually labelled (red line) and used to convert pixel distances between the index fingertip and thumb tip into centimetres.

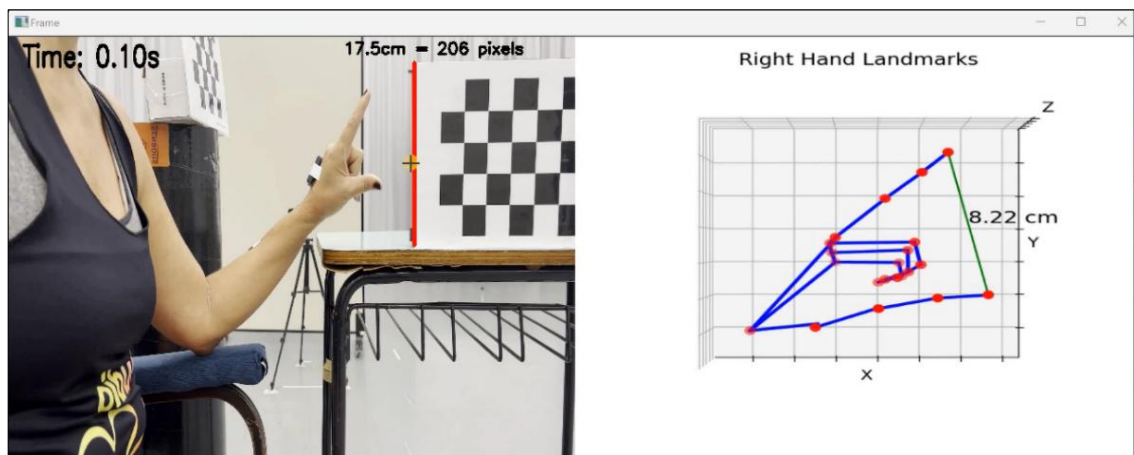


Figure 1.4. Checkerboard positioned in video frames, aligned in the plane of finger and thumb movement, to act as a calibration object. Distance measures derived from the index finger tip and thumb landmark coordinates are subsequently convert from pixels into centimetres.

1.5.2 Techniques to Analyse Movement Data in Finger Tapping Videos

The complex spatial relationships between the fingers and joints combined with baseline shifts of the hand during finger tapping movements have prompted most

studies to focus on the index finger and thumb. Numerous techniques have relied on the relative movements of the index finger and thumb as a baseline measure for constructing motion vectors (Khan et al., 2014; Williams et al., 2020; Li et al., 2022; Zhu et al., 2023). This sequence of distance over time is often analysed in the form of time series signals, where further processing can be applied to derive characteristic features from this raw finger tapping signal – see **Figure 1.5**.

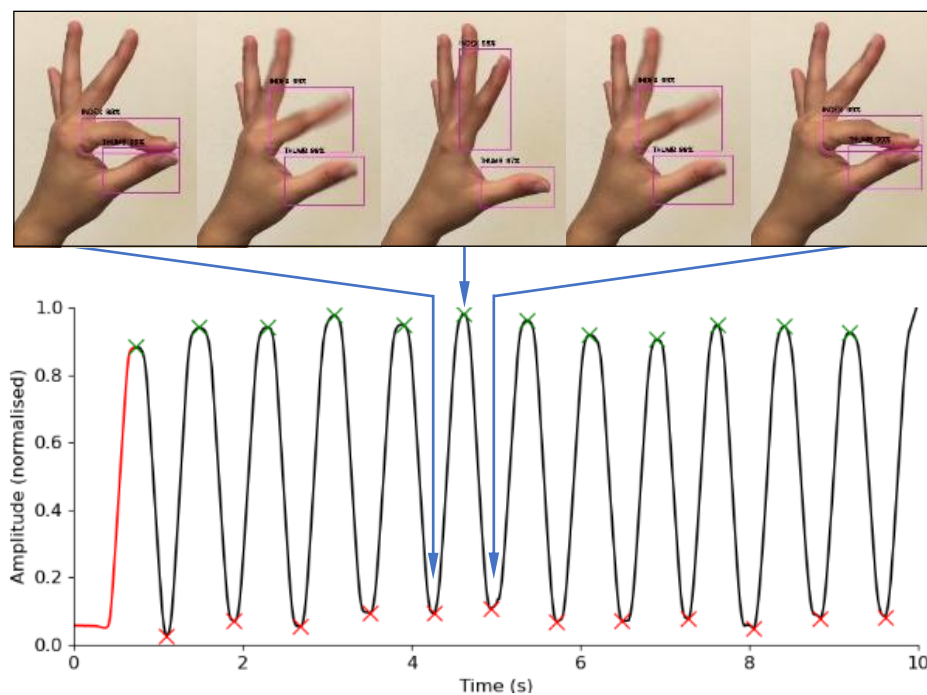


Figure 1.5. Generating time series signals using the distance between the centre coordinates of the index finger and thumb bounding boxes.

In finger tapping movements, maximum flexion involves physical contact between the distal forefinger and thumb and is relatively straightforward to identify due to its clear, measurable endpoint. In contrast, pinpointing peak extensions of the forefinger and thumb introduce more ambiguity in terms of being measured or when being determined through signal processing. This challenge leads to further difficulty in the determination of intermediate movements that occur in between maximum flexion and extension of the index finger and thumb. The determination

of maximum flexion can be reliably detected using contact sensors (Okuno et al., 2006) where contact between finger and thumb is registered or in other cases through signal processing techniques to estimate instances of directional changes. These changes are often inferred from transient shifts in acceleration, where a deceleration and subsequent reversal in movement direction indicate the transition from extension to flexion.

The selection of features from time series signals largely influences how finger tapping motion properties can be characterised. A common set of features extracted through various measurement methods include amplitude, speed and rhythm. Though likely motivated by their prominence in the MDS-UPDRS, their determinations have varied widely across studies. Movement parameters typically include both directly measured data like position or speed, and synthesised data, such as calculated metrics like peak velocity. Signal processing techniques like peak extraction are commonly used to derive relevant MDS-UPDRS movement parameters, while more advanced methods like Fourier series analysis being applied to signals to derive power spectrum of finger tapping movements. Several studies have highlighted the importance of considering the clinical relevance of extracted features (Khan et al., 2014; Wong et al., 2019). As such, feature descriptors are often linked to motion parameters, linking key metrics with clinically significant motor signs.

Motor examinations require an understanding of how symptom presentations deviate from the norm. Although automated assessment methods produce objective measures, when applied in place of the expert eye, they require diverse datasets with varying degrees of symptom severity to 'learn' the relationship between output features and symptom severity classifications. Consequently, two

approaches are typically used, the first simplified approach focuses on establishing the presence of PD using datasets labelled with two health status categories: healthy controls and PD patients (Buongiorno et al., 2019). The second, more advanced approach, uses finger tapping feature outputs to estimate symptom severity. This more granular approach relies on PD patients with assigned symptom severity scores, with features compared against these MDS-UPDRS scores. Another method that does not rely on labelled data, is to identify patterns in movement data through techniques such as clustering. This has previously been used to uncover key motor symptoms that distinguish movement patterns. This approach is useful for establishing important movement parameters and highlighting fundamental differentiators of motor examination signs. Conversely, without clear labels, it is difficult to interpret results in a meaningful clinical context.

Similar to clinician judgments, where various observed motor symptoms are considered to assign a single MDS-UPDRS score, kinematic features extracted from finger tapping videos must also be interpreted to reach a conclusion. The complex nature of parkinsonism requires more sophisticated approaches than indiscriminately amalgamating output features, as each feature's contribution must be assessed for its weight and relevance in symptom characterisation. Previous evaluations have assessed individual output features against MDS-UPDRS scores using statistical analyses such as Spearman correlation (Williams et al., 2020) and regression analysis. More recent approaches have employed machine learning models (Zhu et al., 2023), using an array of features to predict symptom severity classification. The concluding step is to evaluate a measurement method's ability to consistently and correctly grade finger tapping.

This is typically achieved by analysing agreement between model and clinician classifications. Another validation approach is to compare one measurement method with another, such as VC-based outputs with accelerometer data. In determining this, several statistical analyses such as regression analysis, Bland-Altman analysis and ANOVA have previously been implemented.

1.6 Aims and Objectives

The primary aim of this research is to develop a visual computing-based assessment method for evaluating standard video recordings of PD motor examinations including: finger tapping, opening-closing and pronation-supination hand movements. PD is a chronic neurodegenerative condition without a cure and is characterised by hallmark motor symptoms that impair functional movement. Clinical assessments of PD centre on the inherently subjective visual judgment of motor examination signs, a process fraught with challenges. In current clinical practise, routine clinical assessments of PD demand both experienced clinicians and long-term patient motivation. A remote, automated assessment method would support frequent monitoring of motor symptoms, enabling earlier detection of symptom fluctuations. This would provide more opportunities to assess treatment response, increasing confidence in its effectiveness and allowing timely adjustments to be made. In response to the current limitations in clinical practise, the core objective of this research is to develop a comprehensive assessment framework using advanced visual computing and AI techniques that is capable of performing automated, real-time and accurate classifications of PD motor symptom severity. In order to accomplish this objective, the following tasks will be undertaken:

1. Motor symptom presentations in Parkinson's disease

Investigation of the characteristic motor symptoms that prevail over the course of PD, allowing the defining features of the disease to be identified and quantified. This supports the selection of clinically significant movement features to represent movement performance, allowing underlying patterns within the movement data to discriminate between participants with different symptom severity levels.

2. Addressing knowledge gaps

Identification of critical research gaps in the objective assessment of PD motor symptoms. This task requires a detailed literature review of published methods that perform motor symptom severity classifications in PD movement tests.

3. Data Acquisition

Acquisition of a substantial video-recorded movement dataset, including finger tapping, opening-closing and pronation-supination hand movements performed by PD and healthy control participants. The dataset will require the assignment of target class labels for each isolated movement test, corresponding to symptom severity. This will require multiple neurologists, each specialising in movement disorders, to assign videos with MDS-UPDRS scores and individual amplitude, speed and rhythm component scores of the MBRS.

4. Movement data extraction mechanisms

Exploration of different movement tracking approaches and their ability to capture complex hand movements, in both 2D and 3D inference models. This task involves reviewing which reference points of the hand can be used to track movement, the spatial planes in which these movements can be captured and

the granularity of movement information that can be discerned for distinct hand movements.

5. Time series signal analysis

Application of signal processing techniques to extract clinically meaningful movement features from time series signals, derived by applying VC and AI techniques to hand movement videos. This intermediate analysis stage enables the leap from raw video recordings of hand movements to the symptom severity classification of movement data. Accordingly, the extraction of movement features from spatiotemporal data must centre on the core movement abnormalities associated with PD.

6. Symptom severity classification

Experimentation of numerous motor symptom severity classification approaches, varying the number of target classes and class labels. This task is designed to establish the effects of different classification configurations on the discriminatory power of computer-derived movement data. More specifically, this involves examining how the assessment framework's classification accuracy is affected by the granularity of the classification problem, such as group-level classification (PD vs. control) and discriminating between more specific symptom severity levels (MDS-UPDRS scores).

7. Assessment framework design

Incorporation of critical design elements into the PD symptom severity assessment framework. This task requires attention to several aspects: (a) development of a robust and versatile system that supports real-time and video inputs; (b) integration of key evaluation metrics to ensure sudden transient shifts

in gross hand movements, self-occlusions that obscure parts of the hand from camera view and complex articulations of the fingers are identified and monitored by the assessment framework; (c) application of effective signal processing techniques to ensure the accurate reflection of time series signal-derived movement data; and (d) ensuring good model interpretability through the development of well-defined analytical stages, breaking down the process from raw video data to the decision-making involved in symptom severity classification.

8. Evaluation of model performance

Evaluation of the model's ability to accurately and consistently capture ground truth movements. This involves the comparative analysis of kinematic metrics derived using the VC method and those obtained from accurate, established technologies. The effectiveness of these output metrics in characterising parkinsonian movement can be established by examining their relationships with clinician-assigned symptom severity scores. For symptom severity classification, the predictive accuracy can be determined by testing the model on unseen data.

1.7 Thesis Structure

Chapters 1 provides an overview of Parkinson's disease and its hallmark motor symptoms which remain central to diagnosing and monitoring its progression. The implications of visually assessing these motor symptoms are then discussed together with the clinical rating scales used to score motor symptom severity.

Chapter 2 provides a literature review of previously published methods for capturing finger tapping movements, one of the most commonly administered motor examinations for PD. It then investigates analytical approaches used to transform movement data into clinically relevant characterisations of movement

quality. Next, it reviews classification approaches that have been adopted to predict motor symptom severity. Key findings from the reviewed literature are summarised and critical research gaps are identified to guide the research.

Chapter 3 presents a deep learning framework for the objective assessment of the finger tapping test in PD patients as well as younger and older healthy control participants. It details the development of a custom-trained YOLO model to extract spatiotemporal data from standard video recordings and outlines the transformation of this data into meaningful characterisations of movement performance. The chapter includes model experimentation, key system design elements and concludes with an analysis of computer feature correlations with clinician-assigned symptom severity scores.

Chapter 4 presents a comprehensive assessment method using MediaPipe Hands to analyse three PD upper limb motor assessments: finger tapping, opening-closing and pronation-supination hand movements. It details the collection of a video-recorded movement dataset, in which each video is labelled with clinician-assigned symptom severity scores. The methods section outlines the analytical steps comprising the assessment framework including: generation of time series signals using MediaPipe Hands to capture test-specific kinematic features, signal processing, feature extraction, principal component analysis, and classification for symptom severity prediction. The chapter then reports the results obtained using distinct classification approaches, discussing the implications of these findings.

Chapter 5 provides a comparative analysis of pronation-supination angular velocity data derived from two measurement modalities: computer vision and Inertial Measurement Unit (IMU). It details the bimodal data capture protocol

along with the critical data and signal processing techniques used to extract and compare peak angular velocity values. This is followed by a statistical analysis of the agreement between the two systems and a presentation of results across symptom severity scores. The concluding section discusses key findings and potential factors leading to discrepancies between the two measurement methods.

Chapter 6 outlines the key findings of the research, highlighting their clinical implications. The chapter summarises the novel contributions of this research and concludes by relating the key findings to prior work, highlighting the key challenges in developing robust computer vision-based assessments for use in clinical practise.

Chapter 7 informs on future research directions. It discusses strategies for enhancing the existing dataset including the integration of lower extremity movement tests into the assessment framework. It then outlines techniques to improve model explainability, supporting the adoption of computer vision-based assessments in routine evaluations of Parkinson's movement disorders.

Chapter 2: Literature Review

2.1 Study Selection Strategy

As a basis for systematic review, a two-step search strategy was used. In the initial phase of the literature search, the pathology and aetiology of Parkinson's disease were reviewed to establish a foundational understanding of the disorder. Next, the challenges involved in the clinical visual interpretation of PD symptom severity were identified, highlighting the limitations of administering the UPDRS-ME by eye. Finally, the cardinal motor symptoms that are central to the diagnosis of Parkinson's disease were reviewed, this involved exploration of critical examination signs typically used to identify, characterise, gauge symptom severity and distinguish PD from other disorders. This initial clinically focused literature review, conducted prior to the review of objective assessment techniques for PD, aimed to establish a comprehensive understanding of the numerous aspects of PD and its implications for assessment outcomes. Furthermore, this prior knowledge was critical to evaluate the viability of proposed objective assessment methods and the level of clinically significant factors incorporated into their frameworks.

Following the review of clinical concepts, the second phase of the literature review focused on previously published assessment methods that capture and quantify movements during the Parkinson's finger tapping test. The literature search focused on identifying relevant studies using the key search term strategy outlined in **Figure 2.1**, ensuring both data quality and alignment with the research objectives. This structured approach filtered an initial set of 200 studies identified through broad searches down to 42 studies, selected for full review.

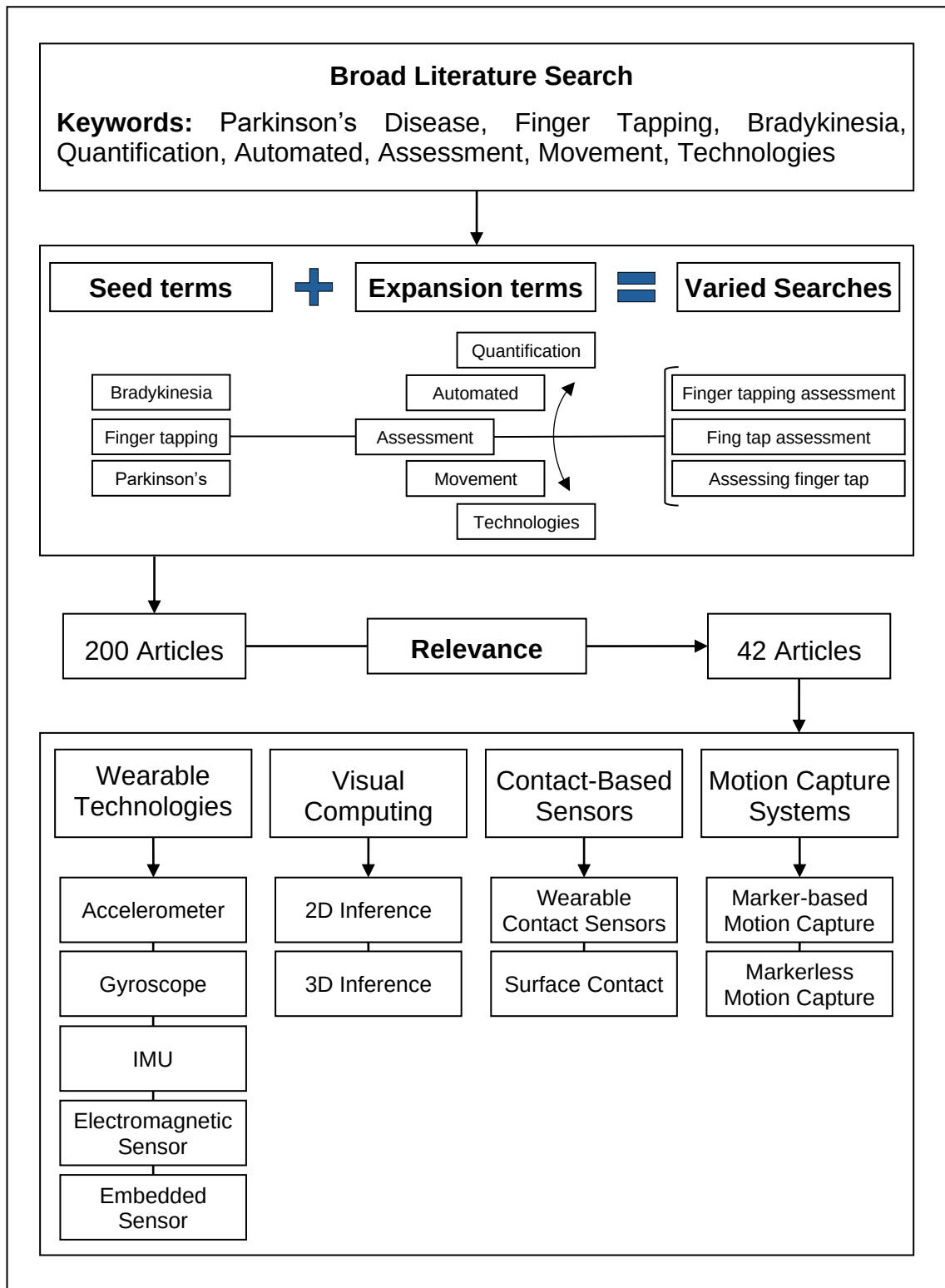


Figure 2.1. Study selection strategy for the literature review of published methods for objective assessment of Parkinson's disease upper limb movement tests.

Study selection was achieved by applying three inclusion criteria: (1) participants were required to have a formal PD diagnosis prior to study participation; (2) studies were required to include analyses of agreement between clinician-assigned scores and outputs of the proposed assessment method; and (3) the motor examinations evaluated had to be performed in accordance with MDS-UPDRS protocols. The studies selected for full review were categorised into four groups according to the movement capture method used: contact-based sensors, wearable technologies, motion capture systems and visual computing methods.

This refined selection now forms the basis for the second stage of the literature review, with a focus on identifying the critical questions that need to be addressed through the review of existing measurement methods used to objectively assess finger tapping. **Figure 2.2** outlines the foundational points that will guide the review of previous work, aiming to establish the strengths and limitations of existing approaches and identify the knowledge gaps that need to be addressed to bridge existing methods with clinically relevant methodologies for assessing and classifying PD symptom severity through the finger tapping test.

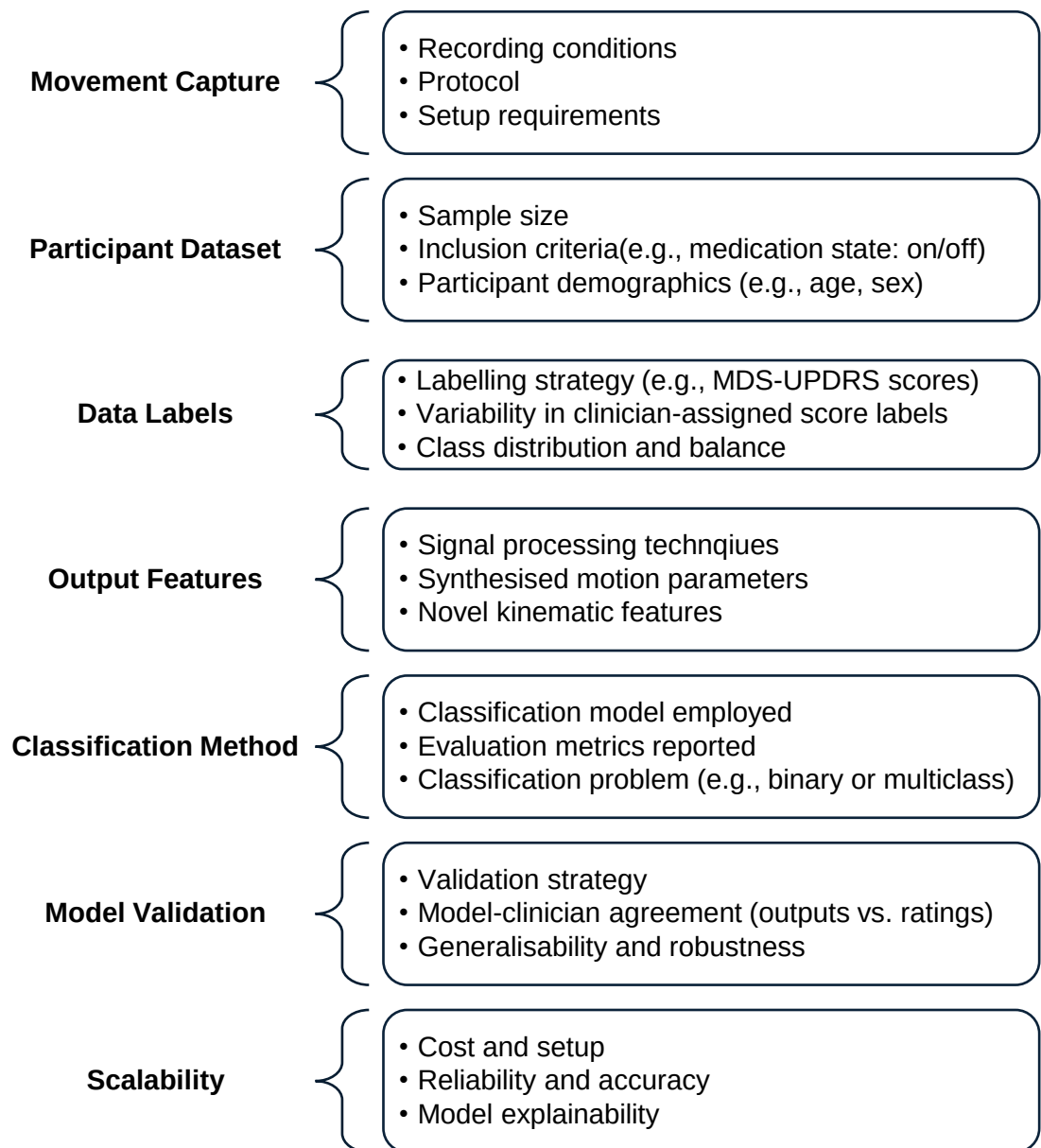


Figure 2.2. Primary questions used to guide the review of current literature to identify existing methods and knowledge gaps in upper limb movement assessments in Parkinson's disease.

2.2 Technology to Capture and Quantify Finger Tapping

A number of studies have employed various technologies to capture finger tapping movements in PD. The finger tapping test involves repeated alternating taps of the index finger against the thumb and comprises two distinct movement phases: flexion and extension. Over the years, a range of existing technologies

have been adapted to quantify these movements including inertial sensors, contact sensors, touchscreen devices, marker-based and markerless motion capture systems, and more recently, visual computing. It must be noted that the measurement method used in prior studies dictates the amount of movement data that can be captured, potentially oversimplifying the intricacies of finger tapping movements. Therefore, factors beyond the accuracy of recording methods are required to evaluate the effectiveness of previous approaches. The subsequent sections of this literature review investigate the capacity of different technologies to capture finger tapping movements within the constraints of their respective measurement methods.

2.2.1 Sensor-Based Technology

Unsurprisingly, wearable technologies represent a large portion of the literature on PD finger tapping quantification, likely due to their well-established use in human movement tracking. Inertial sensors such as accelerometers record changes in motion by measuring linear acceleration (m/s^2 or g) along one or more axes (typically x, y, z axes in modern devices) (Arun Faisal et al., 2019). Several studies have adopted low-cost tri-axial accelerometers to measure the acceleration of finger tapping movements by securing them to the distal interphalangeal joint of the index finger and interphalangeal joint of the thumb (Okuno et al., 2006; Yokoe et al., 2009; Stamatakis et al., 2010; Stamatakis et al., 2013). In the study by Habets et al. (2023), a tri-axial accelerometer was used to develop and validate ReTap, an open-source tool that uses extracted kinematic features of finger tapping movements to automate the prediction of motor symptom severity.

Gyroscopes are inertial sensors that measure the rate of rotation around an axis (degrees/s or radians/s) (Williamson & Andrews, 2001). While the placement of a gyroscope is not as critical as with an accelerometer given that angular velocity is the same for all points on a rotating rigid body (Moreau et al., 2023), gyroscopes are susceptible to signal drift over time. Consequently, previous studies have used Inertial Measurement Units (IMUs), comprised of a triaxial accelerometer and a triaxial gyroscope (six degrees of freedom), to simultaneously record the linear acceleration and angular velocity of the distal forefinger and thumb (Bhattacharya et al., 2015; Rovini et al., 2020). This sensor fusion system allows the signal drift in the gyroscope to be corrected with accelerometer readings, likewise, the accelerometer's high sensitivity to noise and vibration can be countered with smoother angular velocity readings from the gyroscope (Saber-Sheikh et al., 2010). Rovini et al. (2020) used four wearable IMU devices (two SensHand and two SensFoot) to examine finger tapping movements as well as pronation-supination hand movements, resting tremor, postural tremor, toe tapping, leg agility, heel-toe tapping, rotation, and gait. Notably, the ability of IMUs to capture the spatial aspects of finger tapping movements can be enhanced by integrating a triaxial magnetometer for absolute heading estimations, enabling linear acceleration and angular velocity data to be assessed within a directional context (Beange et al., 2018). Li et al. (2020) used a 9 degrees-of-freedom sensor system, which combined a 3D accelerometer, 3D gyroscope and 3D magnetometer, to extract a more comprehensive, drift-free orientation estimation of finger movements during finger tapping. This broader perspective enabled motor coordination of the forefinger to be considered in addition to the acceleration of flexion and extension movements during finger tapping.

In another study, Gao et al. (2018) attached electromagnetic sensors to the forefinger and thumb, with readings from the device (referred to as "PD-Monitor") normalised between - 1 and 1, used to indicate the level of bradykinesia. Electromagnetic sensors generate a magnetic field detected by the sensor, where the separation between the sensor and source (corresponding to the forefinger and thumb) produce output measures based on the intensity of the field changes. It must be noted that using magnetic field strength to measure relative movements between the distal forefinger and thumb prevents the detailed determination of their individual motion properties.

The application of traditional wearable devices for hand movement quantification have been broadened to include smartphones with inbuilt accelerometers (LeMoyne et al., 2010; Barrantes et al., 2017). However, these measurement methods involve attaching smartphones to the dorsum of the hand with an armband or glove, limiting their outputs to tremor-related movements of the hand. Consequently, smartphone-embedded sensors are not suitable for capturing finger joint movements in the finger tapping test due to their size and placement, restricting their use to recording gross hand movements.

Contact sensor-based methods for recording finger tapping can be categorised into two main types: sensors directly placed on the finger and thumb to detect instances of contact between them (Okuno et al., 2006) and devices adapted to act as surface contact sensors using device's capacitive touch screens to register touches (Printy et al., 2014; Lee et al., 2016). Both types of contact sensor-based methods operate on the same basic principle of registering touches and recording the time stamps of these contacts, producing binary time series data. This data consists of on (contact) and off (non-contact) states, which can be used to derive

inter-tap time intervals. Notably, without the integration of other sensors, the tapping rhythm derived from moments of contact cannot distinguish whether variability in time intervals is associated with a change in tapping amplitude or speed. Okuno et al. (2006) used fingerstalls to secure thin stainless steel touch sensors and tri-axial accelerometers to the distal index finger and thumb, capturing finger tapping acceleration together with the precise moments of contact between the index finger and thumb.

Multiple studies that adapt keyboards to register distal finger tapping have been developed to assess upper-limb movements (Giancardo et al., 2016; Trager et al., 2020; Akram et al., 2022). In the study by Akram et al. (2022) the distal finger tapping test was combined with BRadykinesia Akinesia INcoordination (BRAIN), a validated keyboard tapping test, demonstrating a simple, web-based assessment for PD. Typing cadence (also known as keystroke dynamics) has been explored by various research groups but its applications to the medical field is almost non-existent (Giancardo et al., 2016). In contrast to this, Trager et al. (2020), developed an engineered keyboard to not only registers finger taps but the speed of key release, using it to indicate movement rigidity. A clear advantage of surface contact sensors that utilise smartphones is their potential to facilitate the development of smartphone-based applications, enabling widespread, accessible assessments of finger tapping. In several studies, smartphone applications such as “smartphone tapper” (SmT) and “mPower” were designed to factor in motor coordination by registering the pixel position (x, y coordinates) of each forefinger’s tap (Lee et al., 2016; Prince & de Vos, 2018; Arora et al., 2015; Aghanavesi et al., 2017; He et al., 2024). In a study by Wissel et al. (2018), a tablet-based iMotor application was developed, offering two surface contact

tests. The first test registers touches from the distal forefinger on the centres of two concentric circles on the tablet screen to quantify finger tapping movement. The second test registers contact from the palmar and dorsal surfaces of the hand to quantify pronation-supination hand movements.

2.2.2 Vision-Based Technology

Marker-based motion capture systems allow multiple body landmarks to be tracked independently using infrared cameras to track the spatial location of reflective markers that are placed on specific points of the body. In contrast, motion capture systems that capture movement without requiring physical markers to be placed on the body are often termed 'markerless' to distinguish them from marker-based systems. Several studies have employed marker-based motion capture systems to quantify movements in the finger tapping test (Agostino et al., 2003; Růžička et al., 2016; Krupicka et al., 2017). In the study by Agostino et al. (2003), an optoelectronic motion analysis system (ELITE) was used to track a passive reflective marker firmly placed on the last phalanx of the index finger. The test compared isolated (individual) with non-individual finger movements demonstrating that patients had greater difficulty executing individual forefinger taps against the thumb compared to gross hand movements involving all four fingers. Similarly, Krupicka et al. (2017) used BradykAn motion capture system with reflective markers to quantify finger movements, establishing decrement in amplitude as a key output feature for indicating bradykinesia. Another study focused on establishing key metrics as discriminators between PD patients and healthy controls using the Optitrack-V120 system to track finger tapping (Růžička et al., 2016). While highly accurate for detailed movement analysis, configurations like the ELITE motion analysis system consisting of two

infrared ray cameras, passive reflecting markers and a TV image processor, are expensive for widespread implementation in clinical settings and require professional setup to ensure accurate motion tracking.

With the exception of surface contact methods, one commonality of all aforementioned technologies used to measure finger tapping movements is the need for them to be physically and strategically secured to the hand. While wearable devices secured to the hand are unaffected by occlusions, it has been recognised that the human hand is sensitive to environmental influences such as added weight. In a previous study, this impact was investigated using an accelerometer to examine hand tremor power spectra. Notably, for each gram of added mass, the peak frequency of finger and hand tremor decreased by 0.85 Hz and 0.018 Hz, respectively (Kozlovsky et al., 2018). While numerous attempts have demonstrated the full potential of surface contact methods through techniques such as monitoring the contact positions of finger taps on smartphone screens to assess motor coordination (Lee et al., 2016) and even an engineered keyboard that measures the key release speed to infer movement rigidity (Trager et al., 2020); their performance remains inherently limited to short durations of interaction with a surface. These limitations have brought into focus the need for contactless methods that capture the full range of finger tapping movements. Consequently, more recent approaches have used markerless motion capture systems to track finger tapping movements.

The Leap Motion Device (LMD) and Leap Motion Controller (LMC) have been used to quantify finger tapping (Shin et al., 2024; Butt et al., 2018). In these studies, the Euclidean distance and angle between the forefinger and thumb were assessed. The LMD detects fingertips and the length of the metacarpals is set to

zero. It is capable of producing time-series data for finger joint positions and the orientation of the palm (Shin et al., 2024). While these systems offer three-dimensional motion capture, Butt et al. (2018) noted that certain gestures could not be captured due to the LMC's placement. It was further established that these systems are not immune to the challenges presented by complex finger tapping movements, where self-occlusions when the fingers touch or fold over the hand lead to incomplete or inaccurate data. Additionally, LMC's suffers from inconsistent frame rates, posing a particular limitation for capturing fine movements. The Microsoft Kinetic system was explored by Galna et al. (2014) for finger tapping quantification, among several other PD motor assessments. The Kinetic system's performance varied across tests, with assessments involving large gross movements like sit-to-stand demonstrating better accuracy compared to fine motor tasks like finger tapping. This outcome was further constrained by the system's limited spatial resolution (640 x 480 pixels) and 30 Hz sample frequency indicating that although spatiotemporal data can be measured effectively, detection of fine movements associated with bradykinesia remain difficult.

Over the last 10 years, an extensive body of research surrounding VC-based techniques that process, analyse and classify PD symptom severity through the finger tapping test has emerged in an attempt to overcome the limitations posed by traditional methods. In VC methods, the 'motion problem' is nearly invariably handled with the typical approach centred around identifying a search region across consecutive video frames, in contrast, the 'matching problem' presents variably complex challenges depending on the target object (Trucco & Plakas, 2006). This challenge is particularly pronounced in the finger tapping task, where

motion tracking across image sequences rely on distinguishing low contrast anatomical features within a small area like the finger and thumb tips during a ‘pinched’ position. Given the substantial literature and relevance of techniques used to this research, **Table 5** lists published methods that use VC techniques to quantify movements in the finger tapping test, in order of their publication year.

Table 5: Literature Review on Visual Computing-based Methods for Parkinson’s Disease Finger Tapping Quantification and Symptom Severity Classification

Year	Study
2014	(Khan et al., 2014) [1]
2019	(Buongiorno et al., 2019) [2], (Wong et al., 2019) [3]
2020	(Williams et al., 2020) [4]
2021	(Park et al., 2021) [5], (Monje et al., 2021) [6], (Li et al., 2021) [7]
2022	(Guo et al., 2022) [8], (Li et al., 2022) [9], (Yang et al., 2022) [10]
2023	(Amprimo et al., 2023) [11], (Islam et al., 2023) [12]
2024	(Guarín et al., 2024) [13]

Unlike motion capture systems, standard videos lack the depth dimension. A common approach for video recording finger tapping is to capture movement from a lateral (side) view, with the camera positioned closest and providing a direct view of the index finger and thumb. This perspective fixates on the lateral contour of the hand during alternating finger tapping movements, flattening the resulting visual information into a two-dimensional space. Several studies have indicated that alternating finger tapping movements can be sufficiently characterised in 2D, operating on the principle stressed by Polana & Nelson (1996), which suggests that regular repetitive movements provide sufficiently discernible patterns. Conversely, other studies have employed 3D pose estimation techniques to predict the spatial configuration of the hand, asserting that the absence of depth

inhibits the full depiction of complex movements, often underestimating movements that occur perpendicular to the image plane. **Figure 2.3** provides the VC approaches employed by each study in **Table 5**, grouping them into 2D and 3D capture of hand landmarks.

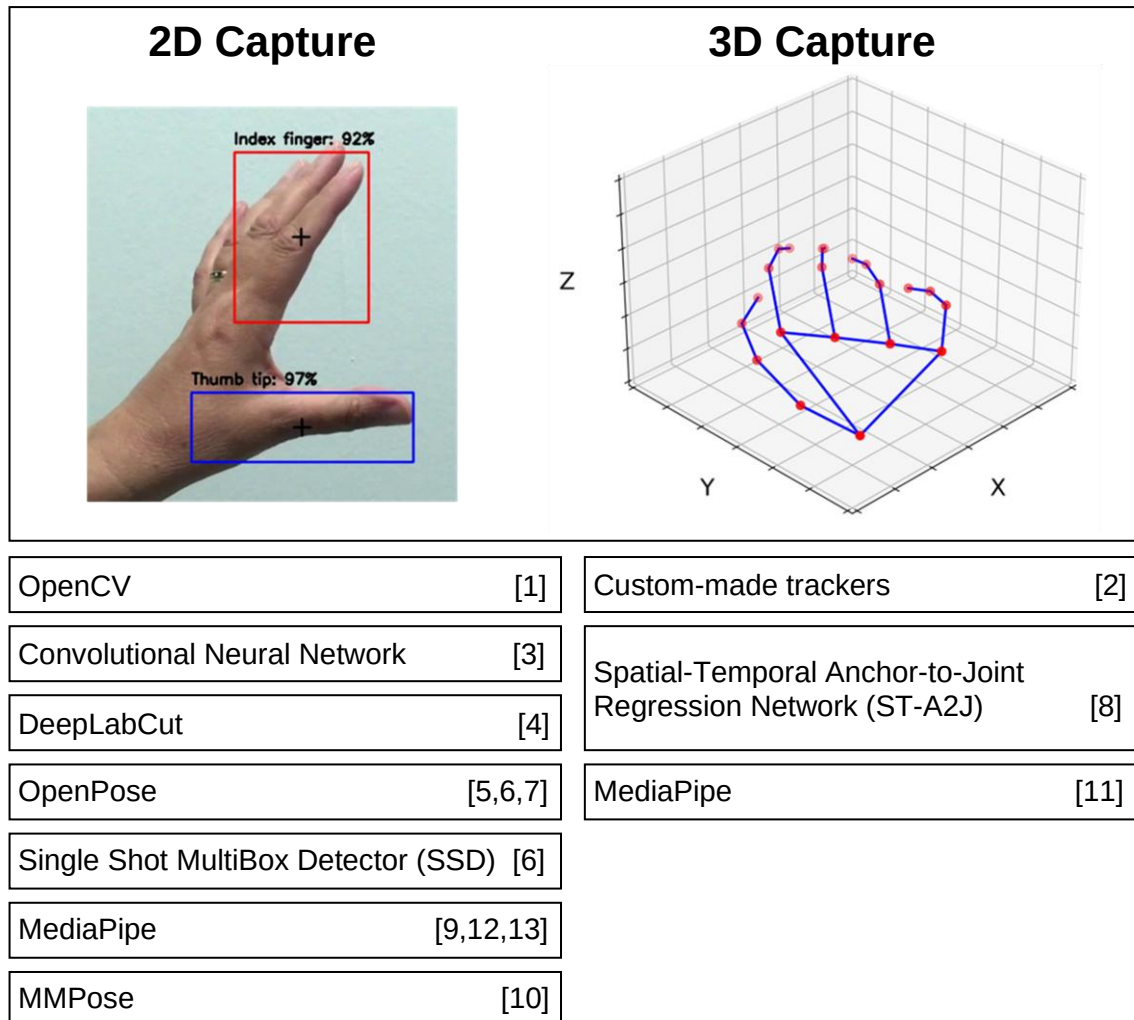


Figure 2.3. Visual computing frameworks used to track finger tapping movements, divided into two and three-dimensional inference models.

Despite the diversity of techniques used across the VC-based literature, there is a considerable overlap in two fundamental analysis approaches: all studies maintain a consistent focus on the index finger and thumb for extracting motion sequence data; all analyses are driven towards performing class separation.

These commonalities likely stem from the broader aim to automate the full finger tapping assessment, made available by the scalable nature of standard video processing. The emphasis on the MDS-UPDRS criteria, more specifically, bradykinesia, led to the consistent computation of amplitude (Khan et al., 2014; Wong et al., 2019; Buongiorno et al., 2019; Li et al., 2021; Park et al., 2021; Monje et al., 2021; Guo et al., 2022; Li et al., 2022; Amprimo et al., 2023; Williams et al., 2020) and in most cases speed, among other meaningful output features. Notably, in the study by Khan et al. (2014), face detection was incorporated into the analysis framework acting as a calibration object to normalise finger tapping amplitude measures, **Figure 2.4**. This additional step was implemented to alleviate inconsistent measures caused by varying camera to subject distances. Islam et al. (2023) negated this potential source of error by using the maximum finger tapping angle, subtended by the index fingertip, wrist and thumb tip landmarks, to estimate finger tapping amplitude, **Figure 2.4**. For distance measures extracted using two reference points (the fingertip and thumb tip), the perceived distance can fluctuate based on how close the hand is to the camera, even without relative finger-thumb movements. This phenomenon, known as perspective distortion arises from objects closer to the camera appearing larger than those farther away due to their increased pixel representation in a frame. **Figure 2.4** illustrates the two contrasting approaches used to negate the effects perspective distortion has on extracting consistently accurate measures of finger tapping amplitude.

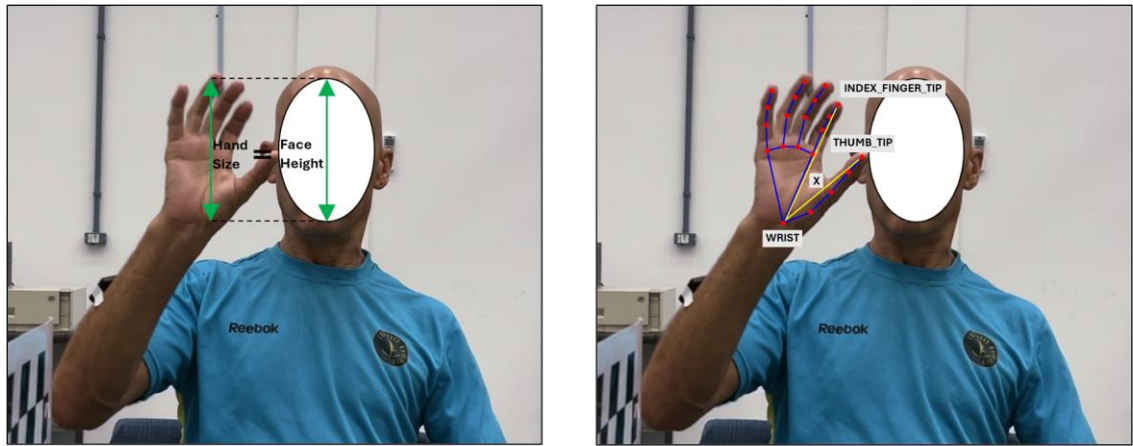


Figure 2.4. Two approaches used to mitigate the effects of varying camera-to-hand distance on finger tapping amplitude. (Left) An illustration of a calibration technique that uses the face to approximate hand length (Khan et al., 2014). (Right) A method to estimate finger tapping amplitude using the maximum angle (x) subtended by the index fingertip, wrist and thumb tip landmarks (Islam et al. (2023)).

2.3 Time Series Signals to Kinematic Features: Methods and Outcomes

The steps following movement data capture play a crucial role in determining the effectiveness of symptom characterisation and classification performance. As such, the subsequent chapters evaluate key aspects of previous work, including signal processing techniques, derived metrics and their clinical relevance, novel feature representations, dataset composition and labelling strategies, and classification methods. This structured investigation is necessary to both highlight and build on effective techniques and to identify knowledge gaps that define the critical focus areas.

2.3.1 Signal Processing Techniques

The critical stage that follows capturing finger tapping movement is typically signal processing, which is usually the pre-requisite for further processing (Fioretti, 1996). It is evident that the initial stage, which involves recording raw finger tapping movements, has varied largely across studies regarding the level of information captured. Studies employing surface contact methods were mostly

limited to registering basic on-off distal finger touch events, resulting in assessments being characterised by binary data. Other approaches utilising electromagnetic sensors, like the one by Gao et al. (2018), generated scaled output measures based on the intensity of the magnetic field changes. While this captures the full range of finger tapping, it lacks the precision needed to capture fine-grained movements, instead, focusing on the pattern of alternating finger taps. More detailed recording methods such as 3D motion capture systems and even VC-based methods employing hand pose estimation models were able to extract metrics like Euclidean distance between the finger and thumb through three-dimensional motion analysis. Despite the shift towards more comprehensive movement capture, most studies begin from a common baseline, with substantial, multidimensional movement data being reduced to a finger tapping time series signal of time (x-axis) against a selected y-variable (often finger-thumb distance). This simplification results in a significant loss of valuable movement data from inception.

It is important to note that the foundational finger tapping signal in most studies is a plot of amplitude against time, where amplitude corresponds to the relative distance between the index finger and thumb, more specifically, their distal tips (Khan et al., 2014; Buongiorno et al., 2019; Williams et al., 2020; Guo et al., 2022). The techniques used to extract features from this signal depend on how movements are intended to be characterised. The repetitive alternating extensions and flexions of the finger and thumb translate into a cyclical signal generating a rhythmic up-and-down pattern. A common signal processing approach is to extract the peaks and troughs of finger tapping signals, corresponding to maximum points of extension and flexion, respectively (Islam et

al., 2023). Although troughs would naturally be expected to have zero values during contact between the fingertip and thumb tip, they are extracted for several reasons. First, they are used to establish all intermediate movements, corresponding to those that occur between the signal peak and trough within each cycle. Furthermore, this is needed to establish the number of full taps executed, defined as a full extension followed by a return to a pinched position. Another important reason is that not all PD patients achieve full contact, this can be caused by impaired coordination, and in some cases involuntary movements like tremor. Finally, depending on how relative distance is recorded, signals often have a consistent offset even during finger-thumb contact. This is a common occurrence in VC-based methods where the landmarks may not exactly reflect the bottom of the finger and top of the thumb, leaving a fixed distance between landmarks even during contact.

Commonly used feature extraction techniques regularly employ the signal module of the SciPy Python package, which includes fast Fourier transforms (FFTs), numerous filters, and some wavelet implementations (Anderson et al., 2010). Tsfresh (Time Series FeatuRe Extraction on basis of Scalable Hypothesis tests) is another frequently used Python package that supports automated time series feature extraction (Guo et al., 2022). Tsfresh provides standard APIs compatible with time series and machine learning libraries such as pandas and scikit-learn and is capable of extracting 794 time series features (Christ et al., 2018). MATLAB is also widely used for signal processing with previous work using functions like 'findpeaks' extract finger tapping extension peaks from signals and classify them into maxima or minima values (Williams et al., 2020).

Given that finger tapping speed is a derivative of amplitude (finger-to-thumb distance), studies have used straightforward calculations like differentiation to derive it. This approach is susceptible to inconsistent results in signals that are chaotic as the speed slope is sensitive to the point of calculation. Williams et al. (2020) calculated the mean rate of change of the normalised finger-thumb distances to derive the speed metric. However, the mean rate of change in distance may not provide a meaningful characterisation of the finger tapping test due to the inherent variability in movement phases. More specifically, each finger tapping cycle involves alternating phases of speeding up, slowing down, stopping, and changing direction, resulting in faster and slower movements cancelling each other out. As such, signal processing techniques must accurately determine signal peaks/ cycles. Sensor fusion systems are advantaged by signal outputs from multiple sources. For example, two studies combined the signals derived from accelerometer and touch sensor (Okuno et al., 2006; Yokoe et al., 2009). The binary signal from the touch sensor was used to identify instances of contact between finger and thumb in the accelerometry signals.

2.3.2 Signal Filtering

A substantial deterioration in the rhythm of finger tapping signals can be observed with increasing PD symptom severity, complicating the feature extraction process. As such, many finger tapping assessment studies filter signals prior to further processing, enhancing the signal by removing noise whilst preserving meaningful characteristic features. Previously, Williams et al. (2020) used a Savitzky-Golay (S-G) filter to remove sudden transient shifts from finger tapping signals caused by DeepLabCut mislabelling. This was done to remove signal noise across consecutive frames that were associated with physiologically

impossible hand movements. The Savitzky-Golay filter has further been used to provide a signal smoothing effect and enhance the accuracy of subsequent analysis of raw skeleton data (Li et al., 2021). Amprimo et al. (2023) applied a low-pass, zero-phase Butterworth filter of order 3 with a 10 Hz cut-off frequency to signals derived from the 3D joint trajectories. Low pass filters, typically in the range of approximately 5-10 Hz have been commonly applied across studies to filter signal frequencies above a defined threshold, often termed the 'cutoff frequency' (Amprimo et al., 2023; Guarín et al., 2024; Islam et al., 2023).

Krishnan & Seelamantula (2013) define Savitzky-Golay filters as “finite impulse response lowpass filters obtained while smoothing data using a local least-squares (LS) polynomial approximation”. The Butterworth filter has been described as a type of signal processing filter that aims to provide a passband with the most uniform frequency response (Somwanshi et al., 2020). Although both Savitzky-Golay and Butterworth can be described as low-pass filters, they function differently: Savitzky-Golay filters smooth data in the time domain, fitting local polynomials to retain signal characteristics, while Butterworth focuses on the frequency domain, attenuating high-frequency noise. **Figure 2.5** illustrates these noise reduction techniques and the trade-off between noise suppression and retaining signal characteristics, with the Savitzky-Golay smoothing the signal while preserving its overall structure and the Butterworth filter being applied with increasing cutoff frequencies (5, 8, 10 and 15 Hz) allowing increasingly higher frequency components of the signal to pass through. In both cases, when applied to finger tapping timeseries data the resulting signals are generally smoother and less chaotic allowing frequently extracted features like amplitude peaks to become more pronounced.

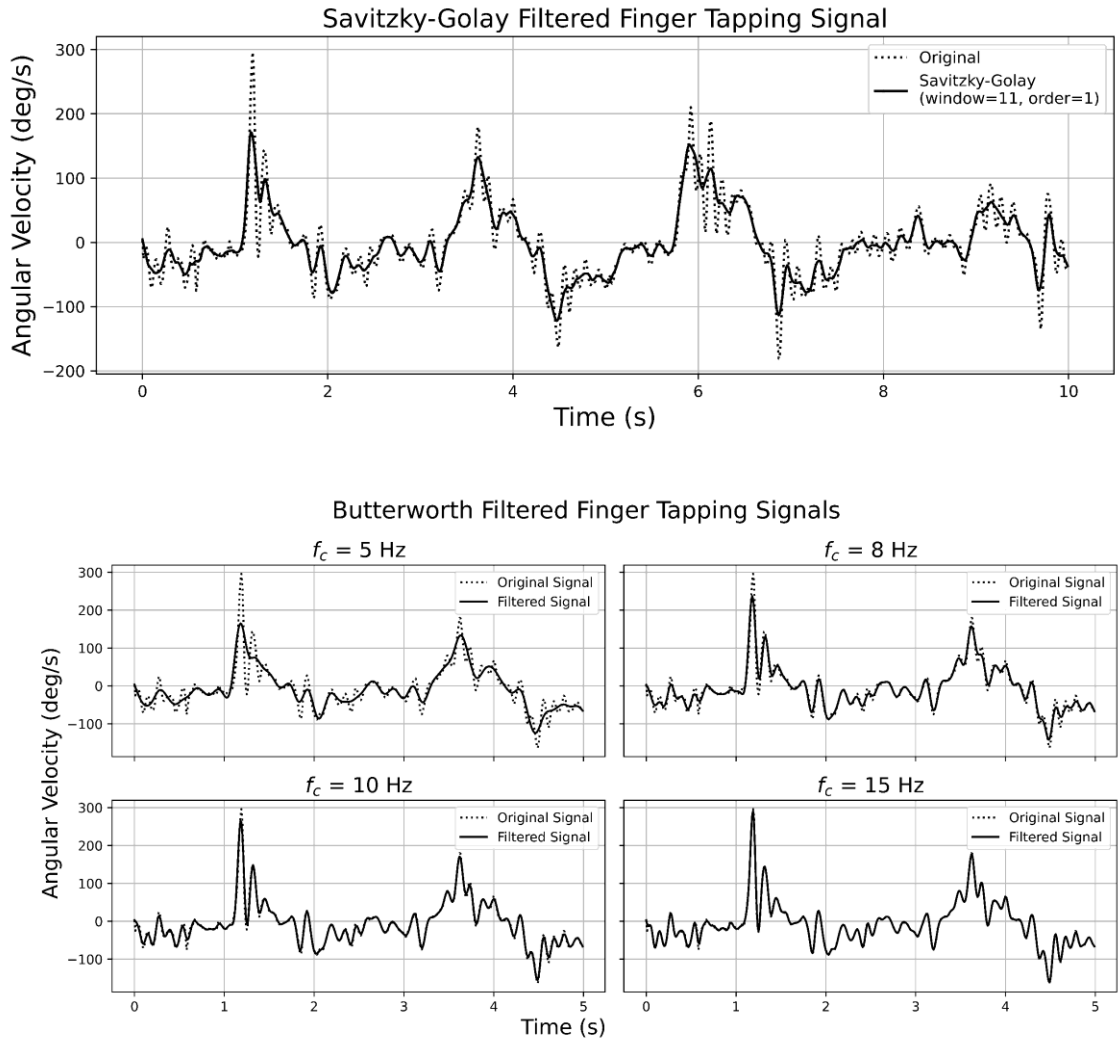


Figure 2.5. (Top) An example of smoothing a finger tapping velocity signal using the Savitzky-Golay filter. (Bottom) An example of low-pass Butterworth filtering (solid lines) applied to a raw finger tapping velocity signal (dotted line) with increasing cutoff frequencies.

2.3.3 Standard Output Metrics

Several factors determine the features extracted from finger tapping time series signals, some of which stem from the depth of information obtained. According to MDS-UPDRS, clinician assigned finger tapping scores are to be determined through observations of key performance measures like tapping amplitude, speed and rhythm. Therefore, it is unsurprising that most studies use some form of finger tapping amplitude to serve as one of the main feature descriptors to

indicate performance. With this in mind, spatiotemporal signals are composed of multiple cycles which can fluctuate considerably in amplitude, frequency and overall complexity over time. For this reason, performance metrics need to summarise the overall quality of movements throughout the test, using these key characterisations to reflect an individual's symptom profile. This has been achieved using a wide range of approaches as shown in **Figure 2.6**.

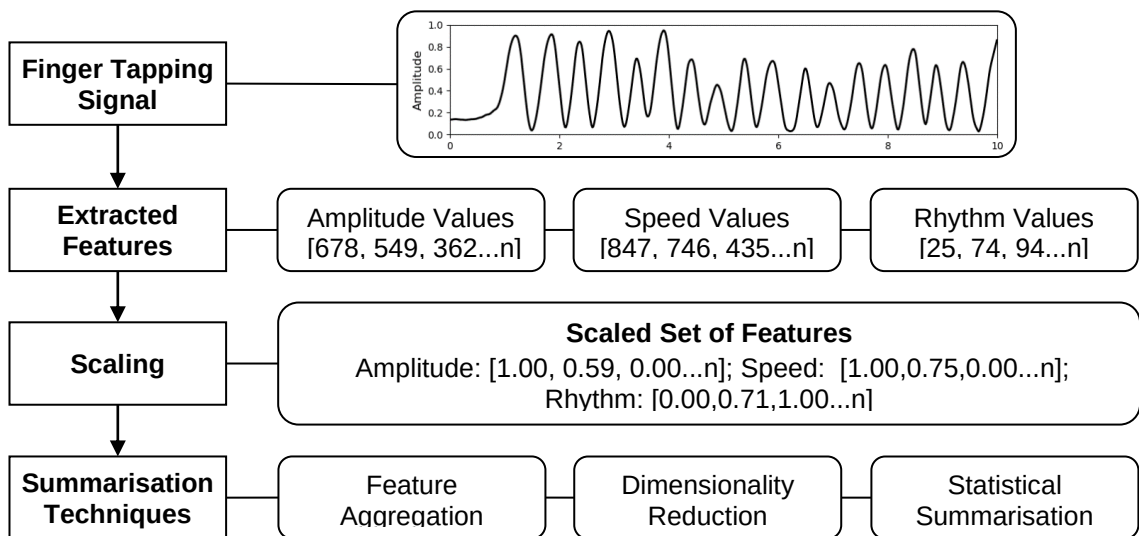


Figure 2.6. Flowchart outlining the process for summarising characterisations of finger tapping using feature extraction from time series signals.

Table 6 highlights that most VC studies recognised both amplitude and speed as primary features, while the derivation of rhythm from time series signals was less commonly reported. It is crucial to point out that identifying which studies extracted the features presented in **Table 6** involved a detailed review that went beyond direct keyword searches. The review considered variations in terminology used across studies i.e. speed and velocity, and in many cases established that different computations of features often led to slight discrepancies between their mathematical representations. This was particularly true for the rhythm metric, where signal frequency was commonly used to derive the rhythm formation of

finger tapping (Krupicka et al., 2017), but through varying definitions. The MDS-UPDRS describes the impairment of rhythm as being marked by “interruptions” or “hesitations” during movement (Goetz et al., 2008). Therefore, several studies have tried to apply this definition by identifying disruptions or difficulties in initiating and sustaining repetitive movement through signal processing. A ‘rolling window’ is often applied to the finger tapping signal to derive the variation of specific signal features across successive time intervals (Espay et al., 2011). Bologna et al. (2016) expressed rhythm as the coefficient of variation of inter-tap intervals, with higher values indicating inconsistencies in repetitive movements. In contrast, a more consistent definition was used for amplitude across all measurement methods, defining it as the distance between the distal index finger and thumb. The same can be said for speed, with most assessments defining it as the amplitude measured over time.

Table 6: Output Metrics for Parkinson’s Disease Finger Tapping Assessment in Visual Computing-Based Studies.

Study	Amplitude	Speed	Rhythm
Khan et al., 2014	✓	✓	✓
Buongiorno et al., 2019	✓	✓	✓
Wong et al., 2019	✓	x	✓
Williams et al., 2020	✓	✓	✓
Park et al., 2021	✓	✓	✓
Monje et al., 2021	✓	✓	x
Guo et al., 2022	✓	✓	x
Li et al., 2022	✓	✓	x
Yang et al., 2022	✓	✓	x
Amprimo et al., 2023	✓	✓	x
Islam et al., 2023	✓	✓	✓
Guarín et al., 2024	✓	✓	x

Other metrics include those that are closely related to or surrogates of amplitude, speed or rhythm. Given their varying computations, their reported feature

descriptions are often based on their intended clinical representation i.e. fatigue (Park et al., 2021; Monje et al., 2021). In the study by Khan et al. (2014), time series signals were divided into distinct time intervals, with key performance metrics like the number of taps being compared between the two segments, **Figure 2.7**. The Cross-correlation between the normalised amplitude peaks were used to indicate arrhythmia, while variation in the number of taps performed between the two time-slots was described as a measure of fatigue.

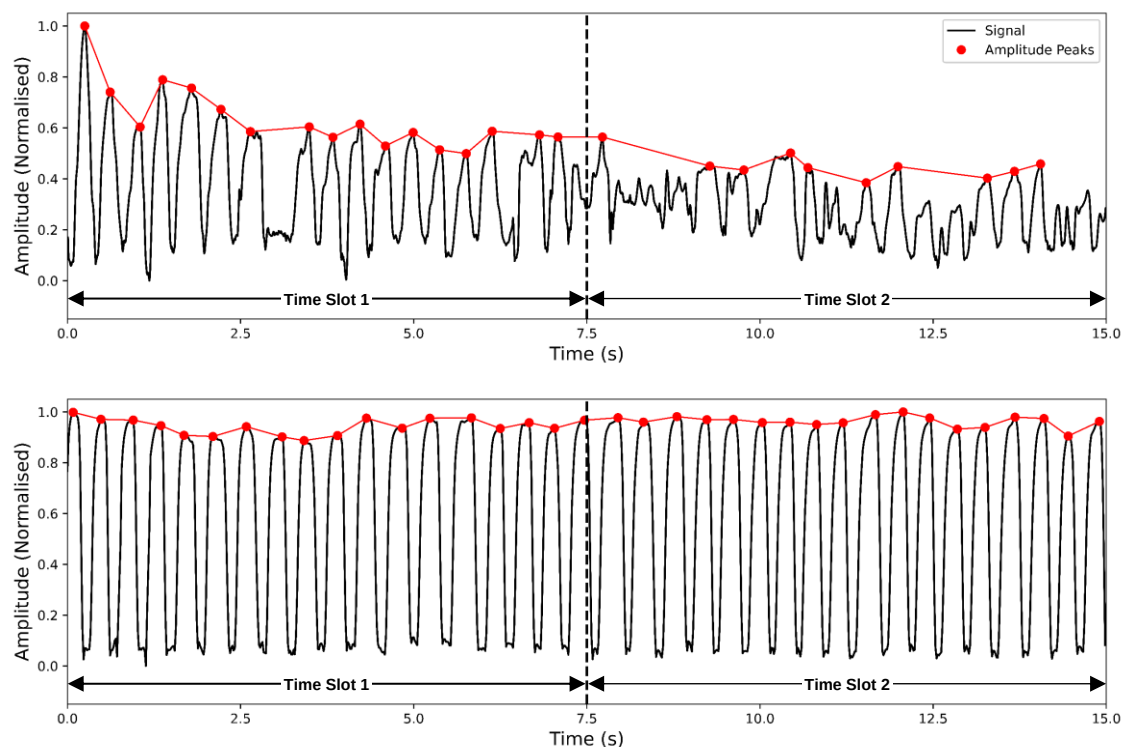


Figure 2.7. An example of time-segmented analysis of finger tapping signals to assess temporal changes in motor performance over Time Slot 1 and Time Slot 2 (Khan et al., 2014). The upper plot shows an apparent deterioration in the signal amplitude and rhythm over time in a Parkinson's disease patient with MDS-UPDRS score 3. The lower plot, from a healthy control with MDS-UPDRS score 0, demonstrates a stable signal across both time slots.

The focus on forming clinically meaningful characterisations of finger tapping prompted most studies to evaluate bradykinesia. As such, one of the primary surrogates for amplitude was its reduction over time, often termed amplitude decrement or 'ampdec'. Most studies derived this measure using a similar time-

slot method to Khan et al. (2014) where the average amplitude was compared across successive time intervals of the FT signal. This bradykinesia characteristic metric was commonly extended to speed reduction metrics, which in effect is directly tied to amplitude decrement. Another frequently reported metric derived from velocity signals was opening/ closing velocity, with an unsurprising dominance across studies that employed accelerometers or gyroscopes (Stamatakis et al., 2013; Yokoe et al., 2009; Krupicka et al., 2017). This metric was generally extracted from velocity signals using the same peak extraction process used to identify amplitude peaks as shown in **Figure 2.8**. All studies measured the number of finger taps performed, regularly comparing them with manual counts to validate the automated output.

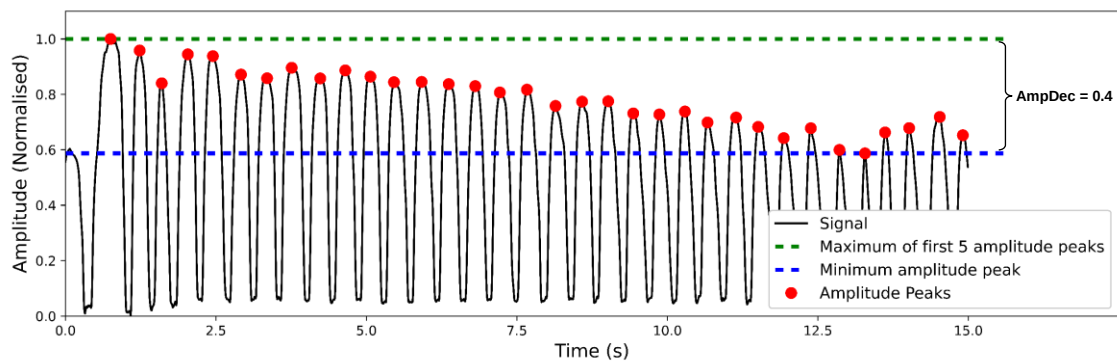


Figure 2.8. Example of computing amplitude decrement (AmpDec) from a finger tapping amplitude signal (Krupicka et al., 2017).

2.3.4 Advanced Approaches: Synthesised and Novel Metrics

In PD motor assessments, a distinct challenge is detecting the temporal deterioration of movements, which is recognised as a key indicator of motor symptom severity (Tinaz et al., 2016). This is especially demanding in the isolated finger tapping test, where its evaluation is typically confined to a ten second period. Though the recognition of gradual reductions in amplitude and speed,

commonly described as the sequence effect, serve as a straightforward marker for PD, advanced approaches have sought to extract more detailed representations of their variations. It must be noted that while the majority of literature provides measures of amplitude, not all analyses investigate the sequence effect (Williams et al., 2020).

Yu et al. (2023) applied piecewise linear regression to fit linear trends to the signal's amplitude peaks. The captured amplitude variations could then be compared between distinct breakpoints as shown in **Figure 2.9**. The use of linear regression lines fit to amplitude data has been previously demonstrated as an effective approach to characterising PD movement, where the line's slope was able to distinguish between PD patients who are on and off medication (Ling et al., 2012). Habets et al. (2023) determined the linear slope for all FT output metrics per “tapping block” emphasising that such representations align with clinically relevant kinematic concepts.

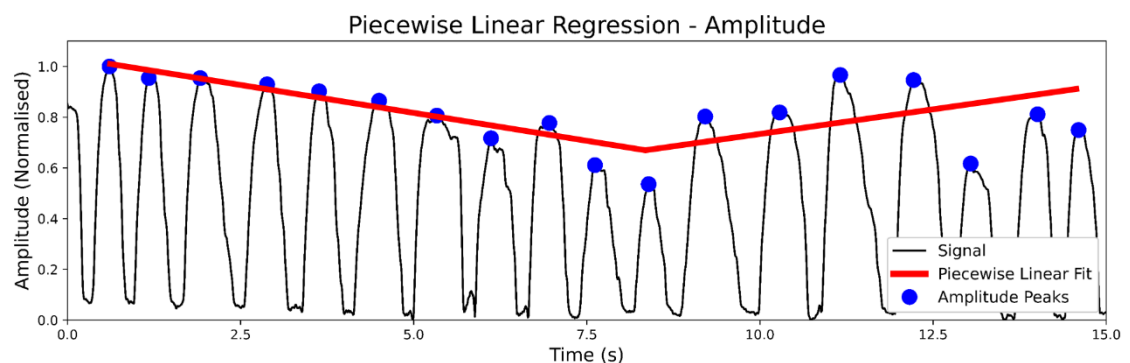


Figure 2.9. Example of piecewise linear regression applied to amplitude peaks in a finger tapping time series signal to identify patterns in movement amplitude over time (Yu et al., 2023).

A study by (Weiss, 1997) demonstrated that PD patients struggle to execute movements effectively, without halts or hesitation, particularly when movements demand accuracy. Detecting the presence of subtle movement halts and

hesitations to indicate underlying motor dysfunction has been previously highlighted as particularly challenging (Monje et al., 2021). A recent study by Yu et al. (2023) noted that these movement interruptions lack detailed definitions and that there is scarcity in assessments that determine these metrics, particularly for repetitive upper limb movement tests. The study proposed a halts metric, where momentary pauses of hand movements were identified by instances of zero amplitude. Hesitations were detected using sudden reductions in velocity, maintaining values above zero but registering those within 20% of the directly preceding and following average amplitudes as a hesitation. Other studies have considered hesitations in their characterisation of finger tapping (Islam et al., 2023; Yang et al., 2022), although did not detail specific thresholds used to identify them. The study by Yang et al. (2022) briefly explained that moments where a patient prolongs a current action are reflected in the flat sections of an amplitude signal's first derivative, which were used to establish the number of hesitations, termed "tapping frozen times". Conclusively, though interruptions to movement may help to highlight difficulties in initiating and sustaining movement, the reliable determination of this feature especially through automated means requires more development, with even more refined methods relying on empirically set thresholds (Yu et al., 2023).

A study by Williams et al. (2020) assessed rhythm formation of FT signals through Fast Fourier Transform (FFT), which analysed the frequency distribution of finger taps. The dominant frequency peak was identified including frequencies within 0.2 Hz on either side, irregular tapping rhythms were associated with more dispersed frequency bands resulting in a broader, reduced intensity of the dominant peak and its neighbouring frequencies. Another investigation using an

accelerometer and VC-based assessment (MediaPipe) in tandem, demonstrated high correlation in estimated movement frequencies of the hand between the two systems (Güney et al., 2022). Dominant frequency was calculated from both measurement modalities, by transforming signals into the frequency domain using FFT, demonstrating strong agreement.

2.4 Motor Symptom Severity Classification

2.4.1 Machine Learning Approaches

In the body of literature, the common objective was to ultimately provide some form of classification. A notable pattern emerges: classification is intended to either identify the presence of PD, i.e. distinguish patients from controls (binary classification) or predict symptom severity in terms of a UPDRS score (multiclass classification) (Amo-Salas et al., 2024). In VC-based assessment methods, the impressive leap from standard video recordings of the finger tapping test to symptom severity classification was achieved through a variety of machine learning approaches, each offering unique advantages and limitations. The common theme across VC-based studies was to develop a contactless, automated, potentially remote and comprehensive (end-to-end) assessment framework, therefore, the implementation of machine learning in their concluding classification stage is expected. Even with device-centric measurement methods, extensive finger tapping datasets are strengthened by machine learning models that can objectively map output features to clinically relevant representations of symptom severity. **Table 7** presents the studies that utilised machine learning techniques to automate the classification of the finger tapping test, providing the classifiers used, the classification problem addressed, the dataset composition and the key performance metrics used to evaluate model performance.

Table 7: Machine Learning Models for Finger Tapping Classification.

Study	Classifier	Classification Problem	Dataset		Performance Measures
			PD	HC	
Khan et al. (2014)	SVM-PUK	Binary Multiclass (3)	13	6	Accuracy
Buongiorno et al. (2019)	Artificial neural network SVM	Binary	16	14	Accuracy Sensitivity Specificity
Wong et al. (2019)	Naïve Bayes Logistic regression SVM-L SVM-R	Binary	20	15	Accuracy Sensitivity Specificity AUC
Li et al. (2021)	Convolutional Neural Network	Multiclass (5)	150	0	Accuracy AUC Precision Recall F1-score
Monje et al. (2021)	Logistic Regression Naïve Bayes Random forest	Binary	22	20	Accuracy AUC
Park et al. (2021)	SVM-R	Multiclass (5)	55	0	Weighted κ Intraclass Correlation Coefficient (ICC)
Yang et al. (2022)	Deep Neural Network	Multiclass (4)	300	0	Precision Recall F1-score
Li et al. (2022)	Fully Connected Network	Multiclass (5)	93	27	Accuracy Precision Recall F1-score
Guo et al. (2022)	k-nearest neighbors Random forest XGBoost SVM-L SVM-R	Multiclass (5)	48	11	Accuracy
Amprimo et al. (2023)	k-nearest neighbors Random Forest XGBoost SVM	Binary	35	60	Accuracy Precision Recall F1-score
Islam et al. (2023)	LightGBM regressor	Multiclass (5)	172	78	Mean absolute error (MAE) Pearson's correlation coefficient (PCC)
Guarín et al. (2024)	XGBoost	Binary	31	26	Accuracy
Butt, A.H. et al. (2018)	Logistic regression (LR) Naive Bayes (NB) Support vector machine (SVM)	Binary	16	12	Accuracy ROC curve (AUC)
Shin, J. et al. (2024)	SVM	Multiclass (5)	45	0	Accuracy Recall Precision F1-score
Habets, J.G. et al. (2023)	Random Forest classification Logistic Regression SVM Linear Discriminant Analysis classifiers	Multiclass (4)	37	0	Mean Prediction Error Confusion Matrix Intraclass Correlation Coefficient (ICC) Random-label Permutation Test
Printy, B.P. et al. (2014)	Random Forest Classifier support vector machine (SVM)	Binary	18	0	classification error ROC curve (AUC)
Arora, S. et al. (2015)	Random Forest Classifier	Binary	10	10	Sensitivity and Specificity Mean Absolute Error (MAE)
Giancardo, L. et al. (2016)	ϵ -Support Vector Regression (SVR)	Binary	5	4	ROC curve (AUC) Sensitivity and Specificity
Lee, C.Y. et al. (2016)	logistic regression	Binary	57	87	ROC curve (ROC)
Aghanavesi, S. et al. (2017)	SVM Linear Regression Regression Trees Multilayer Perceptron	Binary	19	22	Correlation Coefficients
Prince, J. and de Vos, M. (2018)	Deep Neural Network (DNN) Convolutional Neural Network (CNN)	Binary	949	866	ROC Curve (AUC). Accuracy F1 score
He, T. et al. (2024)	Multimodal-multiscale ensemble network (MMSENet)	Binary	225	846	ROC curve (AUC) F1 Score Accuracy Precision Sensitivity

Researchers have experimented with numerous machine learning algorithms to tackle the classification task, and commonly employed traditional machine learning models included Support Vector Machine (SVM) (Khan et al., 2014; Buongiorno et al., 2019; Wong et al., 2019; Park et al., 2021; Guo et al., 2022); Shin, J. et al., 2024); Aghanavesi, S. et al., 2017), Logistic Regression (LR) (Monje et al., 2021; Butt, A.H. et al., 2018; Lee, C.Y. et al., 2016), k-nearest neighbours (KNN) (Guo et al., 2022; Amprimo et al., 2023) and Random Forest Classifier Habets, J.G. et al., 2023; Arora, S. et al., 2015). Additionally, studies frequently explored deep learning models (Buongiorno et al., 2019; Li et al., 2021; Yang et al., 2022; Li et al., 2022; Prince, J. and de Vos, M., 2018).

The majority of studies approached the classification of finger tapping tests as either a binary classification problem or a multiclass classification problem. In studies that used binary classifications, the most common objective was to determine the participant status i.e. PD or healthy control. In a study by Williams et al. (2020), clinician-assigned UPDRS scores from two raters were consolidated into two target classes, representing the severity of bradykinesia as either no/ slight (UPDRS $0 \geq 1$) or mild/ moderate/ severe (UPDRS $2 \geq 4$). Subsequent classification used binary classification to predict one of the two outcomes i.e. UPDRS ≥ 1 or ≤ 1 . This demonstrates an alternative approach to performing binary classification using the symptom severity of participants, rather than using participant group labelling. Another study adopting a similar class aggregation strategy, reported that there were very few participant videos assigned a UPDRS score of 3, therefore, the samples in class 3 were merged into class 2 (Khan et al., 2014). The resulting classification model performed multiclass classification using three target classes: UPDRS 0, 1 and 2. In another study, target classes

were consolidated to four classes (0,1,2,3), with Habets, J.G. et al. (2023) reporting a lack of PD subjects with severe motor impairment. The grouping of UPDRS scores enables a more detailed categorisation of symptom severity, allowing finger tapping videos to be classified into multiple, finer-grained UPDRS classes. This method accommodates datasets that have significant minority classes within the dataset or low separability between classes. Therefore, multiclass classification involving classification into one of five UPDRS scores require large datasets to achieve effective class distinction.

The comparison of results for finger tapping assessments is difficult due to the broad range of machine learning models used and variety of metrics used to assess predictive performance. Furthermore, numerous classification approaches have been used with some focusing on binary classification to distinguish PD patients from healthy controls and others attempting to discriminate the PD participant group from the essential tremor participant group (Barrantes, S. et al., 2017). More detailed approaches attempting to classify on a symptom severity level have also differed in the number of target classes used, despite obtaining labelled data on a five score level. These studies reported a particular difficulty in distinguishing between neighbouring classes like 0 and 1. In such cases, the five UPDRS classes were consolidated into either 2 or 3 broader symptom severity classes such as mild, moderate and severe (Buongiorno et al., 2019; Wong et al., 2019). Some studies explore both binary and multiclass classification approaches to determine predictive performance on their dataset. One study approached this by offering two separate evaluations for predictive performance, with one based on exact UPDRS score and the other considered predictions within one point of the true score as correct (Yu et al.,

2023). The study by Gao, C. et al. (2018) assessed classification performance for each hand individually, asserting that the effects of motor asymmetry and patient handedness should be considered in assessment outcomes. Wissel et al. (2018) examined finger tapping tests through a self-administered tablet-based application, demonstrated that binary classification can be used to predict whether patients are in on or off states of medication. **Table 8** summarises the classification accuracies obtained in each study for both binary (healthy controls vs. PD patients) and multiclass (UPDRS $0 \leq 4$) classification, with the number of target classes outlined for each multiclass classification approach.

Table 8: Model accuracy for binary and multiclass classification.

Study	Group Classification	Accuracy (%)	Multiclass Classification	Accuracy (%)
Khan et al. (2014)	PD vs. HC	95.80	0,1,2,3,4 (5)	88.00
Wong et al. (2019)	PD vs. HC	67.00	Mild, moderate, severe (3)	80.00
Guarín et al. (2024)	PD vs. HC	79.00	N/A	N/A
Guo et al. (2022)	N/A	N/A	0,1,2,3,4 (5)	81.20
Amprimo et al. (2023)	PD vs. HC	90.00	N/A	N/A
Li et al. (2021)	N/A	N/A	0,1,2,3,4 (5)	72.40
Li et al. (2022)	N/A	N/A	0,1,2,3,4 (5)	79.70
Islam et al. (2023)	N/A	N/A	0,1,2,3,4 (5)	50.92
Yu et al. (2023)	N/A	N/A	0,1,2,3,4 (5)	80.00
Barrantes et al. (2017)	PD vs. ET	84.30	N/A	N/A
Gao et al. (2018)	PD vs. HC	93.50 (RH) 88.60 (LH)	0,1,2,3,4 (5)	89.70

2.4.2 Dataset Compositions and Labelling Strategies

The granular nature of assessing finger tapping on a five-point scale relies on a sufficiently diverse dataset. This is so that machine learning models can ‘learn’ the subtle variations across increasing symptom severities and associate them with the correct UPDRS score. Previous studies have linked poor class separation between adjacent UPDRS scores with a lack of representation of mild or severe patients in their dataset (Buongiorno et al., 2019). There are enormous variations in both the total number of participants and the distribution of symptom severity representations in the datasets used to conduct research. **Figure 2.10** shows the participant population for each study, detailing the number of PD patients and healthy controls that were involved in each study. The mean number of total participants recruited across studies was 133 ± 322 (mean $\pm$ standard deviation). The range in the total number of participants varied remarkably from 2 (LeMoyne, R. et al., 2010) to 1815 participants (Prince, J. and de Vos, M., 2018). Seven studies, which are not included in **Figure 2.10** due to their contrastingly large datasets compressing other data in the plot, recruited over 120 participants in their studies, with 144 participants in the study by Lee, C.Y. et al. (2016), 150 by Li et al. (2021), 156 by Gao, C. et al. (2018), 250 by Islam et al. (2023), 300 by Yang et al. (2022), 1071 by He, T. et al. (2024) and 1815 by Prince, J. and de Vos, M. (2018). The ratio of PD patients to controls further varied across studies (1.39 ± 0.92), with a stark contrast in the study by Guo et al. (2022), which recruited more than four times the number of patients compared to controls (48 PD and 11 HC). This data imbalance, more specifically, the lack of representation for more severe cases can significantly compromise a machine learning model’s ability to make accurate predictions for individuals exhibit highly abnormal motor presentations.

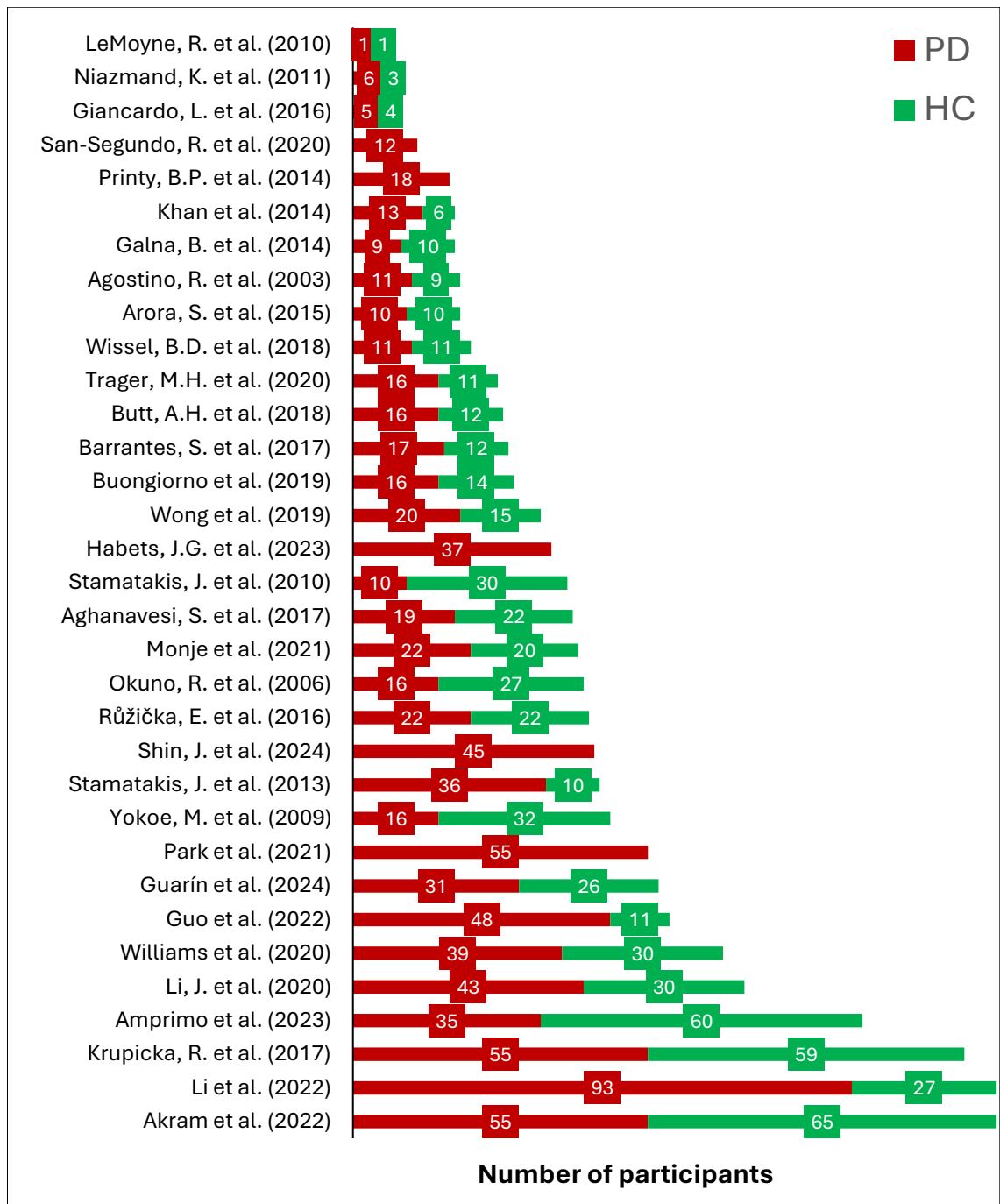


Figure 2.10. Participant compositions: the number of Parkinson's disease (PD) and healthy control (HC) subjects used in existing study datasets that focused on evaluating motor performance through the finger tapping test. Studies are ordered by the total number of participants recruited in their assessment.

In studies designed to classify participants on a group basis, the health status of all participants is confirmed. Patients must be formally diagnosed with PD to

confirm their eligibility for the study and it is ensured that healthy controls are free from any neurological disorders. UPDRS labelled datasets are typically obtained by distributing videos of the isolated finger tapping test to one or more clinician, each video is then blindly scored by assigning a UPDRS score based on the raters visual judgement (0: normal, 1: slight, 2: mild, 3: moderate, 4: severe). The labelled dataset serves two interconnected purposes in studies automating the assessment of the finger tapping test. First, it provides a dataset that can be used to train machine learning models, linking critical features with target class labels. Secondly, it is used to validate the classification model's ability to make correct classifications. This is done by testing the model on unseen data and assessing the agreement between model predictions and clinician scores.

As discussed in **Chapter 1.3.2, Variability in Clinician Ratings**, both inter- and intra-rater variability have been recognised to significantly affect PD assessment outcomes. Therefore, a model's ability to perform classification on finger tapping summary features is strengthened by datasets that are scored by multiple clinicians. **Figure 2.11** provides the number of raters used to assign participants with MDS-UPDRS scores in previous studies. The majority of VC-based studies used 5 or less raters to score finger tapping videos, with one study by Williams et al. (2020) using 22 clinicians, all specialising in movement disorders, to score finger tapping videos. The summary of MDS-UPDRS score distributions across existing datasets are presented in **Figure 2.12**. There is a clear lack of PD patients with moderate (UPDRS 3) and particularly severe (UPDRS 4) symptom severity scores in previous research.

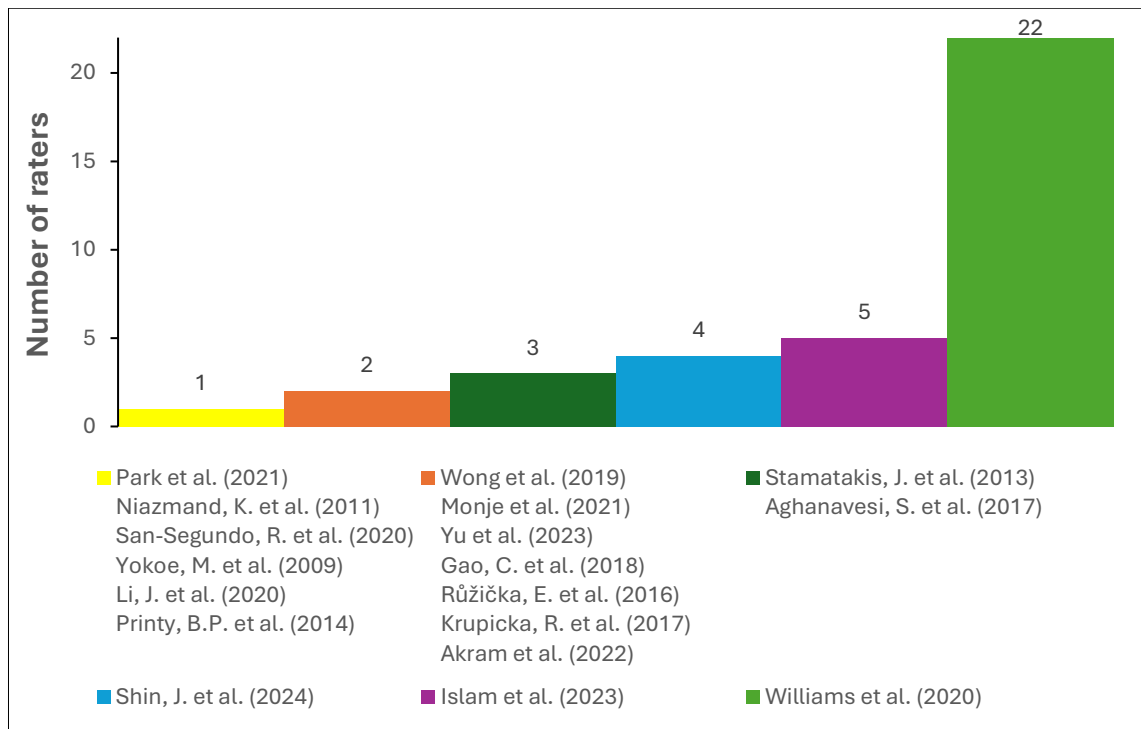


Figure 2.11. The number of raters used to score finger tapping performance across previous studies. Symptom severity scores are based on rater or clinician-assigned UPDRS scores that are used to validate predictive performance in proposed assessment methods.

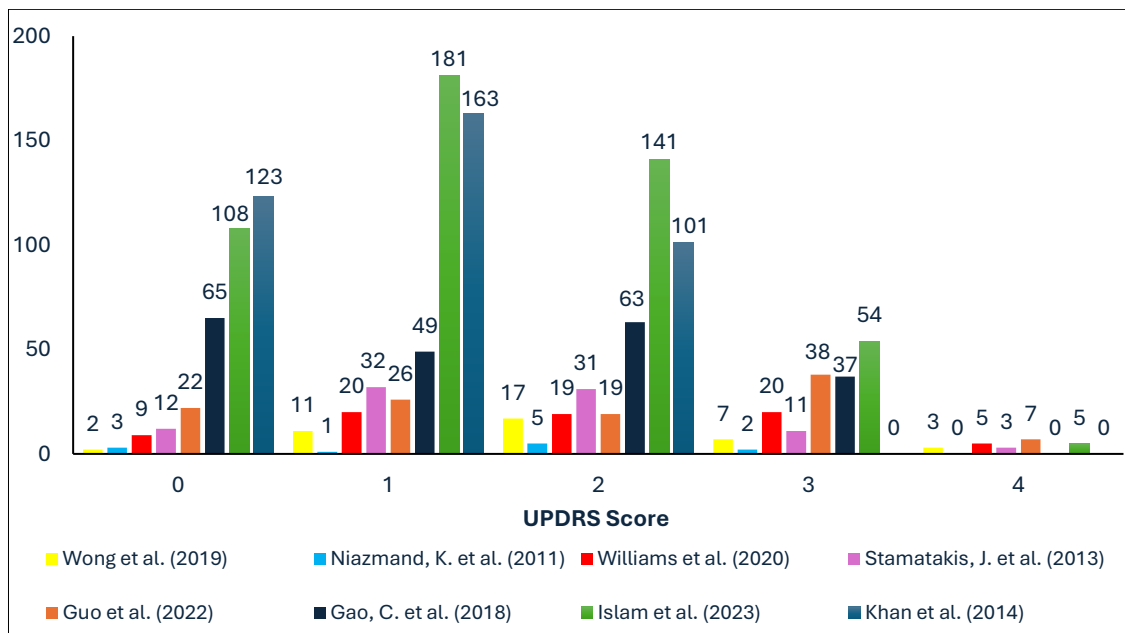


Figure 2.12. The distribution of assigned UPDRS scores in studies that perform multiclass classification for objective finger tapping assessment.

2.5 Summary of Key Findings

In summary, several measurement methods have been employed to capture movement during the Parkinson's finger tapping test. Most studies emphasise objective assessment as the primary motivation for developing frameworks that quantify this upper limb test. Although the majority of previous work share the goal of using objective output features to perform some form of classification, a few exceptions have adopted a more limited approach, concluding evaluations with movement characterisations through key features (Okuno, R. et al., 2006). There is a notable divide in the number of output features across studies, contact-based methods tend to capture a more limited set of outputs, with movement features solely based on moments of contact. In a study that developed a web-based keyboard-tapping test, the absence of data relating to distal finger movement amplitude limited the assessment's ability to fully assess bradykinesia (Akram et al., 2022). Furthermore, it was acknowledged that the variability in keyboard types (key sizes and resistance) across participants could have affected test outcomes. Contact sensors alone do not capture the detailed kinematic features required to characterise PD, although they demonstrate strong potential in providing reliable temporal data for touch events of the forefinger (Yokoe et al., 2009).

Other approaches, including sensor-based devices, motion capture systems and VC techniques extract a broader range of features that consider all movement phases of the alternating finger tapping test. Despite the promising potential of these previous methods to discern meaningful motion parameters, the requirements for implementation vary massively. With sensor-based methods like accelerometers and gyroscopes, the placement of sensors are critical to

recording accurate data that reflect ground truth movements. Gyroscopes are susceptible to sensor drift while accelerometers are highly sensitive to noise, therefore, both devices are commonly used in combination to generate six degrees of freedom movement data (Saber-Sheikh et al., 2010). Several techniques were used to attach these devices to the distal index finger and thumb such as fingerstalls and gloves, which included IMUs consisting of a triaxial accelerometer and a triaxial gyroscope (Guidolin et al., 2021). Despite the miniaturisation of sensors, each gram of added mass to the human hand has been proven to increasingly affect movement, revealing a fundamental flaw in sensor-based examinations of finger tapping (Kozlowszky et al., 2018). This is especially the case in assessments that use embedded sensors in devices like smartphones to capture movement, LeMoine, R. et al. (2010) noted that the iPhone used in their study had a mass of 133 g, which was significantly higher than the mass of other wireless accelerometer systems.

Studies using marker-based motion capture systems were able to derive clinically meaningful parameters such as bradykinesia from finger tapping tests. Krupicka et al. (2017) demonstrated this by determining metrics like amplitude decrement using the BradyKan motion capture system. Another study highlighted the potential of these systems by effectively discriminating PD patients from healthy controls. Two studies explored a motion capture systems comprised of the Leap Motion Device and Leap Motion Controller to quantify finger tapping (Shin et al., 2024; Butt et al., 2018). In these methods, detailed motion properties such as the Euclidean distance and angle between the forefinger and thumb were recorded without the need for physical markers. Despite the positive step towards contactless motion capture, these systems still require specialist equipment

including infrared cameras and image processors, hindering their widespread deployment. Furthermore, infrared cameras are not immune to the challenges presented by complex finger tapping movements with Butt et al. (2018) reporting that certain gestures could not be captured, leading to compromised motion data. Systems like the Leap Motion Controller further pose a particular limitation for capturing fine movements due to their inconsistent frame rates.

Considering all the points discussed, existing methods overwhelmingly point towards the more recently established VC domain in this application as the most viable direction for development. That being said, VC-based studies also present substantially varying approaches for assessing finger tapping using standard video recordings. Spatial data has been captured in both two and three-dimensions to generate spatiotemporal signals of this repetitive movement task. In both cases, most studies demonstrated that core movement abnormalities associated with PD could be discerned from video data. Subsequent to data capture, the common approach was to analyse spatiotemporal sequences, with most studies focused on extracting key summary features from computer-generated finger tapping time series signals. The majority of studies used the movement amplitude of the distal forefinger and thumb as the foundational time series data to derive clinically relevant metrics. In the body of literature, there has been a lack of VC-based studies that address the inherent perspective distortion associated with the camera position relative to hand movement plane. This distortion has only been adequately addressed in two studies that made adaptations to their assessment framework in an attempt to mitigate this source of error. Khan et al. (2014) applied concepts of the golden ratio called the “proportions of man” which assumed the hand length to be equal to the face

height. This anthropometric principle was used to normalise finger tapping amplitude signals, thus adjusting for variations in the distance between the hand and camera. Additionally, a moving average was applied to finger tapping signals to align the signal over the horizontal zero-axis. This established a calibrated signal baseline that accounts for gross lateral movements of the hand. However, the study did not include any PD patients with moderate or severe (UPDRS: 3 or 4) symptom severities. As such, the techniques used to normalise and calibrate signals were not substantiated with hand movements that exhibit more complex displacements across video frames. The study further reported that one of the observers scored finger tapping tests with prior knowledge that the patients were in the advanced stage of PD, introducing significant bias into the entire research. In another study, perspective distortion was addressed right from the outset of the detection stage. Islam et al. (2023) extracted the virtual angle formed by the distal forefinger, wrist and distal thumb landmarks, using it to generate time series signals with finger tapping angle as the y-variable. The use of angle instead of the widely used finger to thumb distance assumed that the three segment lengths used to derive angle measures would vary proportionately during relative movement between the camera and hand.

Post et al. (2005) emphasised the need to evaluate clinician ratings and procedures prior to undertaking longitudinal studies of disease progression or clinical trials. While most studies focused on integrating critical components of the MDS-UPDRS into their assessment framework, there is a notable scarcity in detailed discussions that address the numerous aspects involved in participant scoring procedures. Instead, the reported outcomes of model performance were generally addressed through predictive performance or correlation between

computer outputs and clinicians, assuming clinician assigned MDS-UPDRS scores to be the ground truth. This approach assumes both: conformity in clinician scores and importantly, that observer interpretation was performed under the same conditions that exist in real-world clinical setting. More specifically, there was no mention of the explicit instructions given to clinicians in terms of basing their visual judgement of finger tapping performance on the first observation. While there is an enormous body of research focused on developing standardised PD motor examinations for critical movement assessments in the UPDRS-ME, there is currently no standardised protocol for scoring video recordings of the finger tapping test for research purposes. The distribution of finger tapping videos allows for a more detailed observation, with the option to replay and pause videos. This offers a level of observation that is not available to the expert eye, particularly in fast-paced clinical settings. There was also a lack of mention regarding the level of knowledge clinicians had about participants and whether the videos presented to them were confined to hand movements or showed the entire body, prior to clinician scoring. In the cross-sectional study by Gao, C. et al. (2018), raters were not blinded to the participants' diagnoses, potentially introducing bias. Similarly, Khan et al. (2014) reported a low true positive rate (53%) in the normal (UPDRS: 0) class, linking it to potential bias caused by one of the raters observing head movements in addition to the finger tapping test. This broader context can reveal a number of examination signs that may not be apparent in an isolated view of the hand or may not manifest in the upper limb at all. That being the case, asymmetry in the right and left hands of participants was another area of focus with limited evaluations.

The presentation of PD motor symptoms are widely understood to manifest asymmetrically, meaning that symptoms are typically more pronounced in one side of the body compared to the other (Barrett et al., 2010). In the research conducted by Miller-Patterson et al. (2018), 109 PD patients were observed at least twice over a five-year period and it was established that motor asymmetry in PD patients is both evident at disease onset and persists throughout the entire progression of the disease. In the study by Guarín et al. (2024), the assessment was centred around “What the trained eye cannot see” with a particular focus on incorporating asymmetrical motor presentations into the classification process. Furthermore, the proposed classification approach involved calculating the ratio between the right- and left-hand output features for each participant. This indication of the level of asymmetry was then passed to three independent classification models which predict the probability of an individual belonging to the PD or healthy control group. The study highlighted the importance of assessing asymmetry in PD assessments and reported a notable drop in classification accuracy when the movement side was not considered, with accuracy ranging between 40% and 72%. However, the study included only 31 PD patients, and their symptom severities were not disclosed. This, combined with the significant drop in classification accuracy when asymmetry was excluded from the analysis, suggests that although asymmetry may be a credible distinguishing feature when motor impairment is considerably more pronounced on one side, it may be less effective in patients that present with less contrasting movement performance. This could also be attributed to model overfitting where even slight discrepancies between right- and left-hand features result in false positive predictions i.e. a healthy control predicted to be a PD patient.

The challenge presented in achieving high correlation between objective output features and clinician ratings through the finger tapping test has been previously associated with the crude nature of the MDS-UPDRS criteria (Akram et al., 2022). The development of this discreet integer scoring system designed to standardise the visual examination of PD was tailored to match an observer's visual capacity. Therefore, training classification models using MDS-UPDRS scores as target class labels for fine-grained movement data that is beyond the limits of the human eye, poses major difficulties for the decision-making process of a classification model. Notably, the agreement between computer-derived finger tapping features and the more specific subscores of the MBRS (amplitude, speed and rhythm) have only been evaluated in one study by Williams et al. (2020). Nonetheless, the extracted features used to characterise finger tapping movements must provide strong patterns within each class, distinct from neighbouring symptom severity scores. This is a complex task that necessitates detailed feature extraction, which is reliant on a diverse dataset that captures a wide range of movement abnormalities. **Figure 2.10, Figure 2.11 and Figure 2.12** demonstrate the enormous variation across previous datasets formed by participant populations, the number of clinicians used to score finger tapping videos and the distribution of clinician-assigned MDS-UPDRS scores across participants. Complicating matters further, the level of experience among raters has also varied within previous research with the study by Islam et al. (2023) using two specialists and three non-specialists to score participants, while the study by Aghanavesi, S. et al. (2017) reporting significant inter-rater variability among clinicians, potentially due to the lack of training for consistent scoring. Islam et al. (2023) noted that their proposed methods frequently predicted MDS-UPDRS scores of one for healthy individuals with a clinician assigned scores of

zero. Their classification model misclassified 63% of ground truth zero symptom severity scores, averaging a prediction error of 0.58. Moreover, low prediction accuracy was also obtained for finger tapping videos with severity 3 and 4. For this reason, previous studies have also explored consolidating target class labels, with neighbouring scores like zero and one being classed as ‘mild/ slight’ (Buongiorno et al., 2019; Wong et al., 2019). Another study implemented a similar concept which considered misclassifications within one level of the true MDS-UPDRS score correct, this approach resulted in an increase of 13.3% in prediction accuracy compared to the original five-score classification method (Yu et al., 2023). Unsurprisingly, these multifaceted factors make gauging the true potential of assessment methods exceptionally challenging. The need for substantial datasets has even been brought to attention by studies that lacked control groups, stressing that the full spectrum of motor severities, including normal movement, are needed to facilitate clear decision boundaries between assessments (Printy, B.P. et al., 2014). In conclusion, irrespective of the classification problem, accurate characterisation of clinically relevant movement features must remain central to research, this ensures that regardless of the datasets obtained to substantiate predictive performance, the foundations of object assessments are built on accurate representations of movement.

2.6 Critical Research Gaps

With the growing prevalence of PD (Naghavi, 2019), the development of objective assessment methods that evaluate PD motor examination signs have seen notable progress. That being said, the overarching research aim, across the studies reviewed, remains vague in terms of their intended benefits to clinical assessments of PD, as do the methods by which this could be achieved. One

possibility for this is the emphasis on attaining high classification accuracies of symptom severity to indicate assessment performance. This presents a critical research gap to devise a robust, automated, real-time and remote assessment method that enhances assessment outcomes for PD patients. These factors would facilitate more frequent monitoring of disease progression resulting in earlier detection of motor symptom fluctuations. Routine intervention further provides more opportunities to assess treatment response, increasing confidence in its effectiveness and allowing timely adjustments to be made.

Only one VC-based study in the literature review extended their analysis to include several upper limb tests (Monje et al., 2021). However, the dataset used to validate this proof-of-concept study consisted of only 42 participants (22 PD and 20 controls), with the patient cohort primarily consisting of early-stage PD. Therefore, the classification problem was limited to a broader group-level task, discriminating PD patients from healthy controls. This presents a clinically significant knowledge gap to ascertain whether upper limb bradykinesia presents varyingly across different PD motor examinations such as finger tapping, opening-closing and pronation-supination hand movement tests. This could provide insights into how variations in distinct motor examinations influence outcomes in the classification of symptom severity. Additionally, there has been limited investigation into how alternative class labelling strategies, such as consolidating neighbouring MDS-UPDRS scores into broader target class labels, impact the model's discriminatory power in distinguishing between these symptom severity levels during the classification task.

Chapter 3: Real-Time Video Quantification of Parkinson's Finger Tapping Using a Deep Learning Framework

3.1 Introduction

The primary aim of the current study was to provide an assessment method that objectively measures finger tapping using standard video recordings. Deep learning algorithms have the potential to provide contactless assessment techniques for clinical movement analysis. As such, this chapter propose a model framework that incorporates custom trained object detection algorithm, You Only Look Once (YOLO), to extract domain specific features of finger tapping. The proposed approach to assessing finger tapping using clinically significant variables provides a strong characterisation of movement dynamics. Importantly, input-versatility by use of digital video is fundamental to this system's design as it makes application in more natural setups and clinical settings possible. This approach has the fundamental strength of operating in real-time, generating critical performance measures of finger tapping from standard video inputs. This has obvious strengths for deployment in clinical settings over previous studies. Furthermore, there is still a need for a visual Computing (VC) assessment method that is substantiated by a diverse clinical rating dataset including clinician-assigned scores from the Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) and subscores of the Modified Bradykinesia Rating Scale (MBRS). It is hypothesised that the presented assessment method will produce output features that achieve statistically significant correlation with clinical rating scores and with capacity to discriminate between the PD and control participant groups.

3.2 Data Acquisition: Study 1

3.2.1 Ethical Review (1)

All participants provided informed written consent. The work was approved by the North of Scotland Research Ethics Committee, United Kingdom Health Research Authority (IRAS project ID 256116).

3.2.2 Protocol: Video Recording Finger Tapping

Video recordings of finger tapping were acquired by Dr Stefan Williams, a consultant neurologist at Leeds Teaching Hospitals. Videos were recorded using a GoPro Hero 5 camera placed on a tripod which was positioned approximately one meter away from each participant's hand. Participants were seated, and the elbow was stabilised by resting on the arm of an armchair. It was ensured that only the participant's forearm and hand were captured in frame, see **Figure 3.1**. They were then instructed to tap their dominant hand's index finger and thumb together "as quickly and as big as possible" for at least ten seconds. This was then repeated with their non-dominant hand. Videos were recorded at 60 frames per second with a 1920 x 1080 resolution using the digital camera. Recording conditions consisted of a generally plain background behind participants with only ambient lighting. Except for trimming videos to a uniform ten seconds, no extra efforts were made to enhance recording conditions i.e. background subtraction, and the VC-based analysis was performed on the raw 10 second finger tapping videos.



Figure 3.1. Finger tapping sample frame recorded with a GoPro Hero 5 camera placed on a tripod, capturing only the participant's hand and forearm.

3.2.3 Dataset 1: Clinical ratings

The study investigated videos from three participant groups: thirty-seven PD patients, twenty older controls (OC), aged 65 or older, and nine younger controls (YC), aged under 65. Participants produced a finger tapping video from each hand except for five PD patients and one OC who produced videos from only one hand, which resulted in one hundred twenty-six videos in total (69 PD hands + 39 OC hands + 18 YC hands). From the one hundred twenty-six video recordings, one hundred four were processed successfully by the automated assessment method. All participants in the PD group had previously been formally diagnosed with PD prior to participation in the study. Twenty-one consultant neurologists across the United Kingdom, all specialising in movement disorders, were asked to score thirty videos each (randomly distributed) from the 126 rapid finger tapping videos. All clinicians were blinded to the diagnosis/ participant group and independently rated finger tapping videos using the MDS-UPDRS and MBRS criteria. There was a median of five raters for each video. **Figure 3.2** provides the breakdown of clinician-assigned MDS- UPDRS scores for the participant dataset,

with the top bar chart displaying the score distribution across the PD, YC and OC participant groups and the bottom bar chart displaying the score distribution across the PD and control groups, where younger and older control labels were combined into a single control group. The separate clinician-assigned amplitude, speed and rhythm component scores of the MBRS are displayed in **Figure 3.3**.

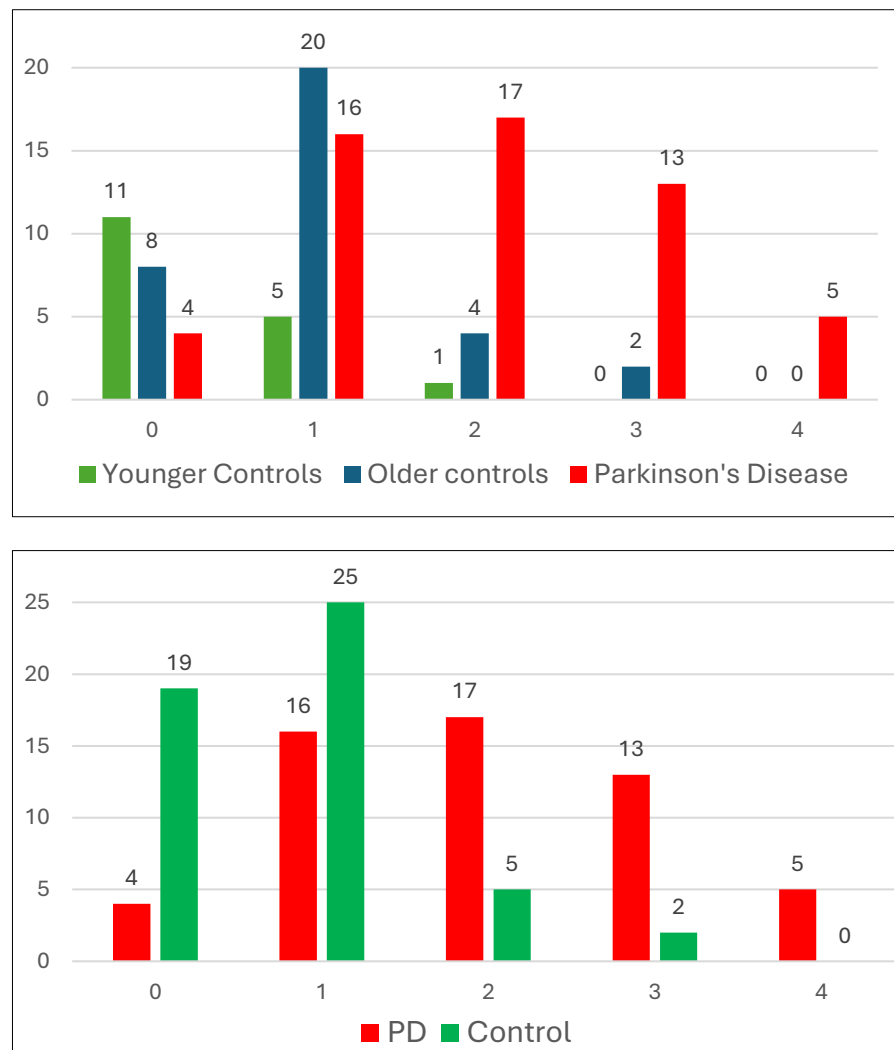


Figure 3.2. MDS-UPDRS Score Distributions. The top bar chart summarises the participant distribution where healthy controls are divided into younger and older controls resulting in three participant groups: younger controls, older controls and Parkinson's disease patients. In the bottom bar chart, the younger and older control groups are consolidated into a single healthy control group. The resulting MDS-UPDRS scores are labelled using the healthy control and Parkinson's disease groups.

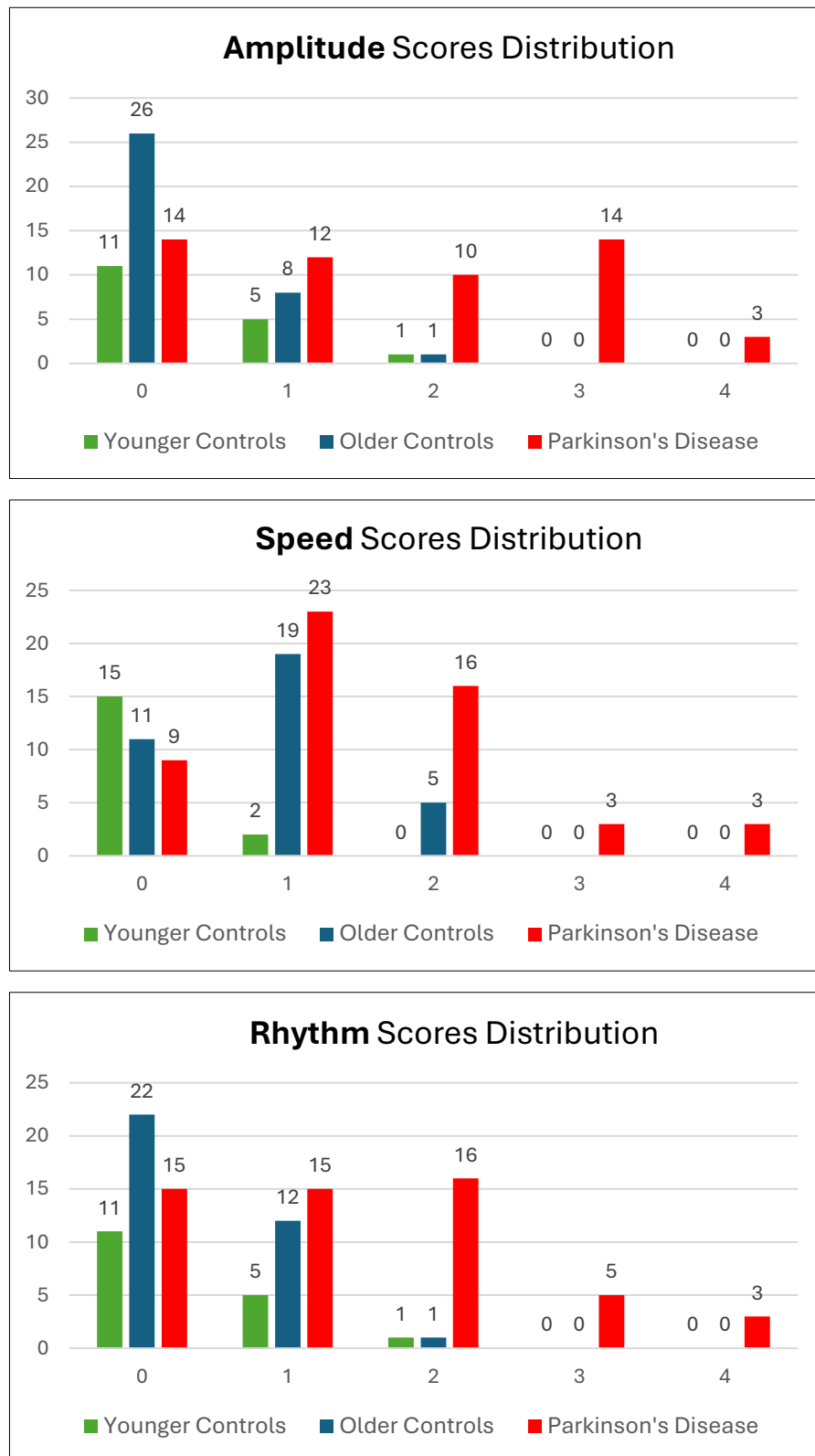


Figure 3.3. Modified Bradykinesia Rating Scale (MBRS) subscore distributions. The median scores of 21 neurologists are provided in separate bar charts for the amplitude, speed and rhythm components of the MBRS. Each bar chart shows score distributions from the younger control, older control and Parkinson's disease participant groups.

3.3 Methods

3.3.1 Image Annotation Process

YOLO models are typically pre-trained on large datasets such as COCO, which include general object categories like vehicles and household items. Therefore, a custom-trained version of YOLO was required to track and distinguish the independent movements of the index finger and thumb in finger tapping videos. The subsequent section outlines the process undertaken to annotate these anatomical segments in image frames extracted from the finger tapping video dataset. First, the scene video filter setting in the VLC player software was used to extract 600 frames from each 10 s video across the 126 rapid finger tapping videos obtained in the study. To incorporate a diverse range of hand orientations for image annotation, 20 image frames were randomly selected from each 600-frame video, producing a total of 2520 annotatable frames, **Figure 3.4**.

Labellmg is a graphical image annotation tool written in python and supports YOLO format. It is a user-friendly tool that allows for easy image annotation which is required to generate labelled data and consists of using the computer mouse to manually create bounding boxes that confine the desired characteristics of the index finger and thumb in each image. Annotation then occurs by associating each label with a corresponding index finger or thumb class. In the context of this research, the focus is on the annotation of the index finger and thumb in images. The aim is to generate a dataset of labelled images that can be used to train the model to detect and identify these segments in rapid finger tapping videos. By using Labellmg, the process of annotation is made easy, allowing for a large dataset to be generated relatively quickly and efficiently. Once the annotation is complete, a text file of YOLO format is generated for every image. The file

contains information about each object class followed by the corresponding x, y coordinates of the upper left vertex of the label box and the width and height dimensions. This information can then be used to train a machine-learning model to detect and identify the segments within successive video frames.

The Index finger and thumb were the two classes annotated in each frame and were labelled by creating a bounding box from the finger and thumb metacarpophalangeal joint to the finger and thumb tip, see **Figure 3.4**. The finger tapping videos were captured from a side profile, hence labelling the entirety of the index finger and thumb maximises the identifiable features incorporated in the training and testing images. To simulate real-world conditions, image frames with background noise and varied lighting were included in the training dataset to ensure the model can cope with the noise commonly found in clinical settings.

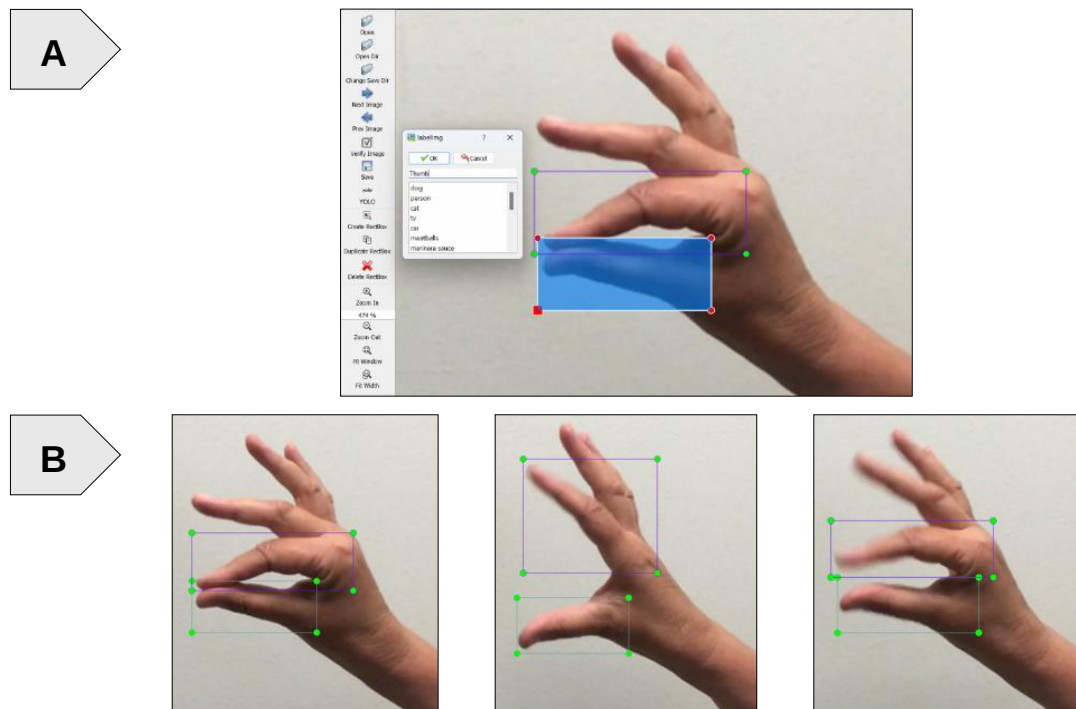


Figure 3.4. Image annotation process using Labellmg. (A) Annotation of the index finger and thumb; (B) annotation during varying levels of extension and flexion movements of the index finger and thumb to diversify the annotated image dataset used to train the YOLO model.

3.3.2 Object Detection Algorithm

In this research the deep learning-based object detection algorithm Tiny-YOLOv3 was employed for the object detection model. The deep learning-based approach utilises a single convolutional network capable of predicting multiple bounding boxes simultaneously. The YOLO object detection model developed by Redmon et al. (2016) was first introduced as a groundbreaking approach to object detection. As the name suggests (You Only Look Once), YOLO handles object detection in a single, unified process that happens at 'first glance', mimicking the way a human instantly recognises, identifies and locates an object at first sight. The model frames object detection as a single regression problem, predicting both spatially distinct bounding boxes and the associated class probabilities directly from the full image in a one-step evaluation. The original YOLO architecture consists of a single convolutional neural network (CNN) comprised of 24 convolutional layers followed by two fully connected layers. While Tiny-YOLOv3 is comprised of seven convolutional layers and six pooling layers. YOLO is implemented using the Darknet framework, an open-source neural network. This unified architecture is extremely fast, with the base YOLO model capable of processing images in real-time at 45 frames per second (Redmon et al., 2016). The selection of YOLO as the data extraction mechanism for finger tapping movements was based on its simplistic pipeline and incredibly fast detection performance, making it possible to process videos in real-time.

Objects are detected by dividing the input image into a grid, predicting bounding boxes and class probabilities directly through a single network pass, and then using confidence thresholds to identify and localise objects in real-time, see **Figure 3.5**. Subsequent advanced versions of the YOLO model have since been

developed including: YOLO V2, YOLO V3 YOLO V4 and YOLO V5 (Jiang et al., 2022). The YOLO model was trained on the COCO (Common Objects in Context) dataset, which contains a large variety of images and object classes. Importantly, its framework supports customised object detection, allowing users to manually annotate and label objects within images. The custom model can then be trained and tested on a custom dataset to detect features like the index finger and thumb across video frames; during model inference, it outputs 2D coordinates (x, y) of bounding box centres, forming the basis of object tracking across videos frames.

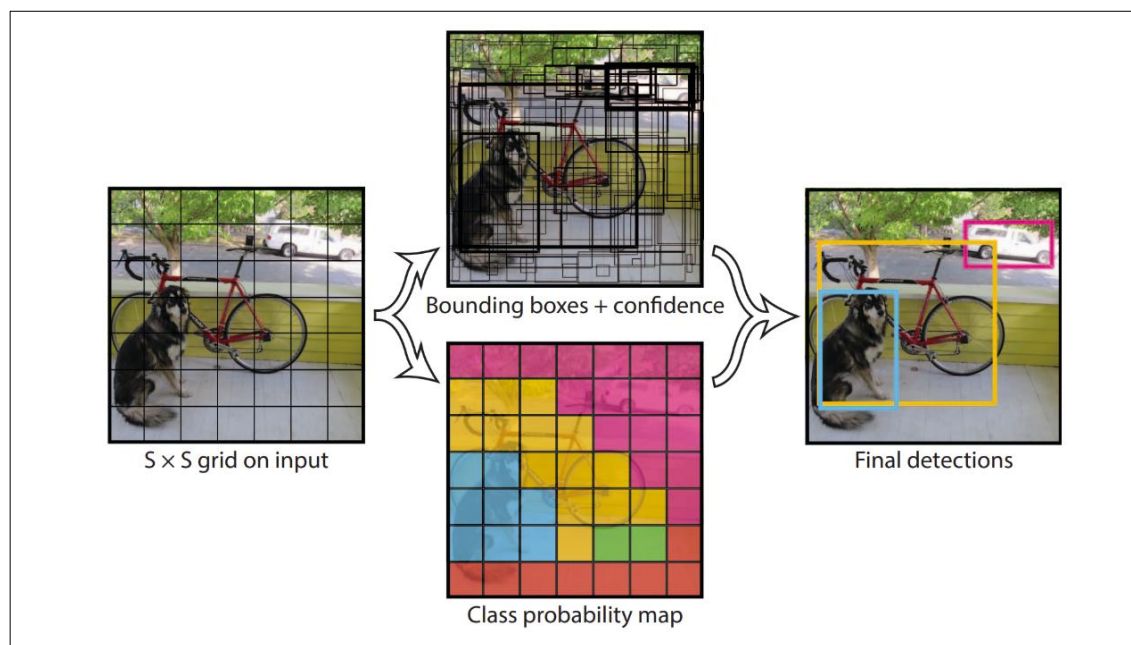


Figure 3.5. The YOLO model treats object detection as a regression problem. Input images are divided into an $S \times S$ grid, with bounding box predictions generated for each grid cell along with corresponding confidence scores for each box and class probabilities. The predictions are encoded as an $S \times S \times (B * 5 + C)$ tensor (Redmon et al., 2016).

3.3.3 Model Framework

The assessment system was developed in Python using the dedicated Integrated Development Environment (IDE) PyCharm. Initially, critical details about the input specifications are recorded including the video frame rate used to generate the

time variable and the input spatial resolution. YOLO then processes incoming frames by evaluating the whole image and making simultaneous predictions of what constitutes the index finger and thumb. Finally, the index finger and thumb bounding boxes, corresponding class ID and detection confidence are set to display for videos and real-time inputs to give a visual representation of how the rapid finger tapping motion is tracked.

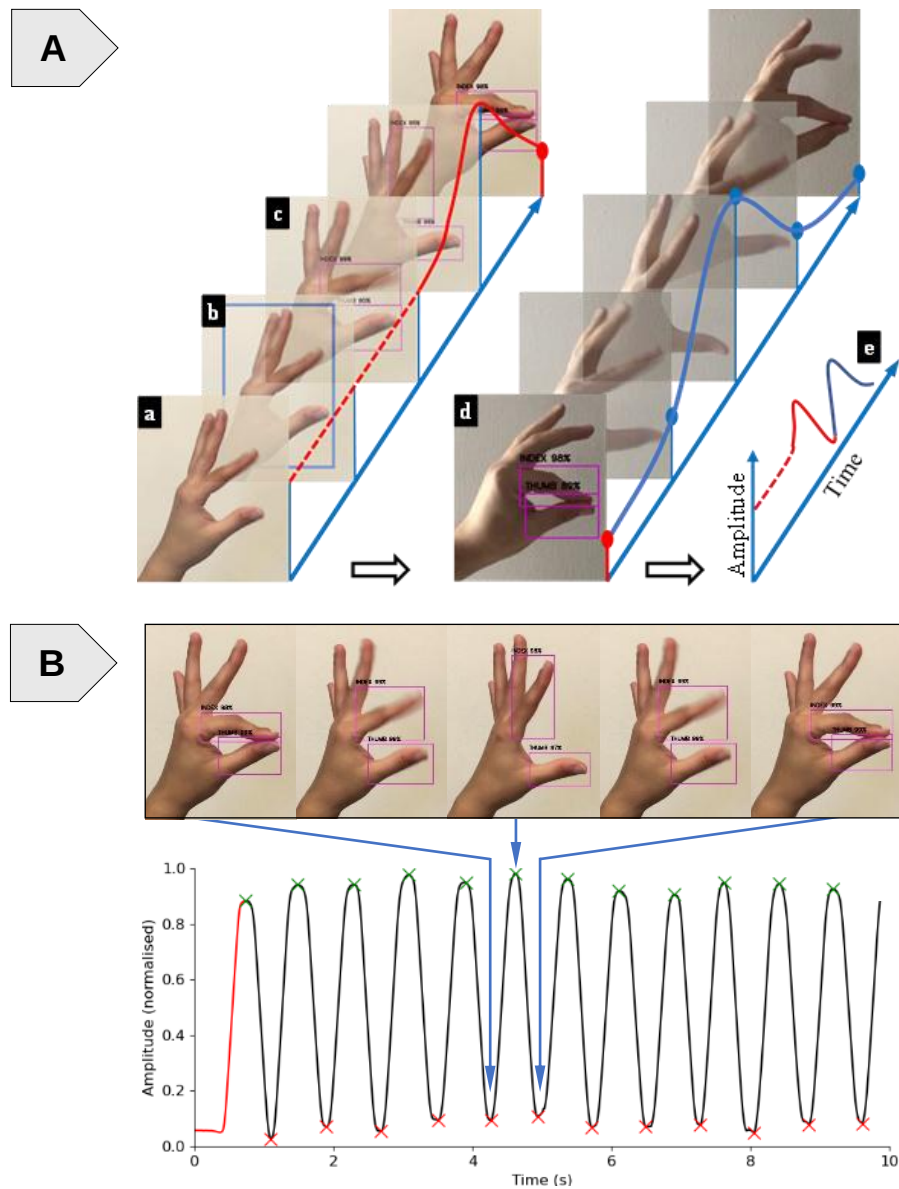


Figure 3.6. (A): Generating finger tapping waveforms: (a) raw video input, (b) hand detection, (c) index finger and thumb detection, (d) finger tapping onset and (e) time series signal frame. (B): Resulting Finger tapping time series signal.

The motion analysis compares the bounding box coordinates in the current frame to a previously referenced frame to deduce movement, see **Figure 3.6**. The standardised pixel displacement between the centre coordinates of the bounding boxes is stored for each successive frame with its corresponding timestamp. Such frame-by-frame detection ensures fine movements are captured. Coordinates are taken from the centre of each bounding box rather than the default upper left vertex to minimise movement readings resulting from a change in bounding box size.

3.3.4 Time Series Signal Analysis

This study presents twelve extracted movement features used to evaluate finger tapping. The features given in **Table 9** were extracted from the finger tapping waveform of each participant. **Figure 3.7** illustrates the foundational finger tapping amplitude time series signal that was used to derive all movement features. The pixel distance between the index finger and thumb bounding box was recorded for each video frame. Python was used to apply the time variable (x-axis) to time series signals by dividing the time between two consecutive frames ($F_i - F_{i-1}$) by the input video's frame rate (FPS), using it to express the cumulative frame number in seconds (**Equations 1 and 2**). The open-source library OpenCV was used to determine the frame rate for each input video.

$$Time (T) = \frac{F_i - F_{i-1}}{FPS} \quad (1)$$

$$T_{cumulative} = \sum_{i=1}^N \left(\frac{F_i - F_{i-1}}{FPS} \right) \quad (2)$$

1. T_n , the number of taps indicates the number of times individuals taps their index finger and thumb together in a ten second video, **Figure 3.7**. The SciPy library used in Python offers a 'find peaks' function which was used to extract

both peaks and troughs. A full tap consists of a trough followed by a peak and another trough.

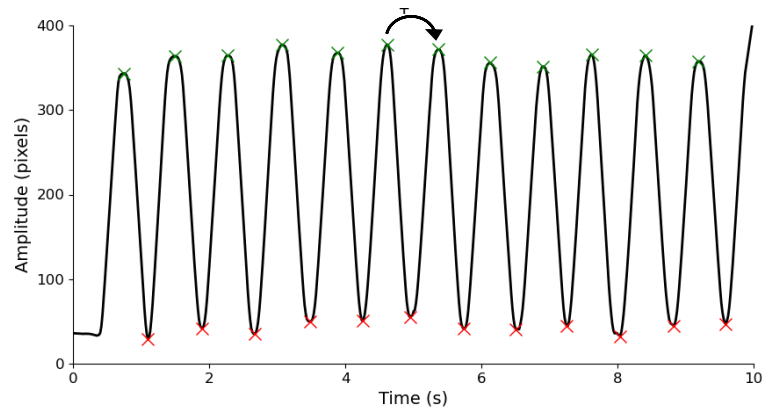


Figure 3.7. The foundational finger tapping amplitude signal used for subsequent feature extraction. The signal is generated by the varying distance between the centre coordinates of the index finger and thumb bounding boxes.

2. TD, tapping distance is a cumulative measure of the pixel separation between the index finger and thumb bounding box. For this, the absolute distance is calculated between adjacent peaks and troughs, **Figure 3.8**.

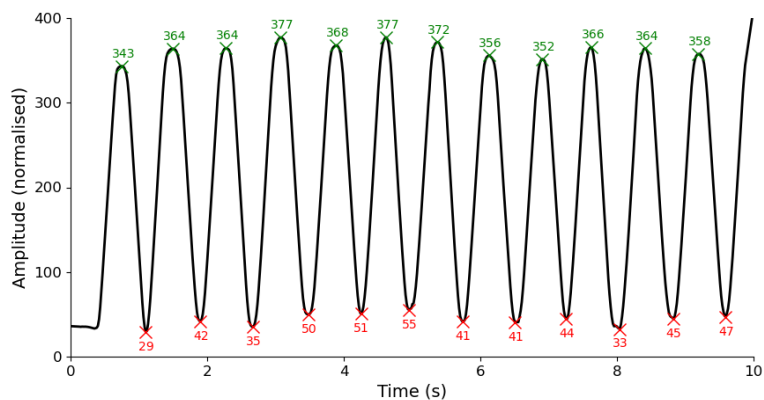


Figure 3.8. Extracting amplitude signal peaks and troughs, representing the maximum and minimum separation between the centre coordinates of the index finger and thumb bounding boxes.

3. TP_{CV} , the coefficient of variation was calculated for the time periods between amplitude peaks, **Figure 3.9**.

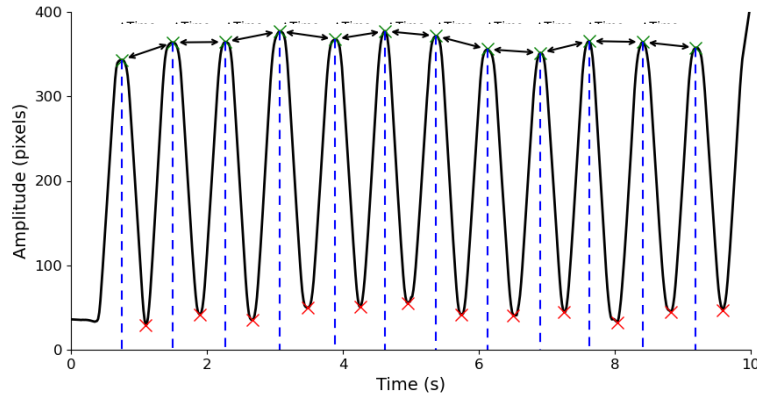


Figure 3.9. The coefficient of variation in the time (x-axis) duration between consecutive peaks are computed. This measure uses the intertap time variability to indicate the regularity of finger tapping rhythm.

4. P_{avg} , average peak amplitude (maximum open hand distance). Peaks in the amplitude graph characterise the maximum separation between the centres of the index finger and thumb bounding box in a full tapping cycle where X_I , Y_I and X_T , Y_T correspond to the index finger and thumb bounding box pixel coordinates respectively (**Equation 3**). Troughs have a non-zero value as a result of pixel distance being calculated from the centre of the bounding boxes. Distance data are normalised to values between 0 and 1 by subtracting the minimum index finger to thumb bounding box distance from each value then dividing by the range (**Equation 4**).

$$Separation (Sep) = \sqrt{(Y_I - Y_T)^2 + (X_I - X_T)^2} \quad (3)$$

$$Normalised\ value, X_i = \frac{(X - X_{minimum})}{(X_{maximum} - X_{minimum})} \quad (4)$$

5. P_{cv} , the coefficient of variation was calculated for amplitude peaks. The blue dashed line in **Figure 3.10** indicates the change in peak extension over the finger tapping task. The coefficient of variation is used to indicate the consistency in tapping amplitude.

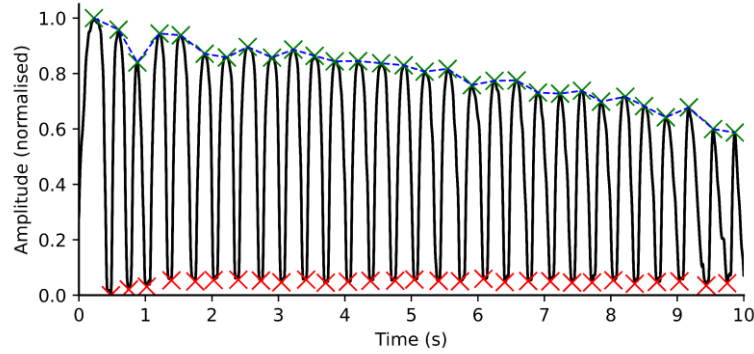


Figure 3.10. The coefficient of variation in amplitude peaks. This measure is used to indicate the consistency in the magnitude of alternating index finger and thumb extension and flexion movements across the 10 second finger tapping test.

- 6-9. Speed (S) is calculated over consecutive frames using the index finger to thumb pixel separation and corresponding change in time (**Equation 5**). **Figure 3.11** shows finger tapping speed and was used to derive both averages and coefficient of variation measures for opening and closing hand speeds. (6) OSavg: average opening speed is the mean of speed values at troughs. (7) CSavg: average closing speed is the mean of speed values at peaks. (8) OSCV: the CV was calculated for opening speed using trough values. (9) CSCV: the CV was calculated for closing hand speeds using peak values.

$$Speed (S) = \frac{Sep_i - Sep_{i-1}}{T_i - T_{i-1}} \quad (5)$$

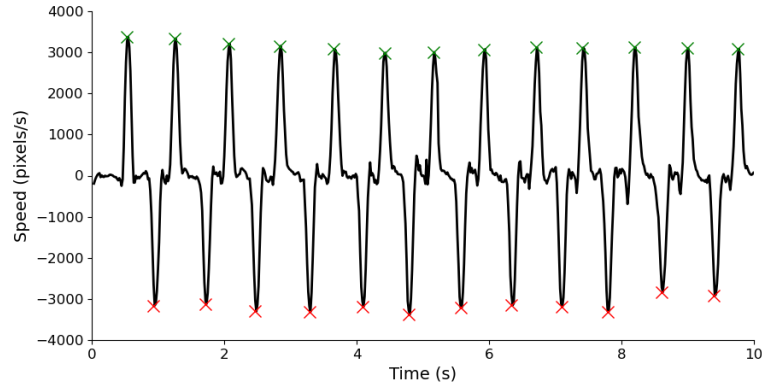


Figure 3.11. Finger tapping speed time series signal, computed using the change in distance between the centre coordinates of the index finger and thumb bounding boxes over two consecutive video frames.

10. R_{avg} , average rhythm is calculated by multiplying the index finger to thumb separation by the speed over consecutive frames (equation 6).

$$Rhythm, R = (Sep_i - Sep_{i-1}) \times (S_i - S_{i-1}) \quad (6)$$

11. A_{avg} , average acceleration is the average of all closing hand accelerations. This refers to the acceleration of the index finger to thumb immediately after maximum extension.
12. H, halts or hesitations indicate the level of disruption in continuous finger tapping. Decrement or variation in amplitude is not considered here and instead difficulty in both initiating and sustaining smooth movement is evaluated. For this metric, the prominence value which determines the 'importance' of peaks is lowered from 0.2 to 0.02. This increases the sensitivity so less prominent peaks associated with halts or hesitations in movement are extracted as local maxima. **Figure 3.12** demonstrates how a window was set to 'roll' over time series signals produced by PD patients and

healthy controls in two second intervals to calculate the coefficient of variation in the number of non-prominent peaks present in each window.

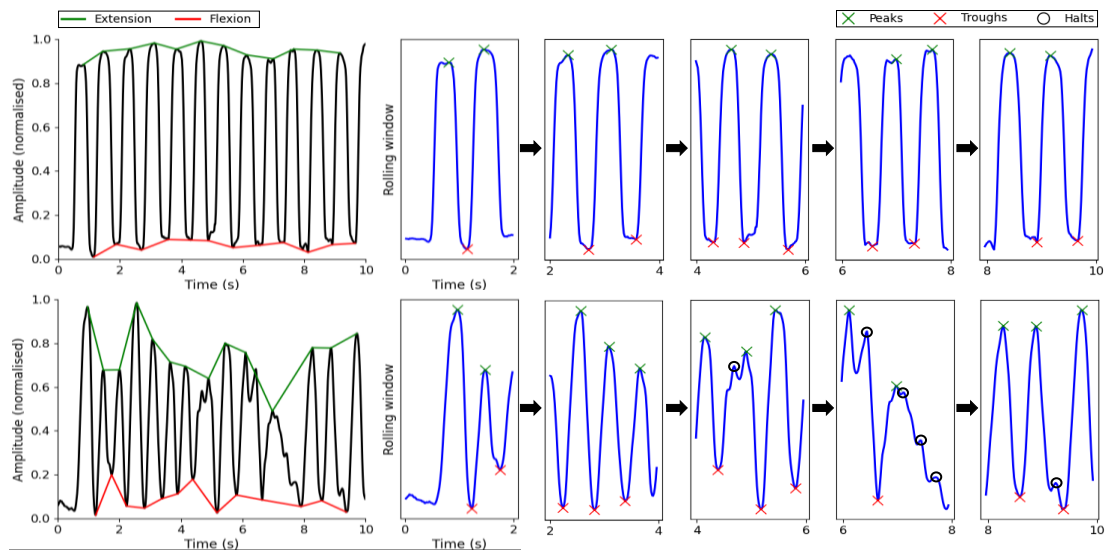


Figure 3.12. Rolling window analysis applied to finger tapping time series signals using python. Two second windows are used to divide the ten second signal into five intervals, registering the number of movement halts detected in each window. The resulting output metric is the coefficient of variation in the number of halts present within each window.

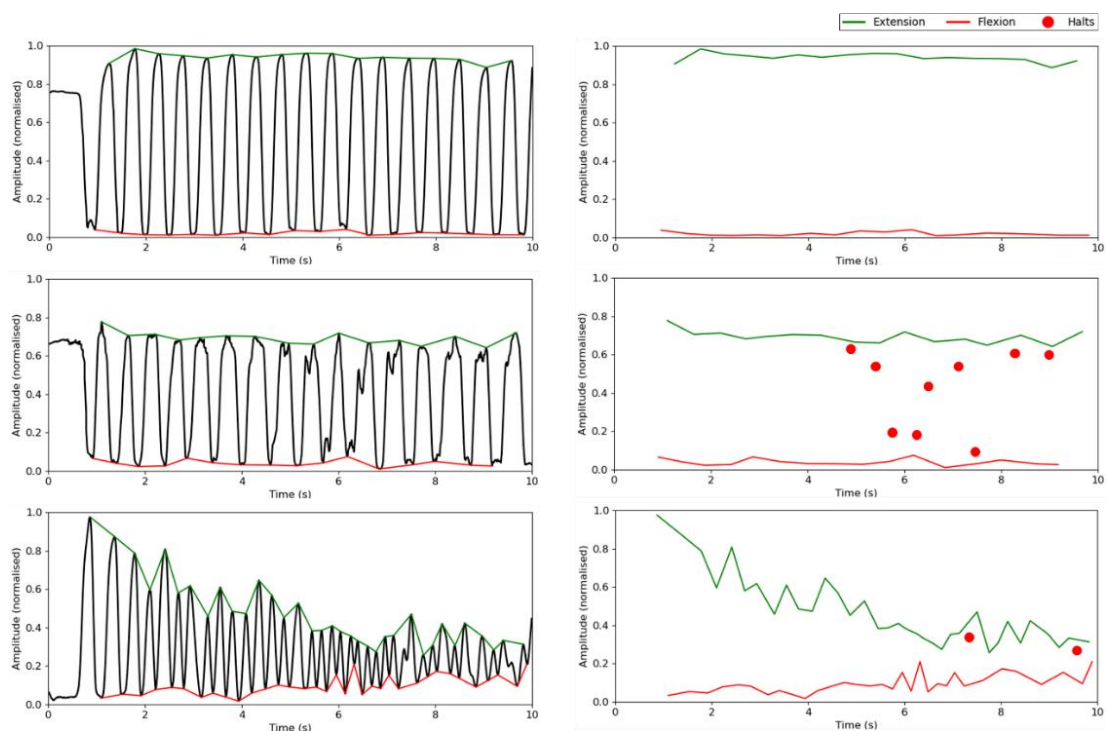


Figure 3.13. Peak extraction from finger tapping time series signals to identify movement halts.

The peak extraction function in the SciPy library used in Python offers a 'find peaks' function. Peaks and troughs were identified using the same methodology hence are referred to as peaks/ local maxima. The SciPy documentation defines a local peak as any sample whose two direct neighbours have a smaller amplitude. Peaks and troughs characterise the waveform composition by forming the periodicity of signals. Local maxima were determined by comparing neighbouring values and enforcing vertical height, horizontal distance and prominence thresholds to acquire desired peak values. Therefore, the find peak function was optimised to identify 'true' peaks among neighbouring maxima associated with halts in movement and distinguish halts or hesitations in movement from physiologically unrealistic transient jumps in index finger thumb bounding box distance between consecutive frames resulting from inaccurate detection. **Figure 3.13** illustrates how the peaks and troughs were extracted from finger tapping waveforms to highlight the consistency of finger tapping amplitude.

3.3.5 Feature Extraction

The twelve movement features extracted from the time series signals outlined in **Chapter 3.3.4, Time Series Signal Analysis** are summarised in **Table 9**, with the corresponding feature descriptions. The subsequent analyses used these output features to perform two subsequent analyses: participant group analysis and correlation analysis between computer features and clinician scores.

Table 9: Extracted Features from Finger Tapping Time Series Signals

Symbol	Feature	Feature Description
1. T_n	Number of taps	Number of index finger-thumb taps over 10s
2. TD	Tapping distance	Cumulative maximum amplitude
3. TP_{CV}	CV of time period	Coefficient of variation of time intervals between amplitude peaks
4. P_{avg}	Average amplitude	Average peak amplitude (maximum open hand distance)
5. P_{CV}	CV amplitude peaks	Coefficient of variation of peak amplitudes (maximum open hand distance)
6. OS_{avg}	Average opening speed	Average opening speed (initiating open hand from 'pinching', speed)
7. CS_{avg}	Average closing speed	Average closing speed (initiating closing hand from maximum amplitude)
8. OS_{CV}	CV opening speed	Coefficient of variation of maximum opening hand speeds
9. CS_{CV}	CV closing speed	Coefficient of variation of maximum closing hand speeds
10. R_{avg}	Average rhythm	Average maximum amplitude multiplied by average speed over 10s period
11. A_{avg}	Average acceleration	Average acceleration of the index finger over 10s period
12. Halts	CV Halts/hesitations	Ability to sustain continuous/ low disruption movement during finger tapping

3.4 Model Experimentation

A pilot study was conducted to evaluate the detection accuracy of the custom trained YOLO model. The objective was to determine the accuracy of the bounding box detection assessment framework in capturing ground truth movements during finger tapping. Twenty healthy participants performed 10 s of finger tapping, this involved alternating extensions of the index finger tapping followed by flexion (tapping the mouse button). A Python code was developed to

record the timestamps of contact events between the distal forefinger and mouse using the pynput library's listener. The finger tapping movements were simultaneously recorded with a webcam and the YOLO model was deployed in real-time to generate a time series signal of the displacement of the index finger's bounding box- see **Figure 3.14**. The signal troughs were extracted from the VC-generated signals, and each cycle's trough was used to represent instances of contact between the distal forefinger and mouse. The time stamps from the YOLO model and the mouse were plotted together for each participant. Subsequent analysis computed the time difference between the timestamps derived from the YOLO model and mouse, for each finger tapping cycle. **Figure 3.15** shows the bland-Altman analysis which demonstrates high agreement in the timestamps recorded by the two methods (YOLO model and mouse). A mean difference of -0.05 s, with 95% limits of agreement ranging from -0.15 to 0.06 seconds was obtained across 200 finger tapping cycles, indicating a small but consistent time lead in the YOLO signal. These slight underestimations are likely connected to difference in how instances of maximum flexion are determined by the two methods. While the mouse records contact times directly through distal finger clicks, the YOLO model derives these moments from generated displacement time series signals of the index finger. Additionally, the YOLO model represents the spatial location of the index finger using its bounding box centre coordinates, while the mouse registers clicks from the distal index finger. Notably, the number of finger tapping cycles, indicated by the number of recorded mouse clicks, derived from the YOLO model exactly matched that recorded by the mouse.

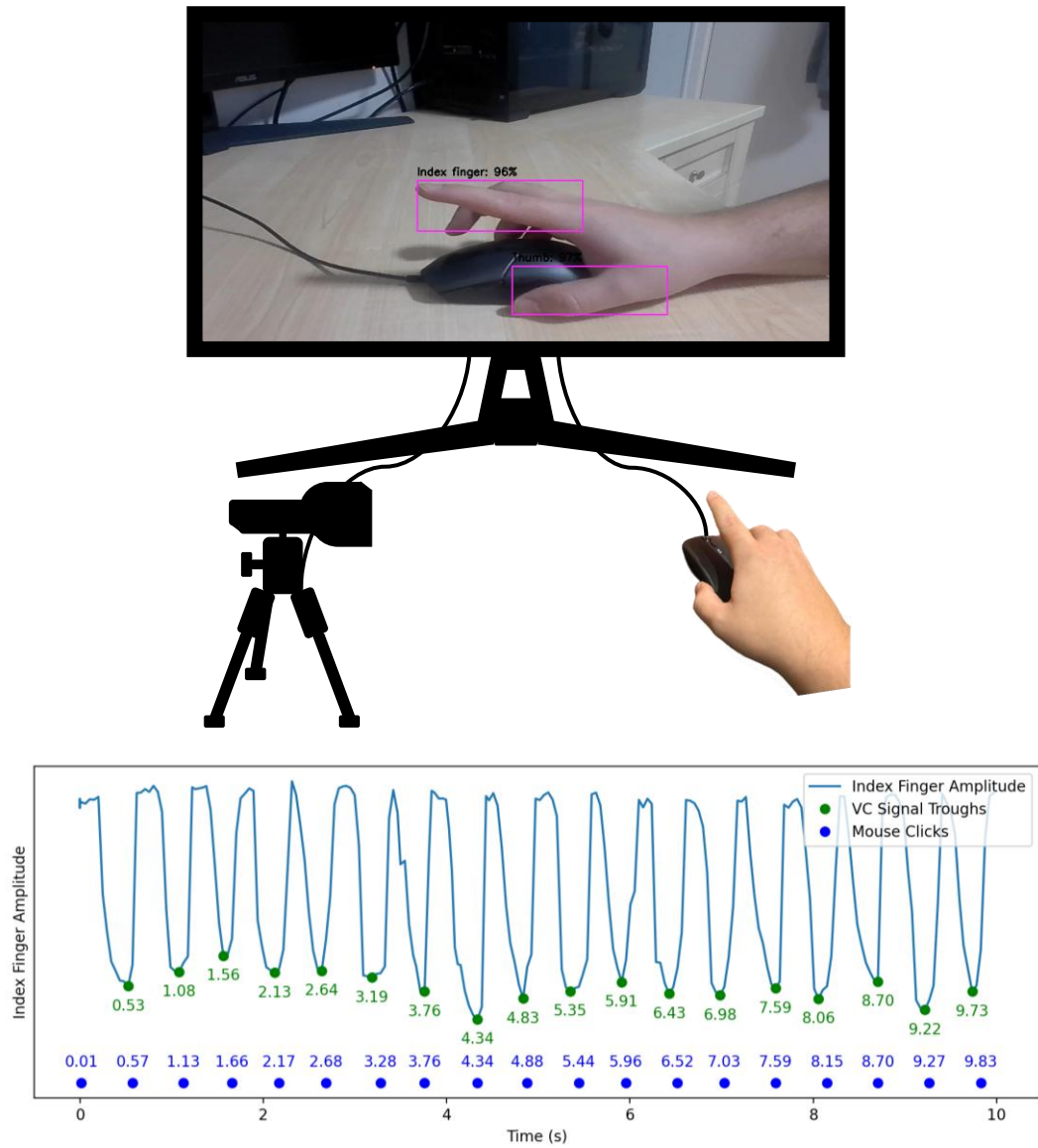


Figure 3.14. Parallel deployment of the trained YOLO model and a mouse acting as a contact sensor to register the timestamps of distal forefinger taps. Subsequent analysis uses the corresponding times of signal troughs derived from the YOLO model (green markers) and the registered mouse taps (blue markers) to compute the coefficient of variation (CV). The CV indicates the level of agreement in the registered times of repetitive maximum flexions of the index finger derived from the two measurement methods.

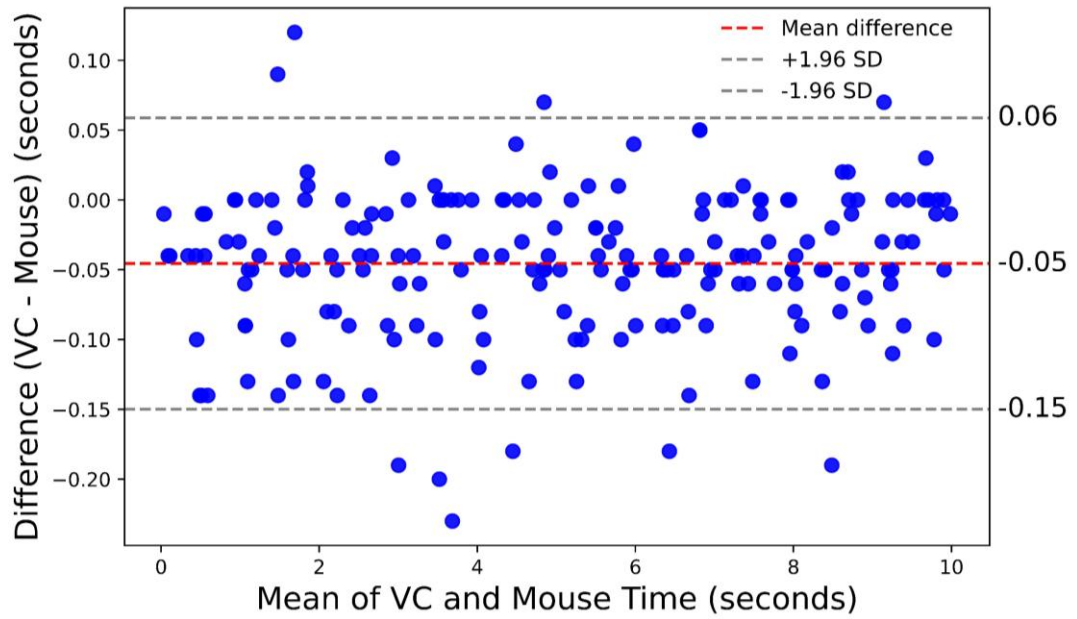


Figure 3.15. Bland-Altman plot showing the agreement between the timestamps derived from custom trained YOLO model and a mouse. The timestamps of distal index finger mouse clicks were directly recorded by the mouse, while the YOLO model generates displacement time series signals of the index finger to extract the timestamps of signal troughs, corresponding to moments of contact.

3.5 Calibration and System Design

Tracking accuracy, time series signal analysis and feature extraction form the three stages of the system design and thus require evaluation to ensure good output performance. Regarding retrospective analysis, the preparation of finger tapping videos involved visually inspecting each video to ensure the entire hand remains in frame. A camera-to-hand distance of one meter was maintained to help enforce this. Normal ambient lighting with a generally plain background was used in the recording of finger tapping videos. Evaluation metrics are used to indicate how accurately the waveforms generated by bounding box detection reflect the actual finger tapping movement. The system design incorporates evaluation metrics detailed in this section to monitor performance accuracy particularly useful for ensuring model performance is not hindered by changes in input conditions i.e. background noise in video recordings. The critical evaluation

stages of the proposed automated assessment framework used to generate computer features of finger tapping are shown in **Figure 3.16**.

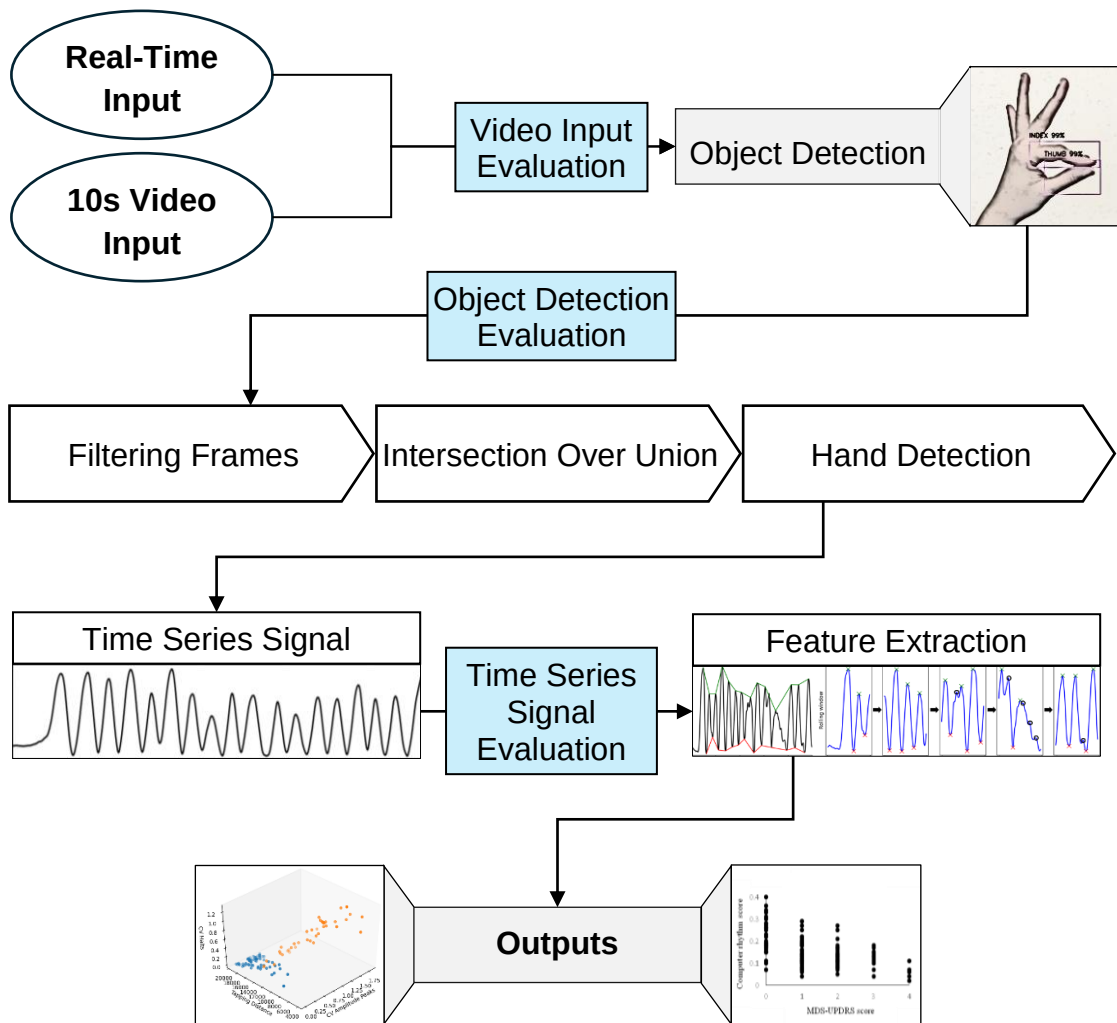


Figure 3.16. The automated assessment framework. The model supports both real-time inputs such as webcam feeds and prerecorded videos. Intermediate processes perform detection of the index finger and thumb, generate time series signals and extract relevant features to characterise finger tapping movement. Subsequent analysis examines computer-derived summary features across varying UPDRS scores and individual MBRS scores.

Evaluation Stage 1. Video Input:

Although the current study performs analysis on pre-recorded videos, the system design supports inputs of both real-time feeds and pre-recorded finger tapping videos. The video input evaluation stage extracts critical information about the

video specification including frame rate, video spatial resolution and video length. Real-time feeds utilise the object detection framework to initiate a ten second recording which starts when the entire hand is detected in frame view for a minimum of one second followed by the detection of finger tapping onset.

Evaluation Stage 2. Object Detection:

The tracking of finger tapping movements involve monitoring the centre coordinates of the index finger and thumb bounding boxes across consecutive video frames. Videos were recorded at 60 frames per second, resulting in 600 frames per video. Consequently, inference of the YOLO model was expected to output 600 displacement values, each corresponding to the distance between the two bounding boxes in each frame. The confidence threshold for detection was set to 70 percent for both the index finger and thumb, this was to accommodate instances of rapid hand movements while still filtering out low-confidence detections to reduce false positives and maintain accurate tracking of finger tapping movements. Instances when either the index finger and/or the thumb were not in frame or fell below the minimum confidence score caused transient jumps in the displacement time series signal. To overcome this, video frames that did not include detection of both object classes (index finger and thumb) were eliminated as shown in **Figure 3.17**. A secondary measure was enforced to ensure that the generated time series signals accurately reflected the entire duration of the finger tapping test, this included setting a criterion to flag instances where 5 or more consecutive frames were eliminated. Furthermore, videos that generated a total displacement array of less than 570 values were marked for review. These measures ensured that subsequent analyses of signal-derived output features were not affected by inconsistencies in sample frequency.

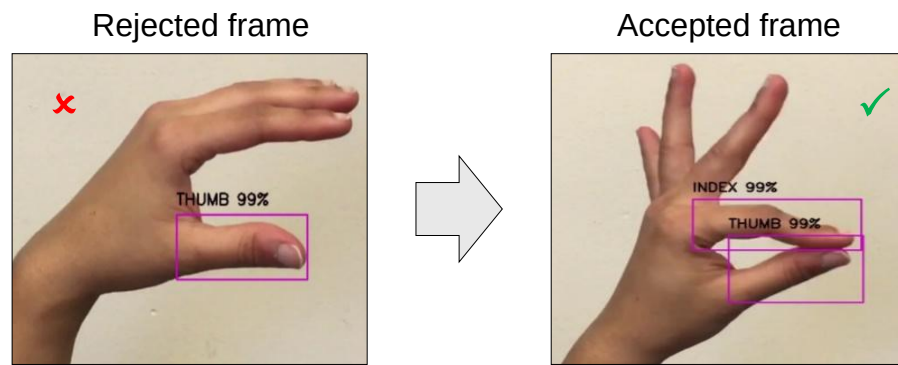


Figure 3.17. Detection validation: video frames are filtered to exclude instances when either one or both object classes are not detected. This is implemented to eliminate transient shifts associated with physiologically impossible movements in time series signals resulting from detection inconsistencies.

Intersection over Union (IOU) is a useful metric used to determine how closely a predicted bounding box mirrors the ground truth, which in this case is the annotated data. As shown in **Figure 3.18**, higher IOU value represent a higher level of overlap (agreement) between the areas of the predicted bounding box and the manually annotated (ground truth) bounding boxes. The mean IoU across all frames of the testing data for both the index finger and thumb was 0.83, indicating a strong overlap between predicted and ground truth bounding boxes.

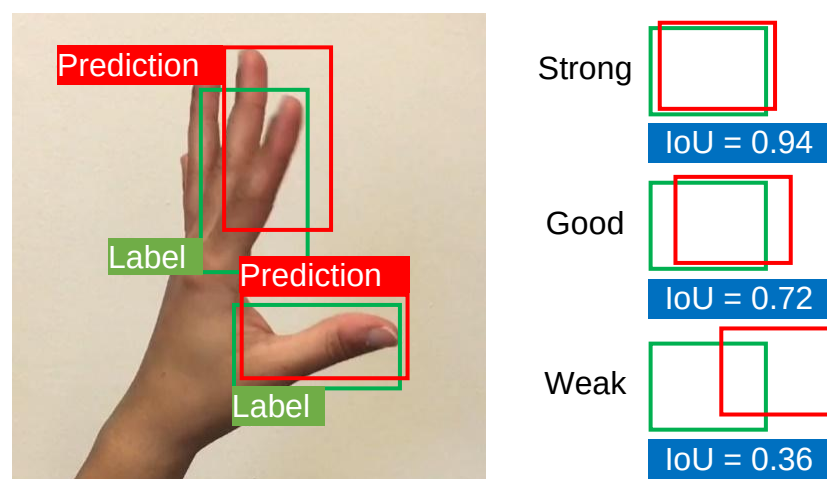


Figure 3.18. Intersection over Union (IoU) to measure detection accuracy of the index finger and thumb.

Incorporating hand detection into the assessment framework allowed out-of-frame instances where the hand moves either partially or completely out of camera view to be detected. This allows instances where self-occlusions result in no detection or partial detection to be distinguished from out-of-frame hand movements. **Figure 3.19** illustrates the detection of the hand in addition to the index finger and thumb to ensure the entire hand is in frame prior to finger tapping onset in real-time inputs.

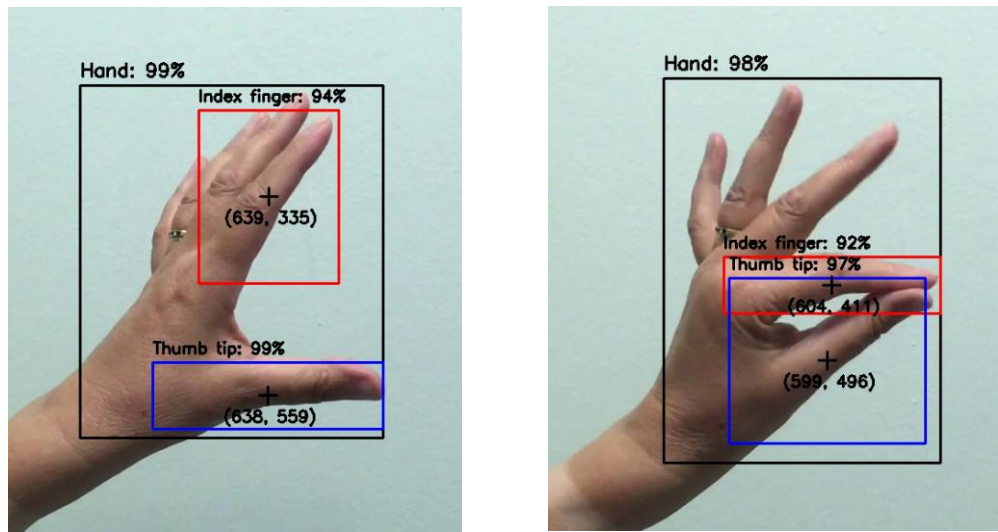


Figure 3.19. Three object class detections. A bounding box that captures the entire hand is used to establish when the hand is in frame during real-time inference. The index finger and thumb bounding boxes are used to generate finger tapping displacement time series signals.

False movement readings can be generated by transient changes in bounding box size during negligible movement. The orientation and position of the fingers can alter the bounding box size which can result in false movement readings. To minimise this effect, centre coordinates (x_I, y_I) and (x_T, y_T) corresponding to the index finger and thumb bounding box respectively are taken. **Figure 3.20** demonstrates the model's ability to track the index finger and thumb despite considerable rotation of the hands movement plane relative to the camera.

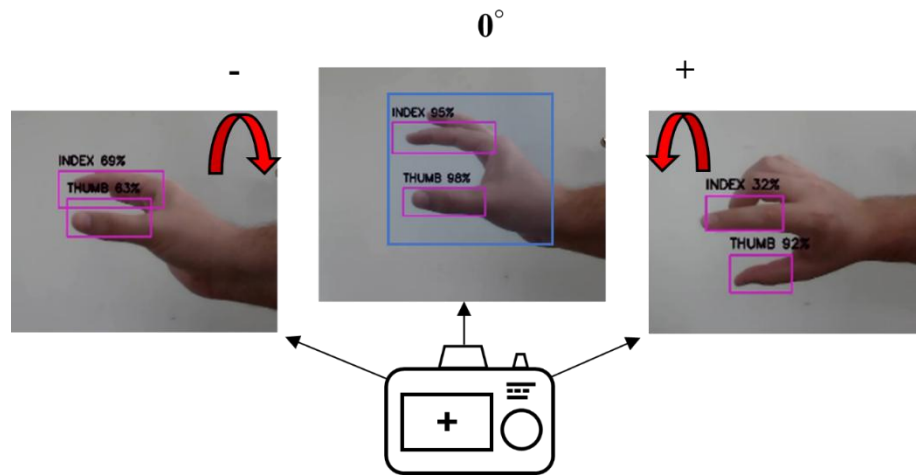


Figure 3.20. Finger tapping movement tracking. The custom trained YOLO model detects the index finger and thumb despite rotations in the movement plane of the hand relative to the camera.

Evaluation Stage 3. Time series signal:

Performance measures were applied to the time series signals to offer a final evaluation stage for output validity. Characteristics of finger tapping movement are extracted from the waveforms, therefore, evaluation metrics applied to time series signals must serve to filter signals that misrepresent movements in the input video. Under normal clinical circumstances, participants will first sit down and stabilise their elbow using an armchair ensuring their full hand is in frame. This adjustment period may become part of the sample recording as the hand may be partially detected moving in and out of camera view. To obtain a valid ten second video sample, the full video duration must consist only of the finger tapping attempt. Therefore, the model framework was designed to initiate video analysis only once the entire hand is detected in frame for a minimum of one second to accommodate for the initial adjustment period. This one second delay is followed by index finger and thumb bounding box detection. No more than one full tap was cut from the beginning of participant finger tapping videos.

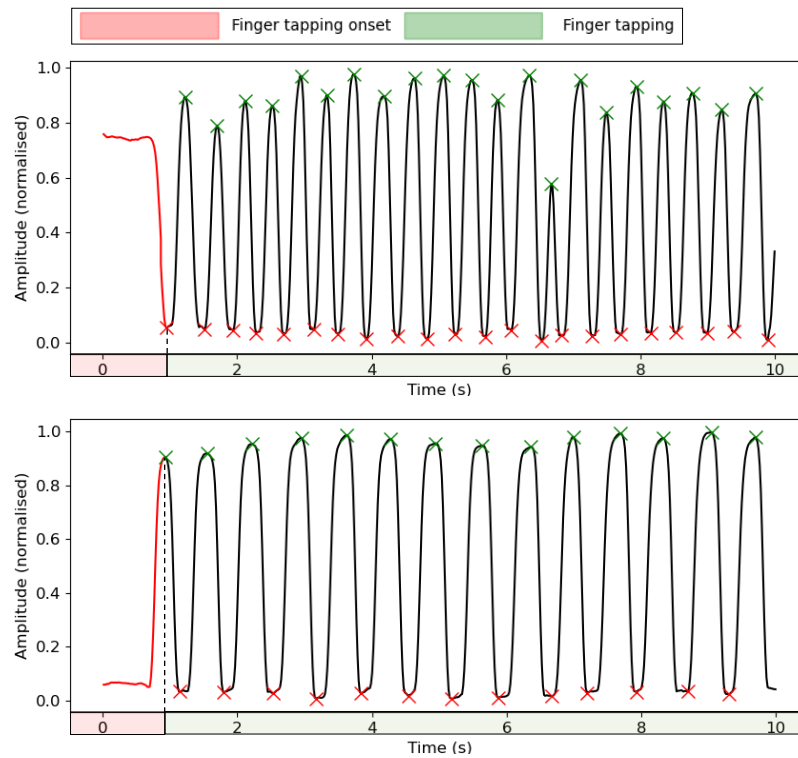


Figure 3.21. Eliminating the adjustment period from time series signals. Peak and trough extraction establishes finger tapping onset in real-time analysis. The top signal represents a finger tapping test that begins with the extension of the index finger and thumb. In this case, the first flexion (signal trough) is used to mark the start of the tapping signal. In the bottom signal, the index finger and thumb are initially detected in a pinched (flexion) position. The first extension (signal peak) is used to establish the start of the tapping signal.

For real-time analysis, a one second delay cannot be applied to the model framework as the onset of finger tapping will vary. To achieve a valid ten second time series signal the peak extraction function is repurposed to recognise a period of low amplitude movement with high baseline shifts. The signal is scanned for noise associated with initial hand adjustment and collection of time series data is initiated by the first prominent peak or trough that satisfies a threshold vertical height of 0.2 relative to its neighbouring peaks. **Figure 3.21** demonstrates two real-time generated finger tapping signals where the first signal is initiated from the first prominent trough or maximum hand flexion. Here the first prominent trough has a height of approximately 0.7 less than its preceding neighbouring

data points. The second signal is initiated by the first prominent peak corresponding to maximum extension of the index finger and thumb. In this case, the height of the first prominent peak has a value of approximately 0.8 larger than the preceding neighbouring data points. The system correctly disregards approximately the first second of the signal in both finger tapping waveforms generated from a real-time webcam recording. Therefore, to achieve a 10s finger tapping signal both real-time recordings would continue until approximately 11s.

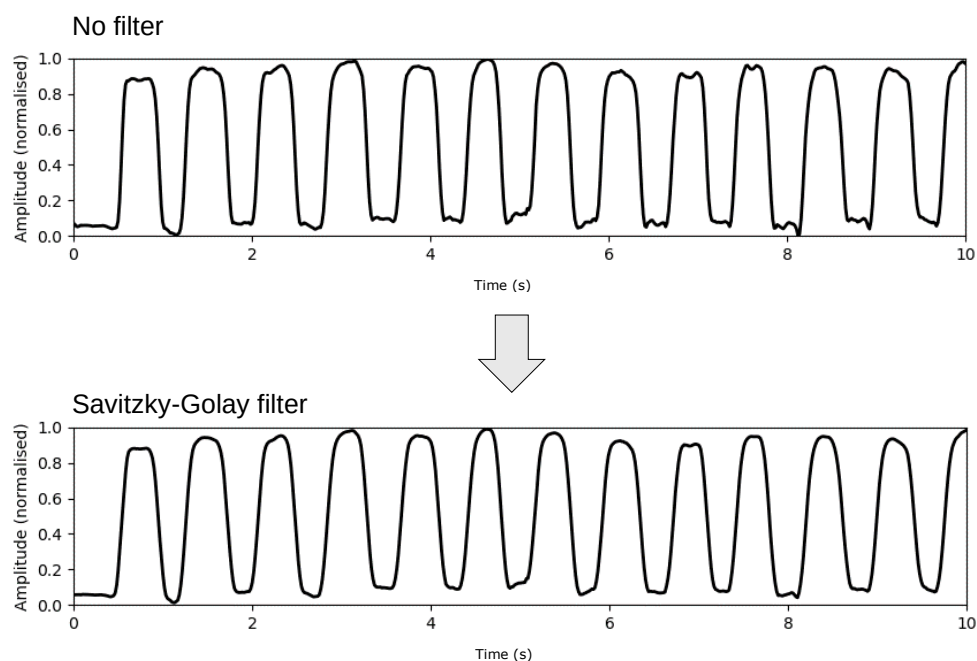


Figure 3.22. Applying a Savitzky-Golay filter to the processed finger tapping time series signal prior to feature extraction. The signal before (top) and after (bottom) applying the Savitzky-Golay filter with a window length of 3 and a polynomial order of 1 in Python. The filter smooths the signal, reducing noise while preserving the characteristic features of the finger tapping movements.

The Savitzky-Golay filter is available in the SciPy library in Python. This digital signal processing filter was applied to finger tapping waveforms to further reduce the effects of transient jumps in bounding box location between successive frames. To avoid eliminating subtle halts in movement, a polynomial order of 1 and a short window length of 3 was applied to avoid excessive smoothing. The

filter was primarily applied to smooth data obtained from finger tapping flexion phase particularly when the index finger and thumb are in contact. **Figure 3.22** shows a PD finger tapping waveform before and after the filter is applied.

3.6 Results

Finger tapping videos were quantified using the twelve extracted features and group averages for the PD patient, younger control and older control participants were calculated for each feature. Three primary bradykinesia related movement features, amplitude, speed and rhythm were used to determine the correlation between computer generated features and clinical ratings. Spearman's rank correlation coefficients were used to assess the correlation between computer features and MDS-UPDRS scores. Additionally, association between the amplitude, speed and rhythm computer features and the corresponding clinician-assigned MBRS subscores were assessed using Spearman's rank correlation coefficients. Since all three movement features affect the total tapping distance achieved during the 10 s finger tapping test, a further correlation analysis was performed between the computer derived tapping distance feature and the MDS-UPDRS scores- see **Figure 3.24**. Finger tapping videos were scored based on clinicians' visual interpretation without prior knowledge of diagnosis, therefore, scores of zero were assigned to one or both hands of some PD patients if movements were considered normal. In total, five videos from the PD group contributed to the twenty-seven subjects with a zero MDS-UPDRS score two of which were the same patient. **Figure 3.23** shows the correlation plots of computer-derived features against MDS-UPDRS scores on the left side, and demonstrates that the same computer features consistently achieve higher correlations against their matched MBRS component's scores, on the right side.

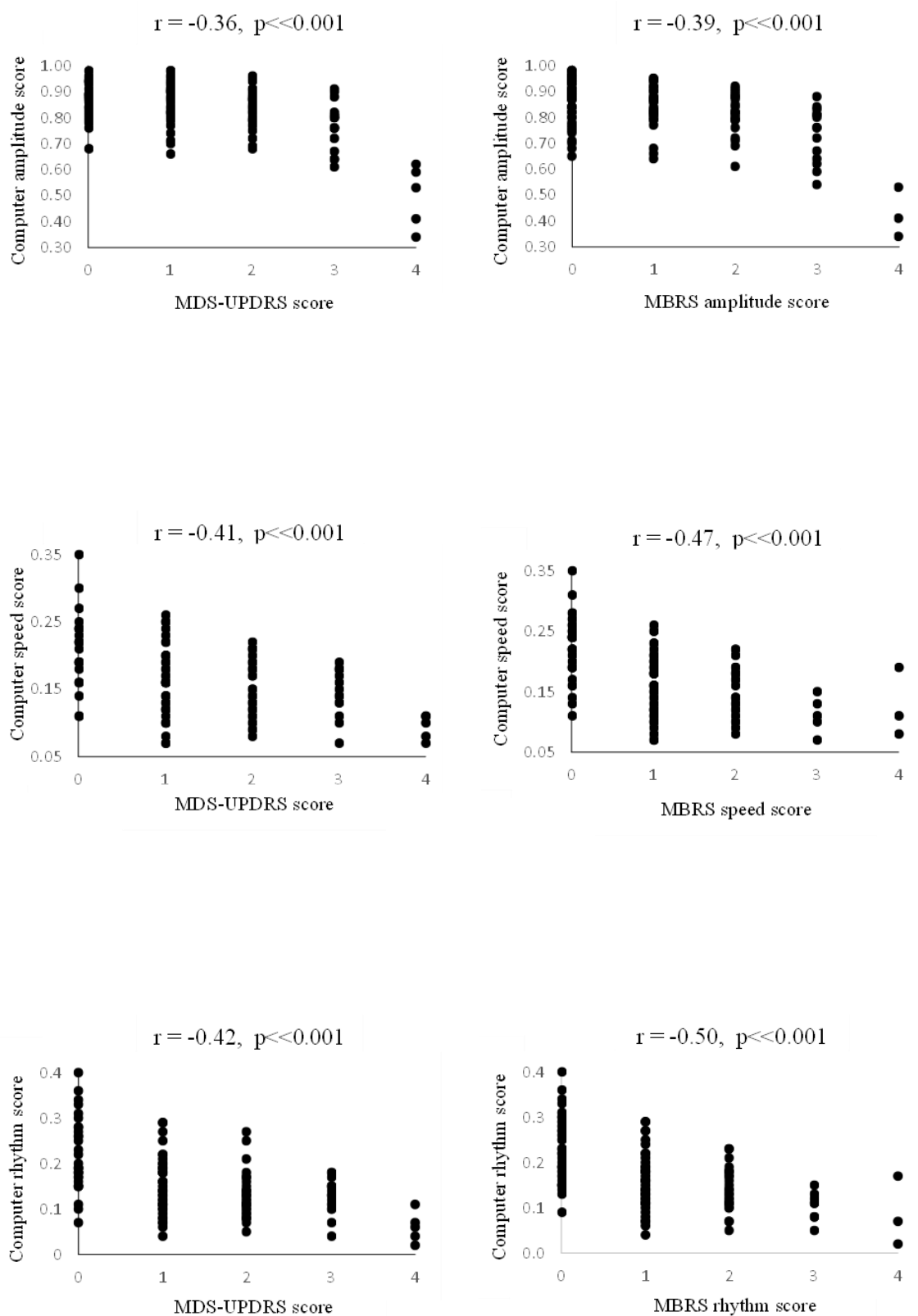


Figure 3.23. Spearman's rank correlation coefficients between corresponding computer extracted features and clinical rating scores. UPDRS and Individual movement components of MBRS (rhythm, speed and amplitude).

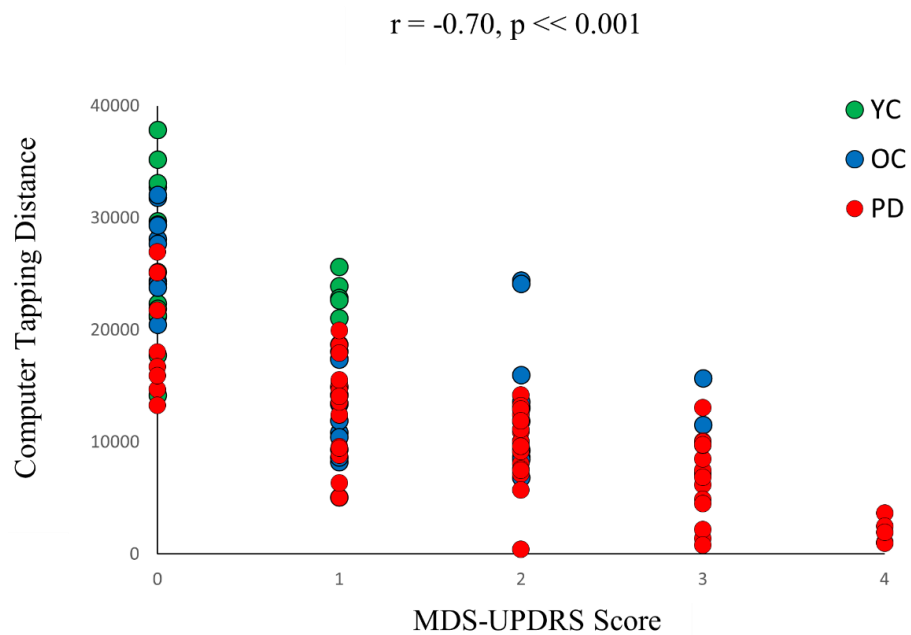


Figure 3.24. Spearman's rank correlation coefficients for total finger tapping distance across the five clinician-assigned UPDRS scores. The distribution of participants across clinician-assigned MDS-UPDRS scores are illustrated by green (younger controls), blue (older controls) and red (PD patients) markers.

An Analysis of Variance (ANOVA) test was conducted to establish whether the finger tapping computer features could discriminate between the PD, younger control and older control participant groups. The ANOVA results indicated that all movement features, excluding the coefficient of variation of the time period (TPCV), were statistically significant, with a p-value less than 0.05. This suggests that these features present meaningful differences across the participant groups. To further evaluate group differences, a Tukey-Kramer post hoc test was conducted. This was to establish pairwise comparisons between the groups for each computer feature, accounting for differences in participant group samples. Notably, the Tukey-Kramer post hoc analysis highlighted statistically significant differences between the younger and older control groups across eight features. **Table 10** provides the twelve computer features used to characterise finger tapping tests together with their corresponding group averages. Additionally, the

results for the ANOVA test and specific pairwise comparisons are provided, highlighting the specific group pairs with notable statistical distinctions for each movement feature.

Table 10: Group results: Parkinson's Disease (PD), Younger Controls (YC) and Older controls (OC). Values with asterisk (*) are normalised between 0 and 1

Parameter			Group			Pairwise Comparison			ANOVA
Symbol	Feature	Units	YC	OC	PD	YC:OC	YC:PD	OC:PD	p-value
1. Tn	Number of taps	N/A	37.42	26.26	22.67	✓	✓	✗	P<<0.05
2. TD	Tapping distance	Pixel	18.26	15.65	8.77	✓	✓	✗	P<<0.01
3. TPCV	CV of time period	N/A	0.29	0.32	0.43	✗	✗	✗	P>0.05
4. Pavg	Average amplitude	Pixels	495	418	308	✓	✓	✗	P<0.05
5. PCV	CV amplitude peaks (regularity)	N/A	0.21	0.28	1.12	✗	✓	✓	P<<0.01
6. OSAvg	Average opening speed	Pixels/s	-3396	-2994	-1553	✓	✓	✗	P<<0.05
7. CSAvg	Average closing speed	Pixels/s	3447	3081	1912	✓	✓	✗	P<<0.05
8. OSCV	CV opening speed	N/A	0.62	0.76	1.10	✓	✓	✗	P<0.05
9. CSCV	CV closing speed	N/A	0.75	0.87	0.95	✓	✓	✗	P<0.05
10. Ravg	Average rhythm	Pixels ² /s	0.67*	0.61*	0.34*	✗	✗	✓	P<0.05
11. Aavg	Average acceleration	Pixels/s ²	0.87*	0.81*	0.64*	✗	✓	✗	P<0.05
12. Halts	CV Halts/ hesitations	N/A	0.08	0.12	1.23	✓	✓	✗	P<<0.01

Finally, the three computer features which demonstrated the highest statistical differences between the group averages, as determined by the ANOVA test, were selected for visualisation. A 3D plot of tapping distance, correlation of variation in halts and correlation of variation of amplitude was used to visualise the discriminatory power of these clinically meaningful features, **Figure 3.25**. In this plot, the data points were labelled by symptom severity rather than participant

groups. This was to determine whether these discriminating features have capacity to reveal distinctions across clinician-assigned MDS-UPDRS scores, which can overlap across participant groups. Participants were categorised into two symptom severity groups, with each data point labelled as either moderate (MDS-UPDRS $0 \leq 1$) or severe (MDS-UPDRS $2 \leq 4$). **Figure 3.25** shows a clear separation between the moderate (blue data points) and severe (orange data points) symptom severity groups. Two corresponding boundary lines were manually plotted to illustrate the separation between the two clusters in the data.

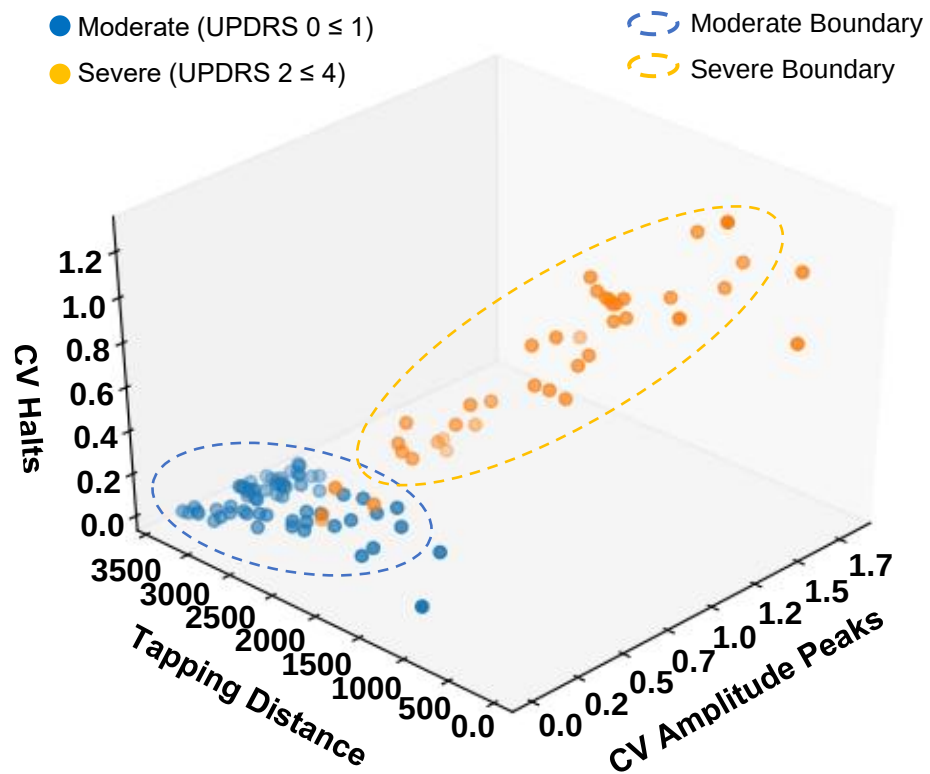


Figure 3.25. 3D plot of coefficient of Halts (CV Halts), Tapping Distance (Pixels), and coefficient of variation for Amplitude peaks (CV Amplitude Peaks). The two manually drawn boundary lines illustrate the distinction between individuals assigned with moderate (UPDRS $0 \leq 1$) and severe (UPDRS $2 \leq 4$) scores.

3.7 Discussion

The study demonstrated that the presented VC-based assessment framework can track the spatiotemporal signal of the index finger and thumb effectively during the finger tapping task. Unlike previous computer vision-based methods used to investigating finger tapping performance, this assessment framework is capable of performing real-time quantitative analysis of the finger tapping test. Optimisation of the system design ensured the system's versatility in dealing with unprocessed video inputs. Two approaches were adopted in the evaluation of finger tapping: establishing correlation between computer output features and clinical ratings using clinically matched features i.e. movement amplitude, rhythm and speed subscores of MBRS, and discriminating participant groups using both matched and novel feature outputs.

Significant correlations were found between the computer output for amplitude, rhythm, speed and tapping distance and their corresponding clinical ratings. This indicates that our automated assessment method provides a strong basis for clinical evaluation of this isolated task. Finger tapping can be observed to a much higher degree than the amplitude, rhythm and speed scoring criteria used to characterise the quality of movement during visual inspection. PD can present complex movement disorders composed of irregular motion such as high speeds in isolated phases of tapping i.e., opening hand phase. Whilst this is apparent visually and can be factored into the judgment of the speed component of MBRS scores; instances of high speed combined with longer periods of slow movement can yield low average computer speed outputs. Therefore, this study has demonstrated that more specific movement parameters such as opening and closing hand speeds achieve a larger difference in average feature output values

between participant groups. A novel feature, the coefficient of variation of the number of halts further demonstrated that there is a statistically significant difference between the mean values in the YC, OC and PD participant groups.

Incorporating performance measures served to both validate output accuracy at consecutive stages of the model framework and to reduce dependency on the video input criteria. The system showed good capability in coping with a diverse range of hands and different hand orientations. The supplementary assessment further demonstrated the system's ability to perform accurate real-time detection in different lighting conditions and with background noise when a webcam was used as the input feed. Design aspects of the system had deployment in clinical settings at the core focus and performance measures gave a good indication of feature representation and accuracy across a range of movement deficits. Moderate to strong correlation with individual MBRS components in addition to MDS-UPDRS scores demonstrated that our computer measures characterise finger tapping movement appropriately.

Finger tapping has been suggested to be one the most difficult neurological tasks to assess (Luiz et al., 2024). The assessment methodology of distributing participant videos blindly without further information on patient status, age or severity ensured that the isolated finger tapping task presented to the computer vision model was similarly reviewed by specialists. All clinicians were encouraged to score videos based on their initial visual impression of participant videos. It must be noted that the visual assessment made by clinicians was limited to a ten second viewing of the isolated finger tapping task. The same videos were used for the computer model's input and real-time recordings were also limited to ten seconds. Establishing correlation with clinical ratings assumes that clinicians are

in agreement with the evaluation of symptom severity and that videos are rated correctly. Therefore, a strength of our research was overcoming inter-rater variability by using a median of five movement specialists for each video to provide a median clinical rating score as a benchmark for assessing model prediction accuracy.

The object detection model relies on centre coordinates of bounding boxes to deduce movement whereas previous studies have adopted specific landmarks of the hands including the finger and thumb tip to track spatial location. This reduces disruption to tracking the index finger and thumb when the hand orientation changes from the ideal side profile or the fingertip and thumb appear to be overlapped in camera view. Furthermore, the approach used to recognise disruptions to rhythm formation during finger tapping provided a good indication of instances where a reduction in tapping frequency is induced by movement halts rather than fatigue. The results indicate that the halts metric achieved a significant difference between healthy controls and the PD patient participant groups whereas the coefficient of variation in time period did not. This implies that variation between the peak-to-peak time period is not sufficient in characterising tapping frequency. It must be noted that in addition to overall symptom severity, changes in movement characteristics such as halts and hesitations can differ between moderate (MDS-UPDRS $0 \leq 1$) and severe (MDS-UPDRS $2 \leq 4$) symptom severities. To reduce the effects of severe Parkinsonian movement deficit on group performance, the coefficient of variation across one second intervals of time series signals was calculated for metrics like the number of halts to target overall movement irregularities in individuals rather than incorporating instances of movement anomalies to group performance.

Chapter 4: Automated Assessment of Parkinson's Disease Symptom Severity using Three Upper Limb Tests

4.1 Introduction

The ability to discern movement data from standard video recordings of finger tapping was demonstrated in **Chapter 3**. Furthermore, the examination signs used to evaluate the underlying motor dysfunction in PD patients was further explored in **Chapter 2.2**. While PD-related movement deficits have previously been captured through the isolated finger tapping test using numerous objective measurement systems (Khan et al., 2014; Wong et al., 2019; Guo et al., 2022; Li et al., 2021; Li et al., 2022; Islam et al., 2023; Yu et al., 2023), only one VC-based study extended their analysis to include several upper limb tests (Monje et al., 2021). In the motor examination section of the MDS-UPDRS, items 3.4 (Finger tapping), 3.5 (opening-closing hand movements) and 3.6 (pronation-supination movements of the hand) share several key elements in that they all assess bradykinesia, focus on the upper limbs (fingers and hand), involve rapid repetitive and alternating movement, are tested unilaterally and conducted while seated. Currently, the frequent use of the finger tapping test in both clinical settings and research stems from two main factors: First, it is widely recognised to reveal clinically meaningful examination signs which serve to indicate an individual's motor performance. Secondly, its time-efficient implementation allows for routine administration. With that being said, incorporating two additional upper limb tests could enhance the assessment with minimal disruption. Since patients perform the finger tapping test seated, the existing camera setup in data collection could be used to subsequently capture the open-close and pronate-supinate hand movement tests.

In a proof-of-concept VC-based study by Monje et al. (2021), it was highlighted that the majority of existing research is constrained to the performance measures derived from an isolated motor assessment. The study aimed to increase the granularity of information by using a combination of the finger tapping, opening closing and pronation-supination tests to produce discriminatory features. However, only group-level distinctions were made, using LR, NB, and SVM to perform binary classification between early-stage PD patients and healthy subjects, with LR achieving a maximum accuracy of 70.37%. Khan et al. (2014) reported potential bias in their study linking it to one of the raters observing head movements in addition to the finger tapping test. This supports the notion that more comprehensive observations of human movement have potential to impact the decision-making process of both clinicians and classification models.

This chapter investigates the use of the finger tapping, opening closing and pronation-supination upper limb tests and their discriminatory strength on varying symptom severity levels. Prior to classification, advanced automated signal processing techniques were implemented to ensure accurate feature representations of each test. Given the addition of two upper limb motor assessments, the clinician-assigned MDS-UPDRS scores which serve as the dataset's benchmark were further analysed using the reliability index ICC.

4.2 Data Acquisition: Study 2

A total of 459 video recordings were collected from 38 controls and 61 PD patients. **Figure 4.1** summarises the videos obtained from the unilateral testing stage. The addition of the opening-closing and pronation-supination hand movement tests were introduced later in the study, which meant that data collection for all three upper limb tests started from control number 34. Hence

control participants 1 to 33 only completed the finger tapping test. A further 3 video recordings were discarded from the healthy control group due to the hand moving out of frame, this occurred for both hands of one participant in the finger tapping test and for another participant's right hand in the pronation-supination test. **Table 11** summarises the demographic information for all participants.

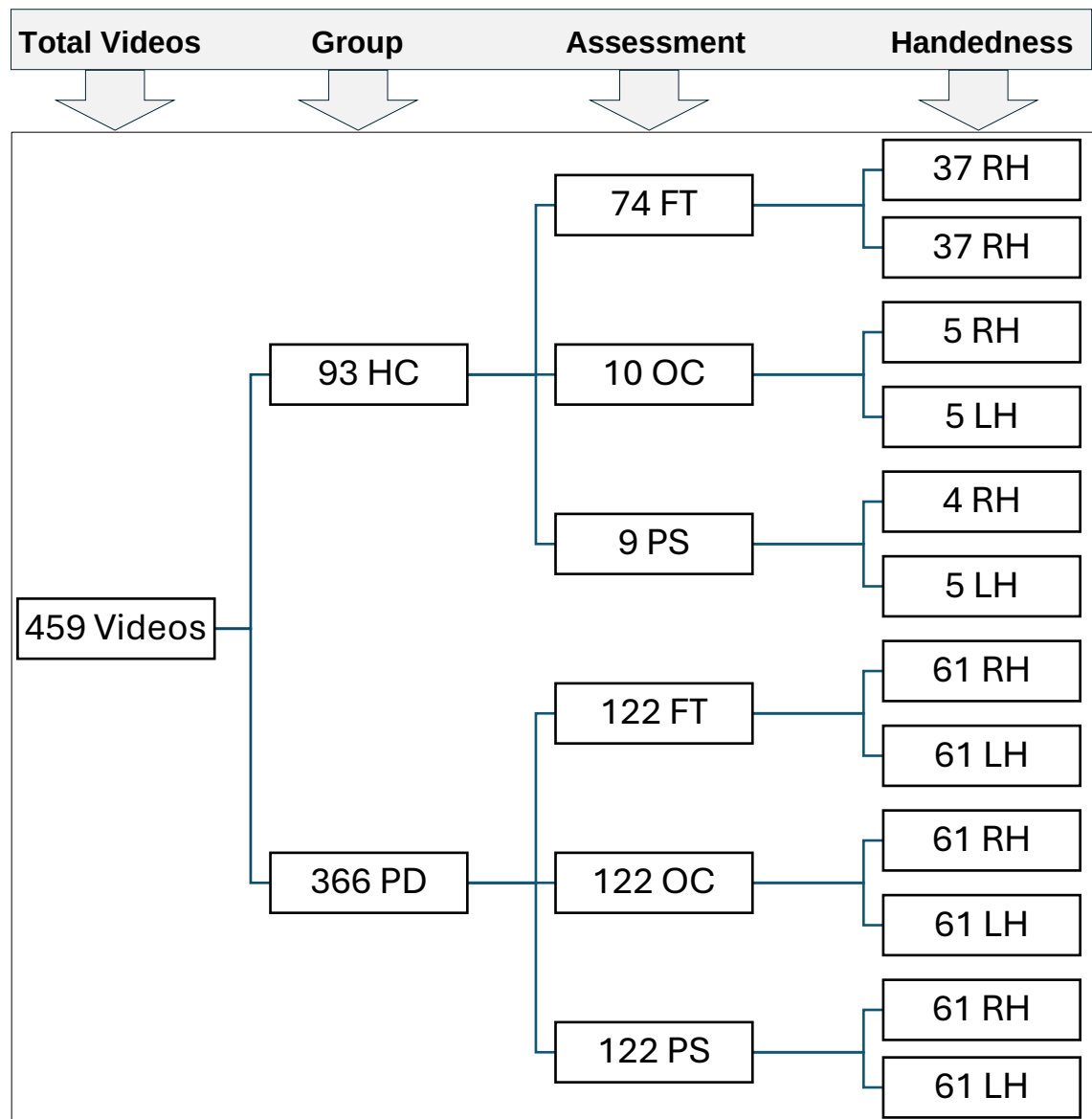


Figure 4.1. Breakdown of the 459 upper limb movement test videos used in the study by participant group and assessment type. The videos are divided into two groups: Healthy Controls (HC) and Parkinson's Disease (PD) participants. The assessments include Finger Tapping (FT), Open-Close (OC), and Pronation-Supination (PS) tests, with the unilateral tests further categorised by right hand (RH) and left hand (LH) tests.

Table 11: Summary of Participant Demographics

Demographics	HC	PD
Number of participants	38	61
Gender: male; female	3; 31	32; 29
Age: mean	70	66
Age: standard deviation	6	10
Handedness: right; left	33; 1	59; 2
Years since diagnosis: mean	N/A	8
Years since diagnosis: standard deviation	N/A	6

4.2.1 Ethical Approval (2)

Data collection for this study was conducted in Brazil and the subsequent analysis was performed locally (United Kingdom). Hence ethical approval was granted by the University of Parana for the collection of participant data overseas and further approval was granted by the University of Bradford to analyse the data locally, as provided below.

HOSPITAL DE CLÍNICAS DA UNIVERSIDADE FEDERAL DO PARANÁ - HC-UFPR:

Ethical approval to collect participant data in Brazil was granted (Approval number: 6.785.485). The Research Ethics Committee (CEP) on Human Research of HC-UFPR, according to the attributions defined in Resolution CNS 466/2012 and in Operational Norm No. 001/2013 of the CNS, manifests its approval of the Amendment.

University of Bradford:

Ethical approval to analyse the data locally (UK) was granted by the Chair of the Biomedical, Natural, Physical and Health Sciences Research Ethics

Panel/Humanities, Social and Health Sciences Research Ethics Panel at the University of Bradford.

4.2.2 Eligibility Criteria for Participant Selection

The study was undertaken in collaboration with three neurologists: Dr. Giorgio Fabiani, Dr. Renata Ramina and Dr. Fernando Spina Tensini (Hospital De Clinicas, Curitiba, Brazil), who advised on appropriate eligibility criteria for both HC and PD patients. For control subjects, the inclusion criteria required them to be age 65 or above and self-report as mildly to vigorously active (as defined by the American College of Sports Medicine (ACSM) physical activity guide-lines in accordance with the ACSM guidelines (Thompson et al., 2013). All control participants were required to be free from neurological problems, injuries or clinical conditions that could affect their ability to perform movement tests, nor taking any medication to address such problems. The exclusion criteria for both participant groups included the presence of any joint pain or muscle injury. Finally, excessive alcohol consumption 24 hours prior to participation was not allowed.

For the PD group, all participants were required to be formally diagnosed with PD prior to the assessment, with knowledge of the year they were diagnosed. Critically, the “wearing off” phase of levodopa is established by the reemergence of PD symptoms as the effects of medication diminish near the dose interval (Liang, 2024). The neurologists advised a minimum washout period of 6 hours to consider patients in an “off” medication state, ensuring motor abnormalities were not masked during movement tests by commonly used dopaminergic medication like levodopa. Therefore, assessment times were organised to accommodate patient’s dose intervals, ensuring they could take medication immediately after

the assessment was conducted. A comprehensive health questionnaire was also administered prior to participation, ensuring there were no further underlying conditions that could affect assessment outcomes were present. All participants were willing to perform all upper limb tests while being video recorded and signed an informed consent form.

4.2.3 Data Collection Protocol: Finger Tapping, Opening-Closing and Pronation-Supination Hand Movements

Following health screening and informed consent, all participants were given an information sheet which detailed the assessments they were required to perform (see **Appendix A**). Recordings were taken using an iPhone 13 Pro Max, placed on a tripod which was positioned approximately one meter away from participant hands. Participants were seated, and the elbow was stabilised by resting on the arm of an armchair. Prior to instructing participants to start the test, it was ensured that their forearm to fingertip was in full camera view to avoid repeat tests. For patients with apparent severe motor impairment, the camera was positioned approximately 1.5 m away from the hand to accommodate potential large transient movements of the hand that may move out of frame. In all video recordings a checker board was placed within the frame to act as a calibration object and was aligned along the movement plane of the index finger and thumb to ensure the derived distance measures could be translated into real-world units i.e. centimetres, **Figure 4.2**. Videos of participants performing the three hand tests for 30 s each were recorded at a resolution of 1080p with a frame rate of 60 frames per second in the order below, starting with the right hand. A minimum rest time of 5 minute was ensured between tests. All participants were asked if they felt comfortable to continue the next assessment.

1. **FT (Table 1):** Participants were instructed to tap their index finger and thumb together “as quickly and as big as possible” until instructed to stop.
2. **OC (Table 2):** Participants were instructed to open and close their hands “as quickly and as fully as possible” until instructed to stop.
3. **PS (Table 3):** Participants were instructed to extend their arm forward and pronate and supinate their forearm alternately as quickly as possible to the maximum degree of rotation that is perceived as comfortably achievable until instructed to stop.



Figure 4.2. Data collection protocol: video recording finger tapping, opening-closing and pronation-supination hand movements. Videos were recorded using an iPhone 13 Pro Max, placed on a tripod approximately one meter away from participant hands.

4.2.4 Clinical Ratings and Inter-Rater Reliability

To obtain symptom severity (UPDRS) class labels for each participant, the 459 video recordings were distributed, with their original resolution and frame rate, to three neurologists (Dr Giorgio Fabiani, Dr Renata Ramina and Dr Fernando Spina Tensini), all specialising in movement disorders. Importantly, participant ID's were removed from videos and distributed in an anonymised form. Clinicians were instructed to score each upper limb test based on their first visual

observation in conjunction with MDS-UPDRS five-point scoring guidelines. The process of acquiring clinical rating scores and using them to label participants is illustrated in **Figure 4.3**. The range in MDS-UPDRS scores across the three upper limb tests are shown in **Figure 4.4** and the score distributions for each test are presented in **Figure 4.5**. The clinician-assigned MDS-UPDRS scores represent 0: Normal, 1: Slight, 2: Mild, 3: Moderate and 4: Severe.

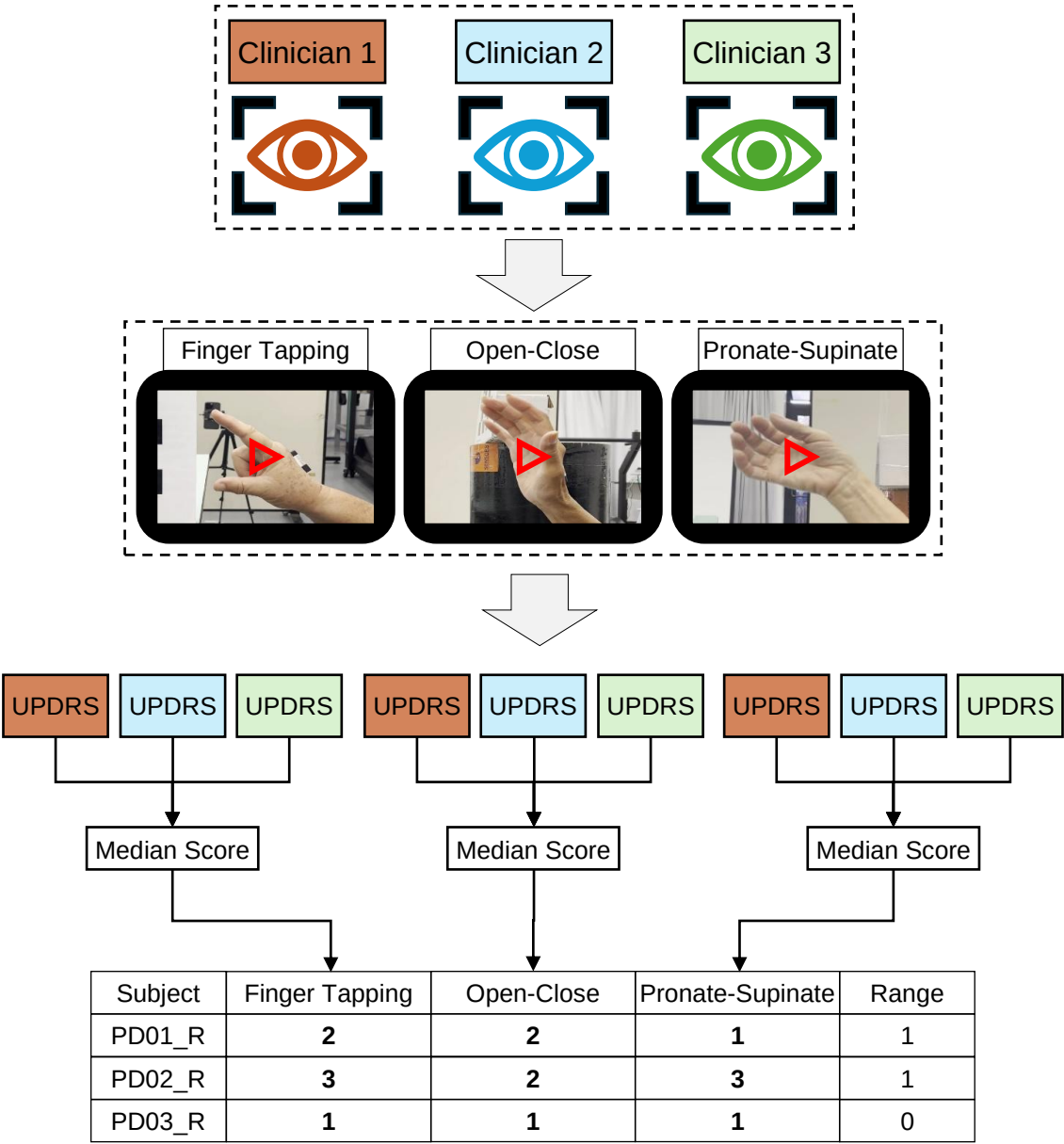


Figure 4.3. UPDRS scoring strategy using three clinicians to grade video recordings of finger tapping, opening-closing and pronation-supination hand movement assessments.

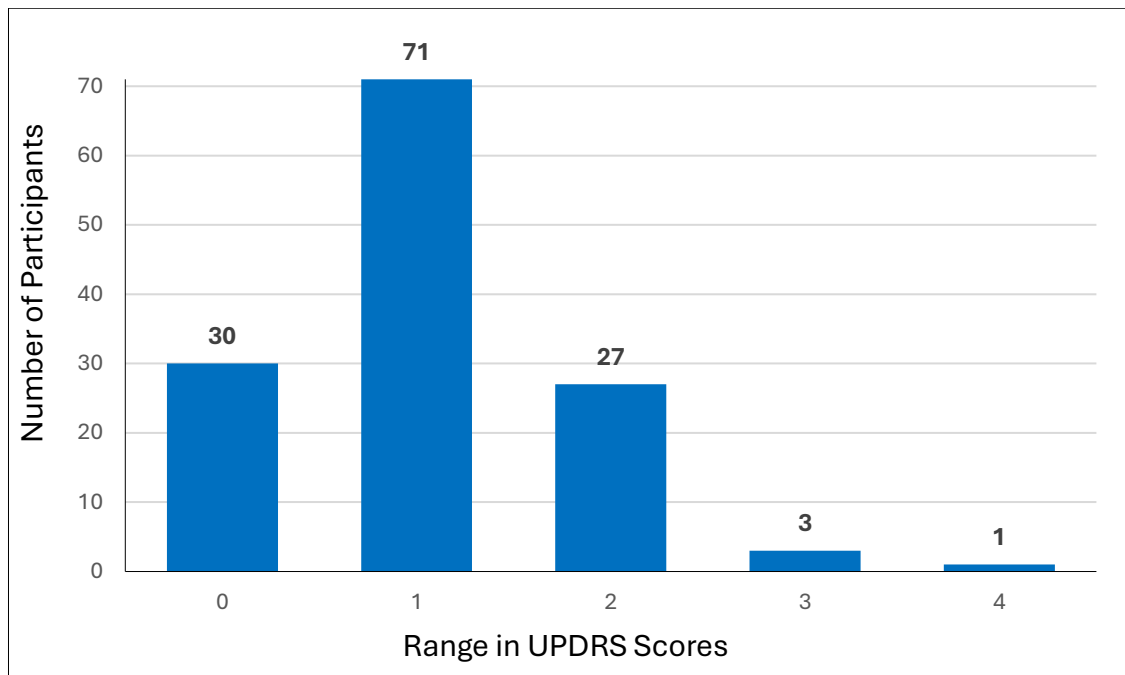


Figure 4.4. The range in clinician-assigned UPDRS scores across the finger tapping, opening-closing and pronation-supination hand movement assessments for all participants.

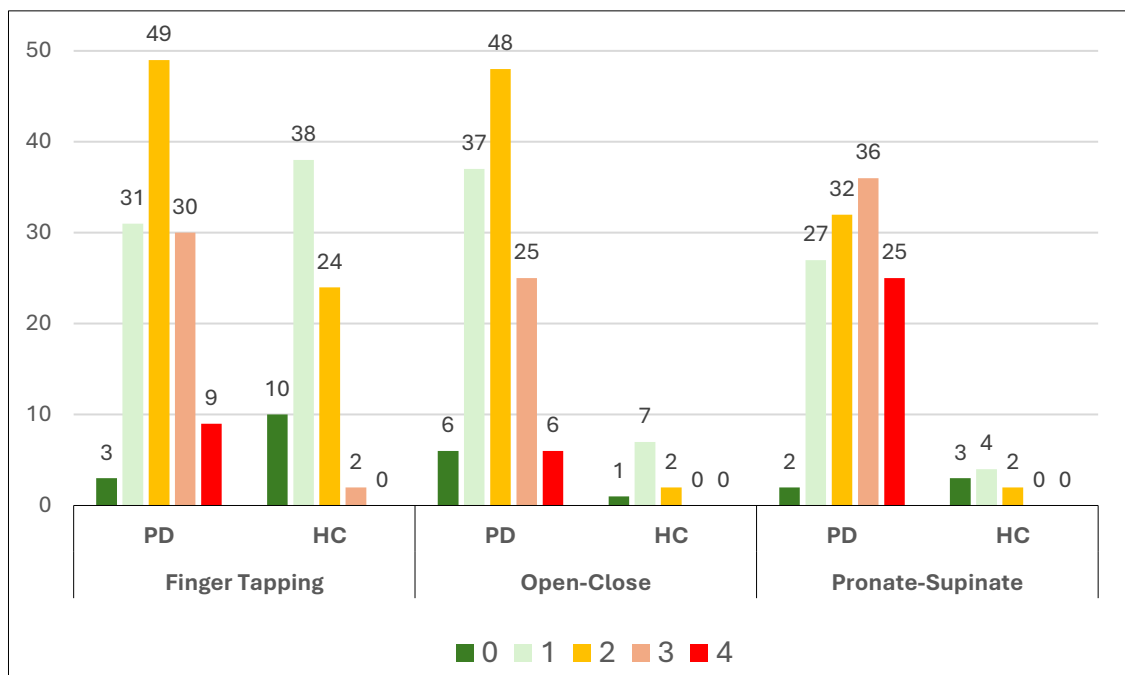


Figure 4.5. Clinician assigned UPDRS scores for finger tapping, opening-closing and pronation-supination hand movement tests. Scores for each upper limb test are divided into two participant groups: Parkinson's disease (PD) and Healthy Controls (HC).

Inter-rater reliability is commonly assessed using the reliability index interclass correlation coefficient (ICC). Koo & Li. (2016) published a guideline on reporting and selecting appropriate ICC models, defining consistency as the extent to which one rater's score (y) can be equated to another rater's score (x) plus a systematic error (c), while the absolute agreement is the level of direct agreement between y and x . Moreover, absolute agreement focuses on whether different clinicians assign identical scores to the same subject, whereas consistency establishes whether clinician scores for the same participant group are correlated with a consistent offset. Therefore, the absolute agreement and consistency of clinician assigned UPDRS scores can be evaluated and compared for each upper limb test using, representing each test's outcomes with these two ICC definitions.

In this analysis, ICC estimates and their 95% confident intervals were computed using the statistical package pingouin in Python. ICC representations are produced independently for both right and left hands for the finger tapping, open-close and pronate-supinate upper limb tests. The model selected was ICC3K, which corresponds to a two-way mixed effects, consistency measure for multiple raters (mean of k raters). Equations 1 and 2 show how consistency (1) across multiple raters and absolute agreement (2) are calculated using the mean square for rows (MS_R); mean square for error (MS_E); mean square for columns (MS_C) and the number of participants (n). **Figure 4.6** shows the ICC results obtained for the absolute agreement (left) and the consistency (right).

$$\frac{MS_R - MS_E}{MS_R} \quad (7)$$

$$\frac{MS_R - MS_E}{MS_R + \frac{MS_C - MS_E}{n}} \quad (8)$$

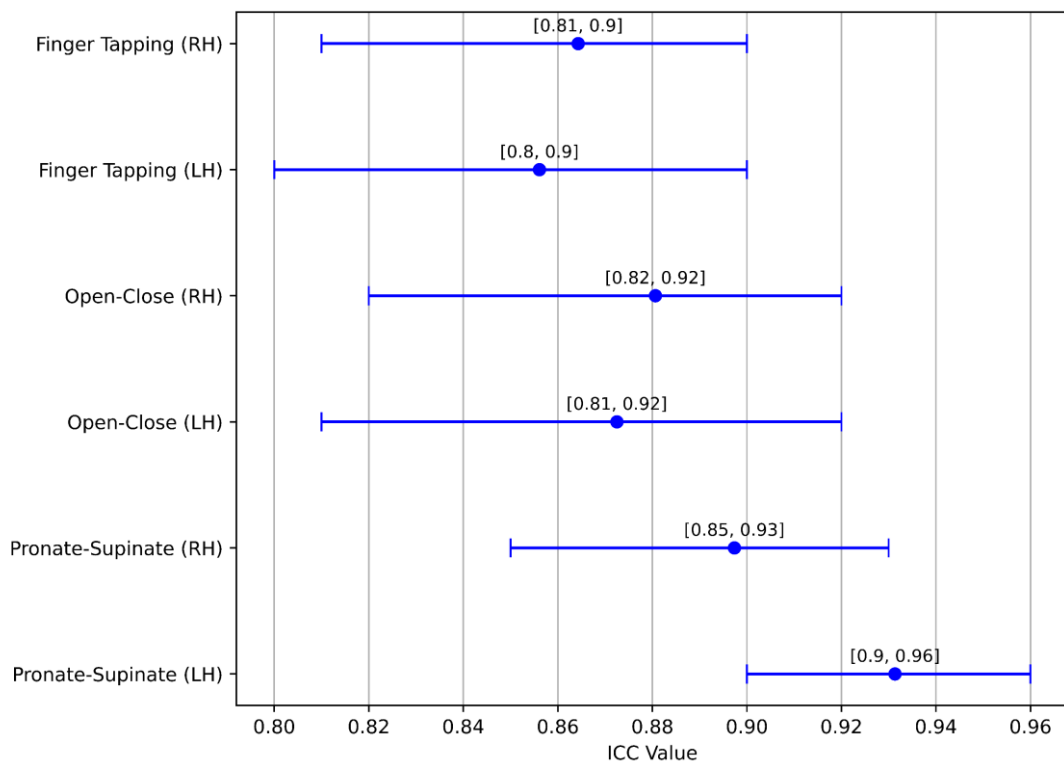
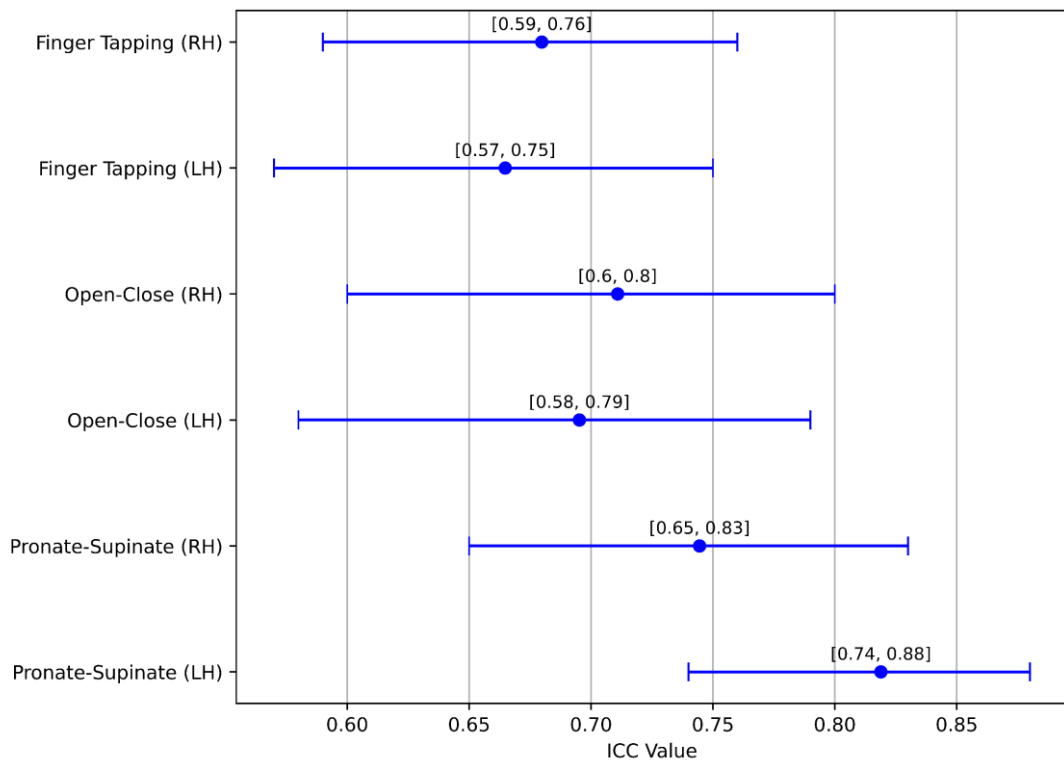


Figure 4.6. Interclass correlation coefficient (ICC) plots for the right hand (RH) and left hand (LF) upper limb movement tests: finger tapping, open-close and pronate-supinate. The reliability indices for the absolute agreement (top) and the consistency (bottom) including 95% confident intervals were computed using the statistical package pingouin in Python.

The absolute agreement in clinician assigned UPDRS scores demonstrated moderate to good reliability. indices ranged from (ICC = 0.679, 95% CI [0.59, 0.76]) to (ICC = 0.819, 95% CI [0.74, 0.88]) corresponding to the left hand tests for finger tapping and pronate-supinate, respectively. The average consistency across raters demonstrated strong to excellent reliability, with values ranging from 0.856 to 0.931. The highest index was for Pronate-Supinate (LH) (ICC = 0.931, 95% CI [0.9, 0.96]), reflecting excellent reliability across raters. The ICC values for all tests were above 0.85, suggesting strong agreement when averaging across the raters.

4.3 Methods

4.3.1 Model Framework

The model framework was implemented using Python and consists of landmark extraction, angle time series signal generation, signal correction, feature extraction from each corrected hand movement signal and symptom severity prediction. The proposed assessment framework is shown in **Figure 4.7** and the detailed steps including the python libraries used to perform this automated assessment are outlined in **Figure 4.8**.

As mentioned, these three upper limb tests share the movement aspect which is that they all involve rapid, repetitive and alternating movement. Therefore, whilst the angle measures were derived based on the landmarks that capture each movement tests gross angular displacements, the features extracted from each test are based on the same kinematic features as shown in **Table 12**. The computation of these kinematic features are detailed in **Chapter 4.3.2**.

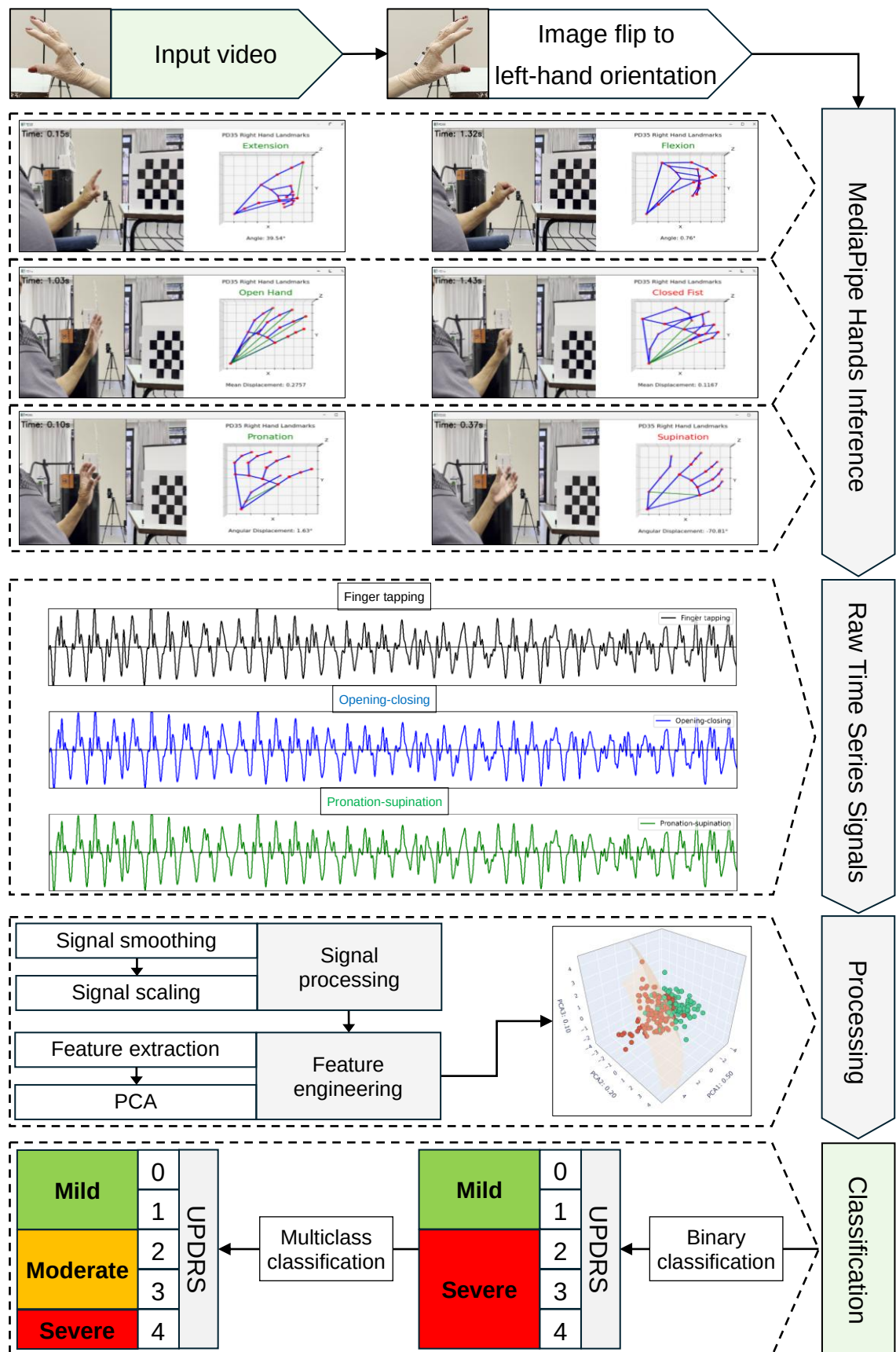


Figure 4.7. Computer vision-based automated assessment framework for grading symptom severity (UPDRS) using the finger tapping, opening-closing and pronation-supination hand movement tests.

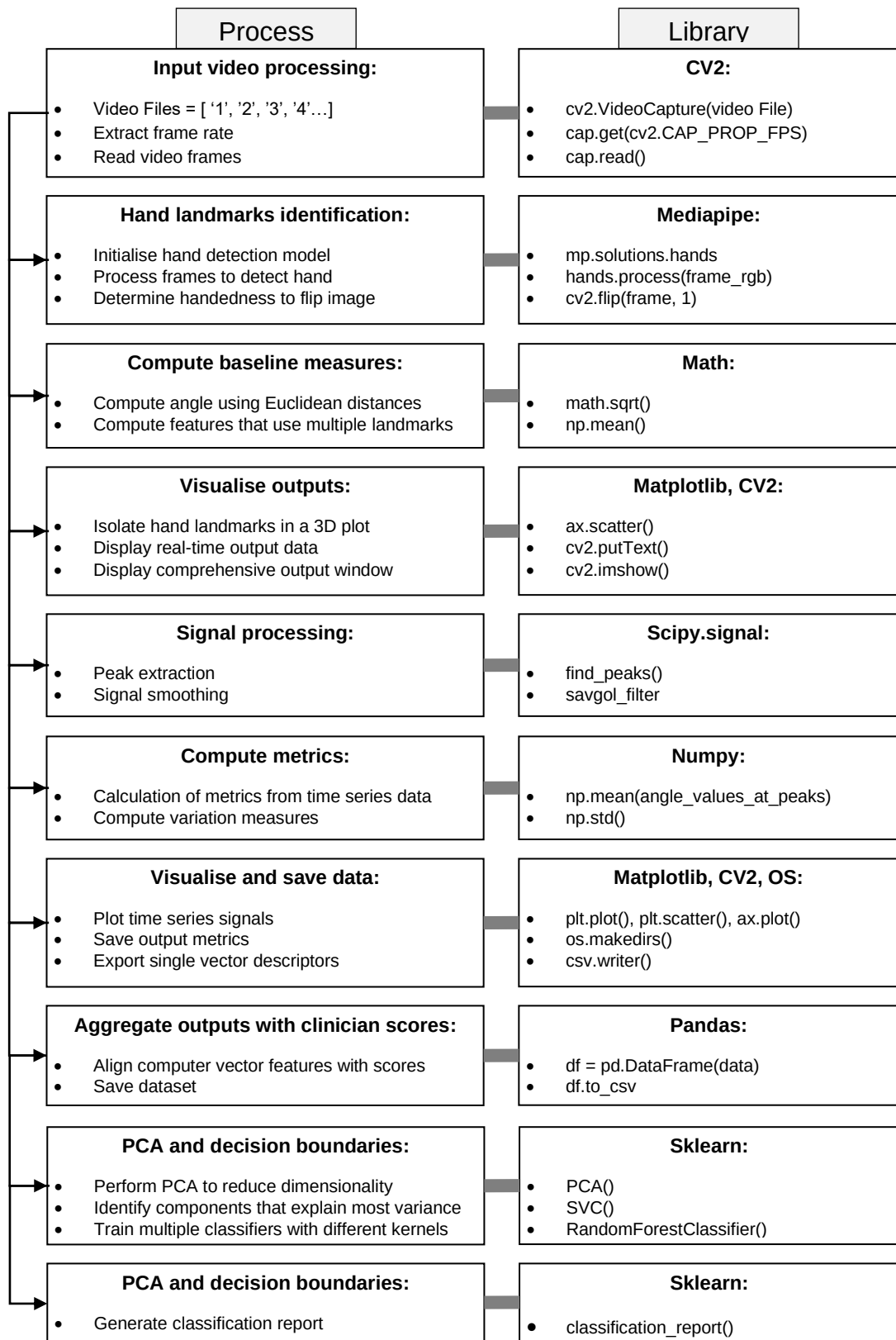


Figure 4.8. Workflow for video processing and feature extraction from upper limb movements using Python. The left column outlines the processes involved in the assessment framework. The right column lists the corresponding Python libraries and functions used in each stage.

4.3.2 Generating Angular Displacement Time series Signals

To mitigate parallax errors associated with varying camera to hand distances and complex orientation shifts during movement tests, kinematic features were extracted from test specific angular displacement temporal signals to form the raw time series signals. **Figure 4.9** illustrates the inherent offset in the signal amplitude derived from the finger tapping angle and the fingertip to thumb tip separation distance caused by misestimations of the two-point distance measure.

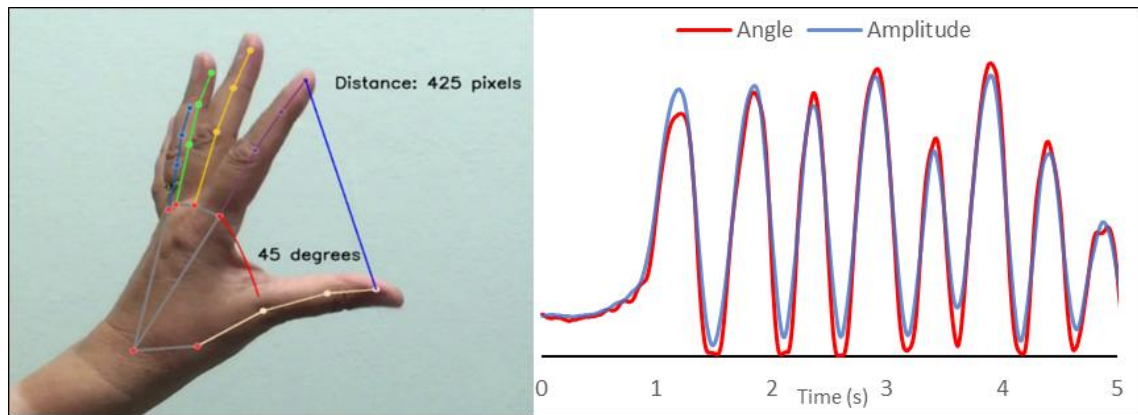


Figure 4.9. Example of the effects of perspective distortion resulting from amplitude measures of finger tapping. The overlaid time series signal illustrates the discrepancy between the angle (red) and amplitude (blue) signals. The angle signal is derived using the angle formed by the coordinates of the index fingertip, wrist, and thumb tip landmarks. The amplitude signal derived from the index fingertip to thumb distance underestimates movement magnitude.

The MediaPipe Hands model was employed to extract key landmark coordinates (x, y, z) for each hand as shown in **Figure 4.10**. The spatiotemporal signals for each movement test were primarily based on angular displacement data. For finger tapping, three Euclidean distances were derived from: the wrist to index fingertip, the index fingertip to thumb tip and the thumb tip to wrist. The length of these three line segments were used to compute the angular displacement, using the angle at the wrist which varies dynamically with changes in the separation between the index fingertip and thumb tip. For opening-closing hand movements,

the fingertip and corresponding MCP joint landmark coordinates of the index, middle, ring and pinky fingers were extracted. For each finger, the angle formed by the fingertip, MCP joint and wrist were used to compute the angular displacement at the MCP joint. The average angle formed by the four fingers were used to represent the angular displacement of opening-closing hand movements across video frames. For pronation-supination hand movements, the midpoint of the line segment formed by the wrist and middle finger's MCP joint was used to capture the angular displacement in the z-axis. **Figure 4.10** illustrates the landmarks and line segments used to compute angle measures and generate angular displacement time series data across consecutive video frames for each upper limb movement test.

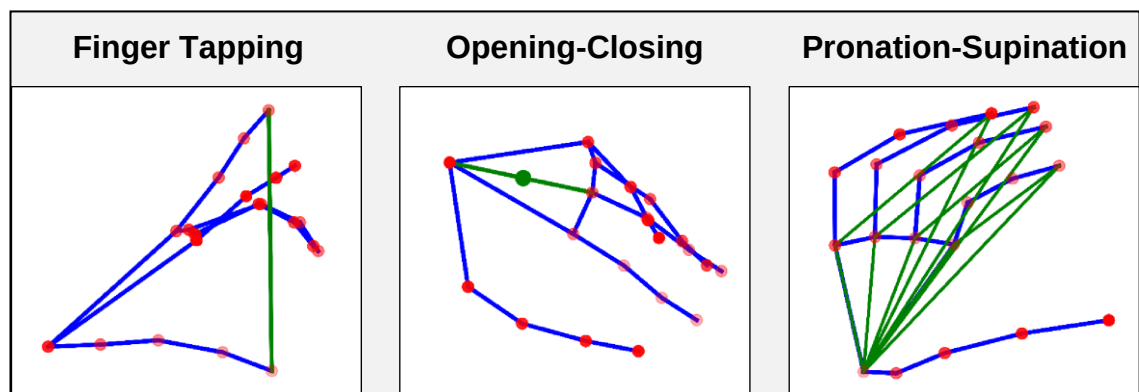


Figure 4.10. MediaPipe Hands model: 3D coordinate extraction of 21 hand landmarks with test specific virtual line segments used to quantify finger tapping (left), opening-closing (middle) and pronation-supination (right) upper limb movements.

An example of the time series signals generated from each upper limb movement test are shown in **Figure 4.12**. It can be observed that the finger tapping signal exhibits sharper, more pronounced peaks in contrast to the smoother opening-closing and pronation supination signal profiles. This can be explained by differences in the type of movements and how they are captured. In each

movement test, the generated time series signal is characterised a cyclic waveform with alternating peaks and troughs. Signal peaks and troughs mark points of inflection and represent distinct movement transitions in each upper limb movement test- see **Figure 4.11**. During finger tapping movements, the index finger and thumb repeatedly move from an inward flexion position (signal trough) to a maximum outward extension position (signal peak). In opening-closing hand movements, the resulting waveform reflects the movements of the four fingers from a closed fist position (signal trough) to maximum extension of the four fingers (signal peak). Finally, in the pronation-supination hand movement test, the generated waveform's peaks represent the points of reversal from a maximally pronated hand (signal trough) back towards a supinated position (signal peak). These critical points were extracted from angular displacement signals to derive key metrics and characterise movement performance for each upper limb test.

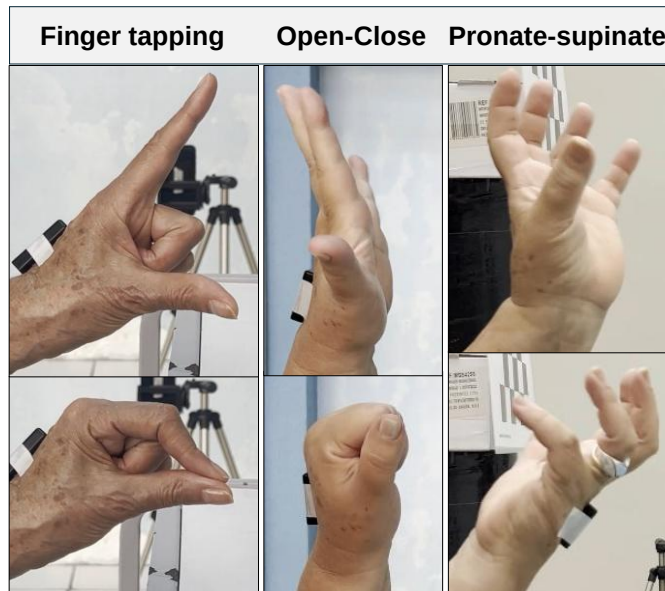


Figure 4.11. Directional changes associated with alternating upper movements. The binary signal representing directional changes during movement (top) is used to validate the number of repetitions derived from amplitude time series signals (bottom) of the finger tapping, opening-closing and pronation-supination hand tests.

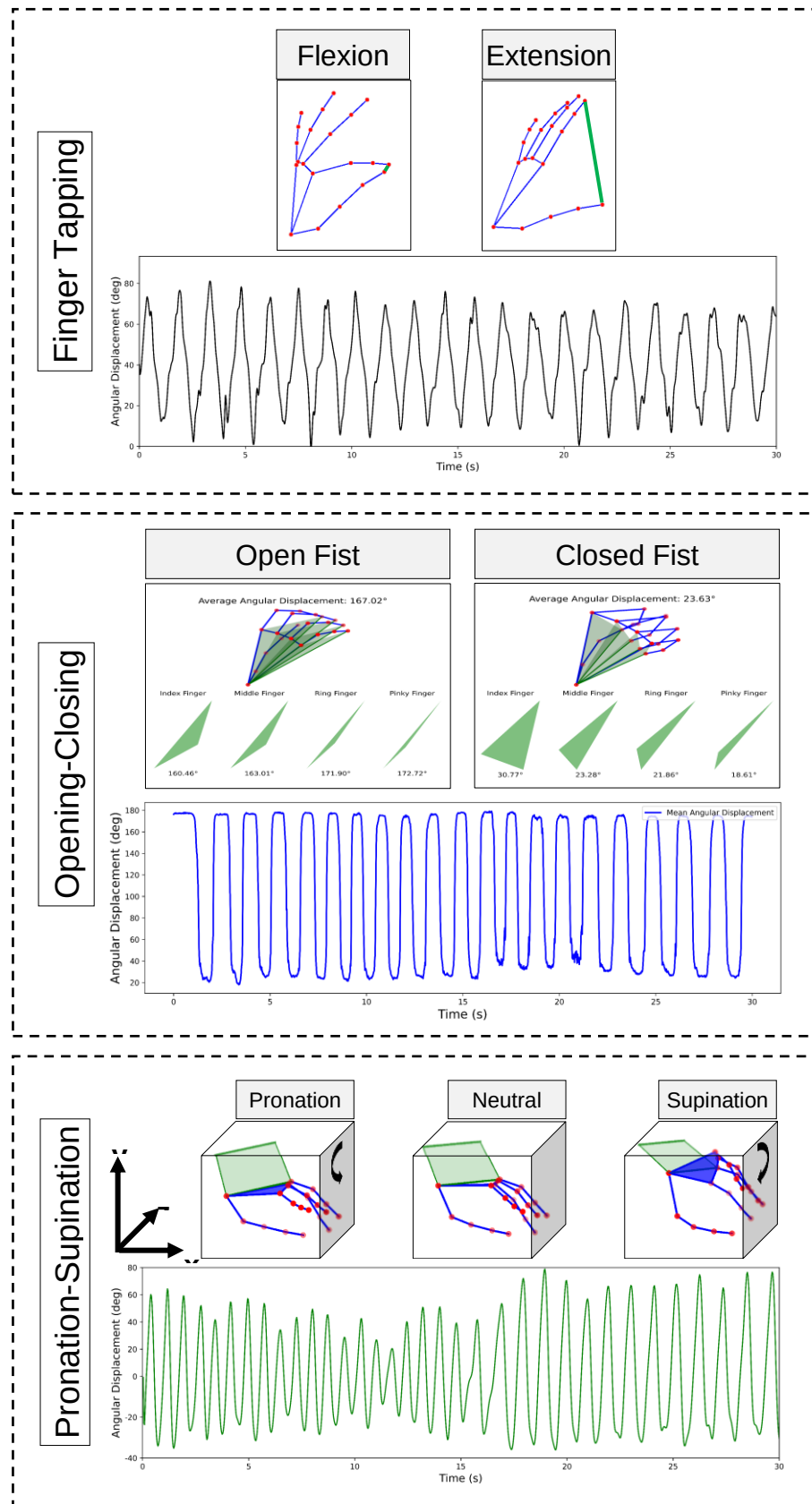


Figure 4.12. Generating angular displacement time series signals using relevant hand landmarks for finger tapping, opening-closing and pronation-supination hand movements.

4.3.3 Signal Processing and Feature Extraction

A low-order Savitzky-Golay filter was first applied to raw time series signals to preserve the signal characteristics. The smoothed signals were then scaled prior to subsequent feature extraction, as shown in **Figure 4.13**. Seven baseline features were extracted from each upper limb movement test by applying the same feature extraction methods. **Table 12** summarises the features extracted from each movement test, outlining the assessment-specific feature descriptions.

Table 12: Test-Specific Features Extracted from Video Recordings of the Finger Tapping, Opening-Closing and Pronation-Supination Hand Movement Tests

	Assessment-Specific Feature Description		
Feature	Finger Tapping (FT)	Opening-Closing (OC)	Pronation-Supination (PS)
Average maximum angular displacement	Average angular displacement at extension	Average angular displacement at open fist	Average angular displacement at supination
Average maximum angular velocity	Average angular velocity at extension	Average angular velocity at open fist	Average angular velocity at supination
Average minimum angular velocity	Average angular velocity at flexion	Average angular velocity at closed fist	Average angular velocity at pronation
Average rhythm	Average finger tapping frequency	Average opening-closing frequency	Average pronating-supinating frequency
Cycle count	FT repetition count	OC repetition count	PS repetition count
CV: maximum angular displacement	CV of maximum angular displacement across FT repetitions	CV of maximum angular displacement across OC repetitions	CV of maximum angular displacement across PS repetitions
CV: Angular displacement path length per movement cycle	CV of angular displacement path length per FT cycle	CV of angular displacement path length per OC cycle	CV of angular displacement path length per PS cycle

First, a Savitzky-Golay filter, available in the SciPy library in Python, was applied to signals to remove sudden transient shifts that occur due to false landmark detections. Signal noise reflects physiologically impossible hand movements so was filtered using a minimal smoothing effect of [3, 1] corresponding to window length and polynomial order, respectively. The smoothed angular displacement time series signals were then scaled between 0 and 1 using **Equation 9**, where x represents each angular displacement value in a signal.

$$x_{normalised} = \frac{x - \min_x}{\max_x - \min_x} \quad (9)$$

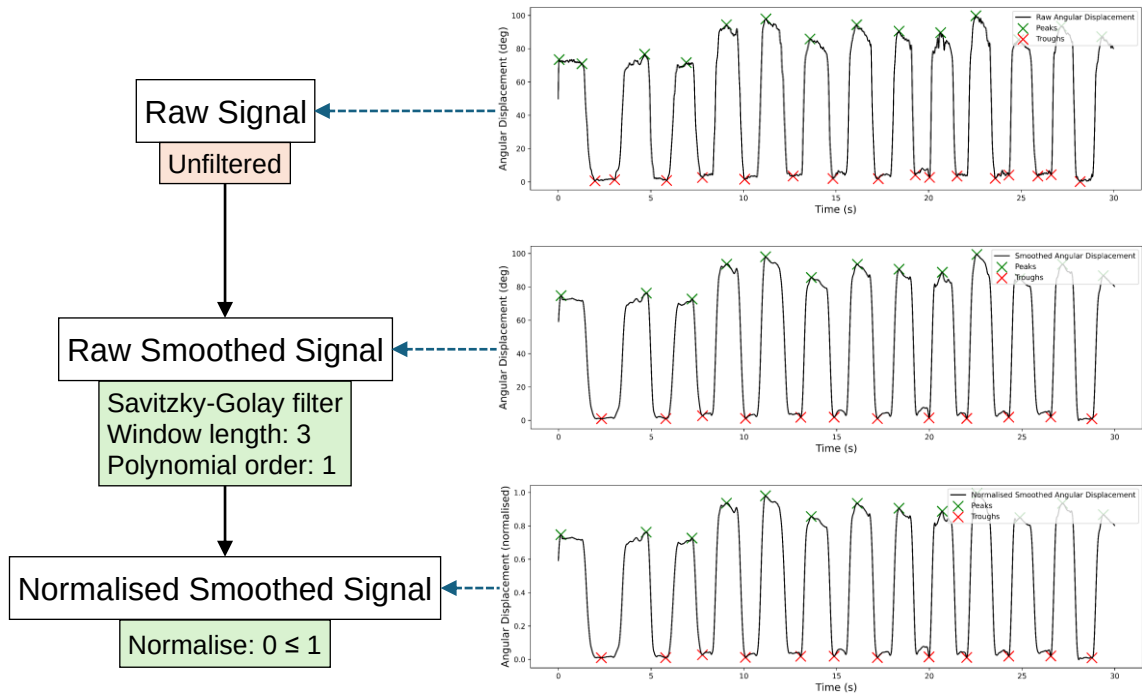


Figure 4.13. Signal processing: applying a Savitzky-Golay filter to raw angular displacement time series signals, then scaling signals for subsequent feature extraction.

The following section outlines the computation of the seven kinematic features derived from angular displacement time series signals using peak extraction. of angular displacement time series signals. The equations used to compute relevant kinematic features are provided.

Average maximum angular displacement: the angular displacement peaks (green x markers) were extracted from the angular velocity time series signal (**Figure 4.14**) and the average peak value was computed across all movement cycles using **Equation 10**.

$$\frac{\text{Sum of peaks}}{\text{Number of peaks}} \quad (10)$$

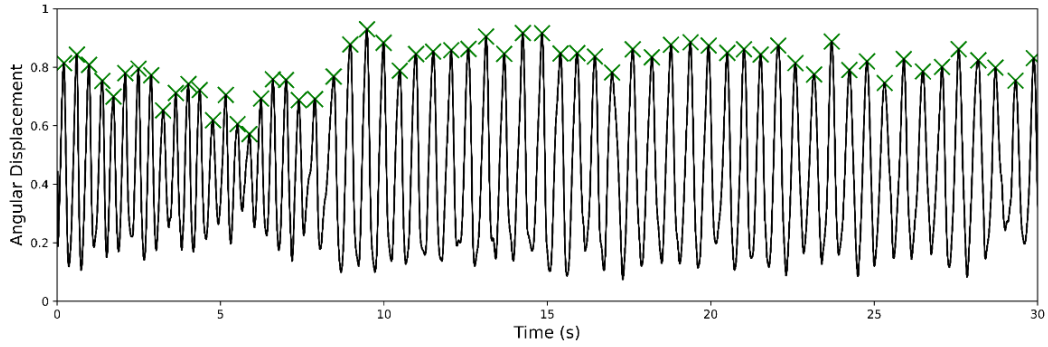


Figure 4.14. Peak extraction from angular displacement time series signals using the ‘find peaks’ function within the SciPy library in Python.

For the angular velocity-derived metrics, an angular velocity signal was first generated (**Figure 4.15**) using the change in angular velocity over each consecutive pair of frames and dividing this change by the corresponding time interval. **Average maximum angular velocity:** the angular velocity peaks (green markers) were extracted and the average value was computed using **Equation 11**.

$$\frac{\text{sum of positive angular displacement peaks}}{\text{number of positive angular displacement peaks}} \quad (11)$$

Average minimum angular velocity: the negative angular velocity peaks (red markers) were extracted and the average value over all movement cycles was computed using **Equation 12**.

$$\frac{\text{Sum of negative angular displacement peaks}}{\text{Number of negative angular displacement peaks}} \quad (12)$$

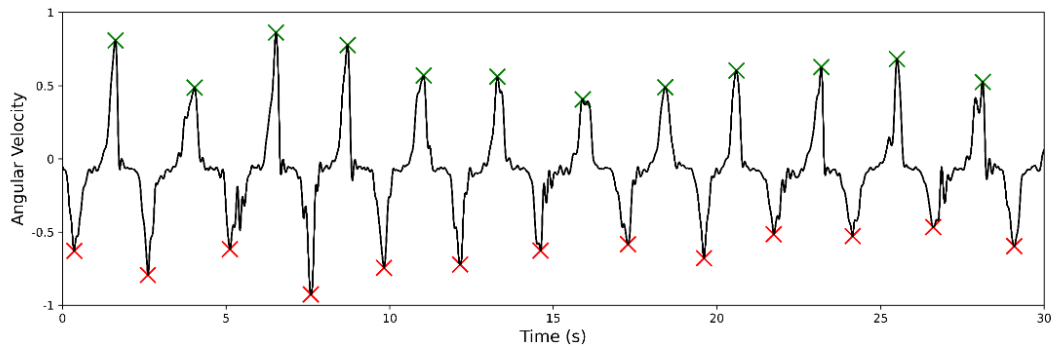


Figure 4.15. Positive and negative peak extraction from angular velocity time series signals.

Average rhythm: the average interpeak times were computed using **Equation 13**, this represents the average time interval between consecutive pairs of maximum angular displacements peaks, **Figure 4.16**.

$$\frac{\text{Sum of interpeak frequency}}{\text{Number of angular displacement peaks}} \quad (13)$$

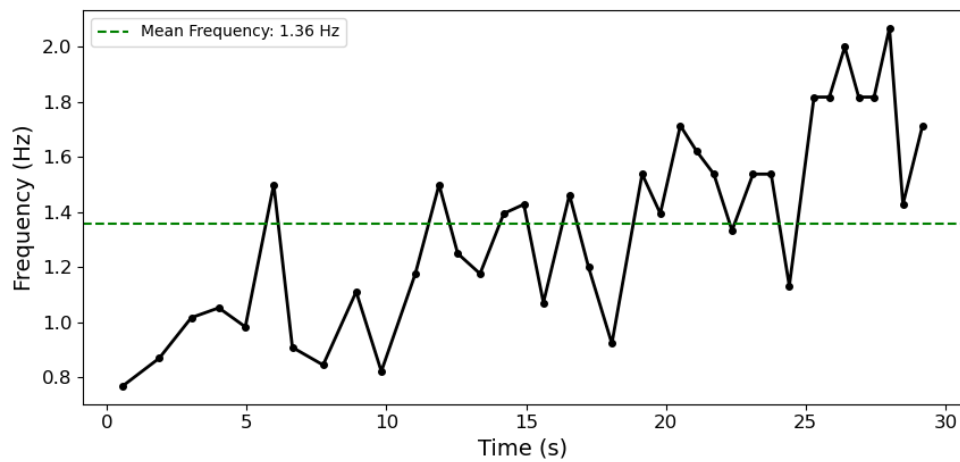


Figure 4.16. Angular displacement Interpeak frequency: the plot shows the change in the time intervals across successive peak angular displacements over the 30 second movement test.

Cycle count: the number of positive peaks were extracted from angular displacement time series signals using **Equation 14**. Only positive peaks that were directly followed by a negative peak were recorded as a movement cycle.

$$\sum_{i=1}^N \text{Positive peaks}_i \quad (14)$$

Coefficient of variation of maximum angular displacements: the maximum angular displacements were extracted from each movement cycle. The coefficient of variation of the angular velocities peaks, represented by the green markers in **Figure 4.17**, were then computed using **Equation 15**.

$$\frac{\sigma_{\text{angular displacement peaks}}}{\chi_{\text{angular displacement peaks}}} \quad (15)$$

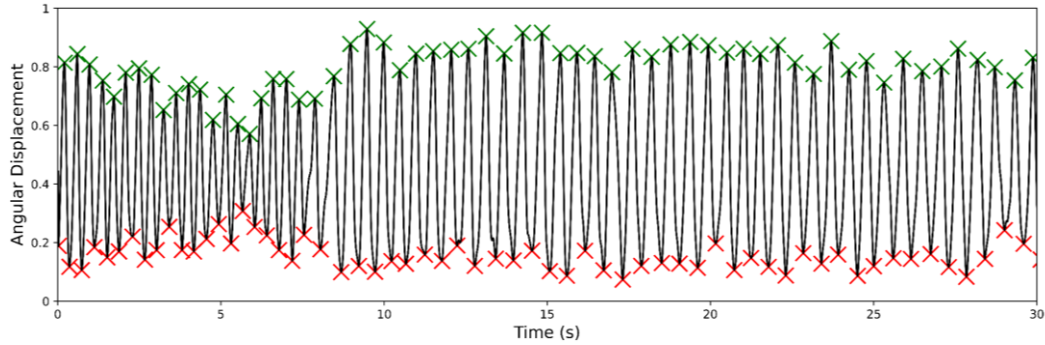


Figure 4.17. Cycle count: the number of positive peaks are used to derive the number of movement repetitions. Peaks (green markers) must be directly followed by a signal trough (red markers) to indicate complete movements. Two peaks directly following each other are not recorded as a cycle.

Coefficient of variation of Angular displacement path length per movement cycle: the total angular displacement path length moved across each cycle was extracted from time series signals. A movement cycle is defined as a signal trough to peak to trough. **Figure 4.18** illustrates the angular displacement path length for each movement cycle by distinguished movement cycles using alternating black and red signal colours. The coefficient of variation of the total angular displacement path length across all movement cycles was calculated using **Equation 16**.

$$\frac{\sigma_{\text{interpeak path lengths}}}{\chi_{\text{interpeak path lengths}}} \quad (16)$$

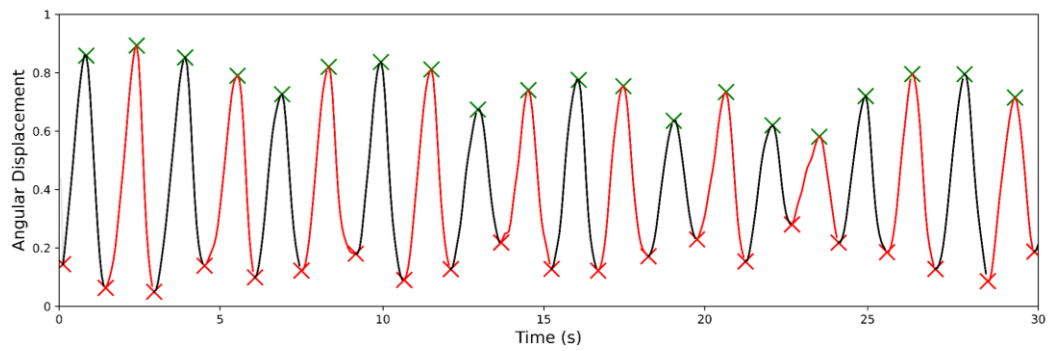


Figure 4.18. Identification of complete movement cycles in angular displacement time series signals to compute the sum of angular displacement path length across movement cycles. A signal trough to peak to trough is used to define a full cycle. Alternating black and red signal colours are used to distinguish between successive movement cycles.

4.3.4 Principal Component Analysis

Principal Component Analysis (PCA) is a common dimensionality reduction technique in data analysis that transforms high-dimensional data into a lower feature space through identifying linear combinations of the original features (principal components) that capture the maximum variance in the dataset. Calculations involving PCA transform features into a new set of orthogonal variables (Kurita, 2020). In this analysis, four distinct approaches were used to analyse the transformed features, using a varying number of principal components and label aggregation strategies used to subsequently perform symptom severity classification.

The first approach in the PCA transformed the time series signal-derived features into two principal components and the second approach used three principal components. For the labelling strategy, the five MDS-UPDRS scores assigned by clinicians were consolidated into both binary and multiclass formats. For binary classification, MDS-UPDRS scores of 0 and 1 were labelled mild and MDS-UPDRS scores 2, 3 and 4 were labelled severe. For the multiclass approach, MDS-UPDRS scores 0 and 1 remained mild, while the severe class was further

split into a moderate and severe class, corresponding to scores (MDS-UPDRS 2, 3) and (MDS-UPDRS 4) respectively. The final classification stage of the assessment used the following four assessment configurations:

1. Two principal components to perform binary classification.
2. Two principal components to perform multiclass classification.
3. Three principal components to perform binary classification.
4. Three principal components to perform multiclass classification.

Figure 4.19 shows the explained variance captured by the first three principal components (PC1, PC2 and PC3) for each of the three upper limb tests: finger tapping, opening-closing, and pronation-supination hand movements. In each bar chart, the blue bars represent the percentage of variance in the data explained by each principal component, while the red line shows the cumulative explained variance which increases with the addition of each principal component. In each test, the cumulative variance captured by the first three principal components was approximately 85%.

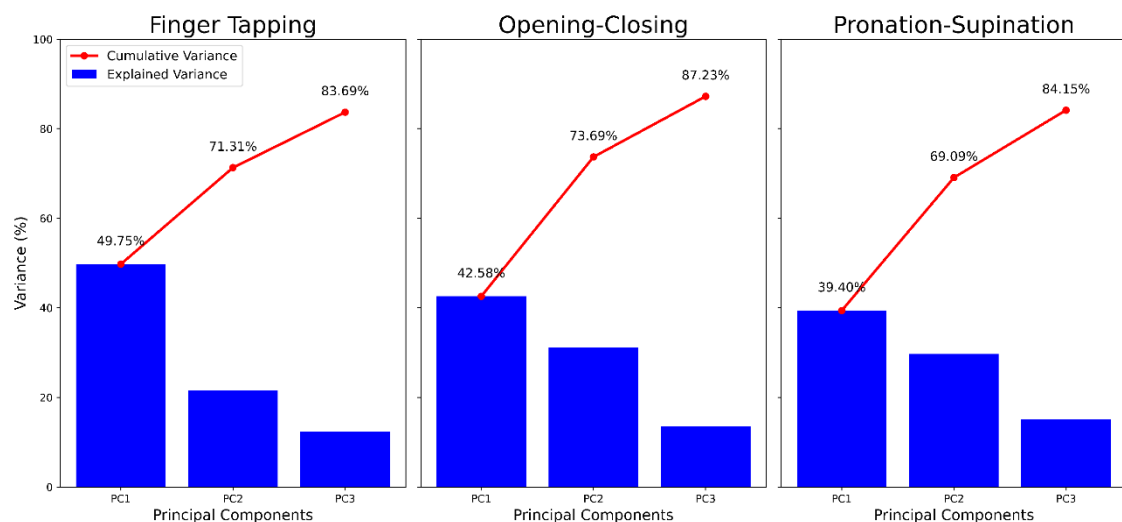


Figure 4.19. Explained variance of three principal components with the cumulative variances for each upper limb assessment.

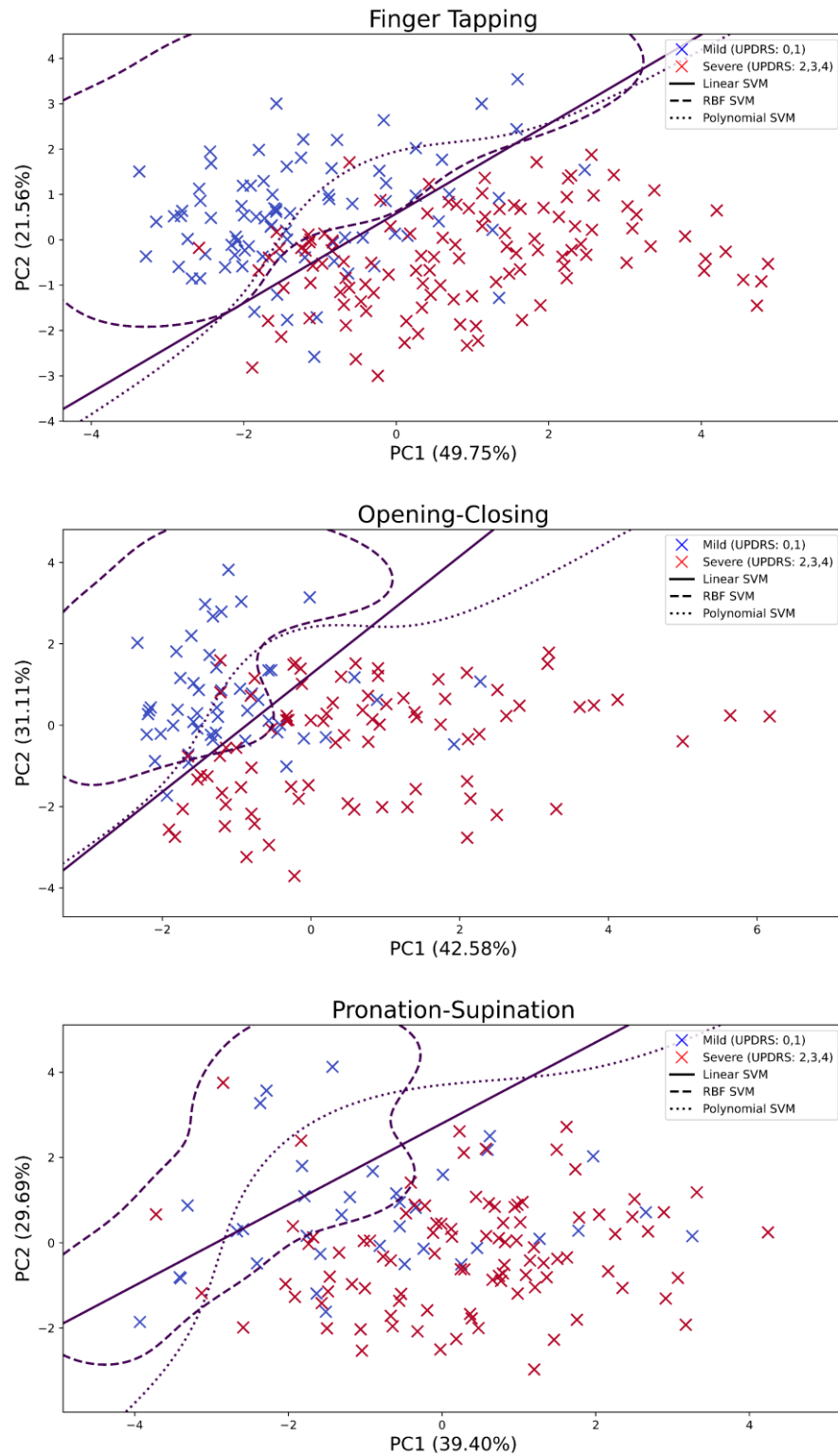


Figure 4.20. Principal component analysis using two principal components. The top plot shows an example of three support vector machine (SVM) kernels used to generate decision boundaries between participants belonging to the mild and severe classes. Decision boundaries are applied to distinguish two symptom severity classes: mild and severe, corresponding to UPDRS scores ($0 \leq 1$) and ($2 \leq 4$), respectively.

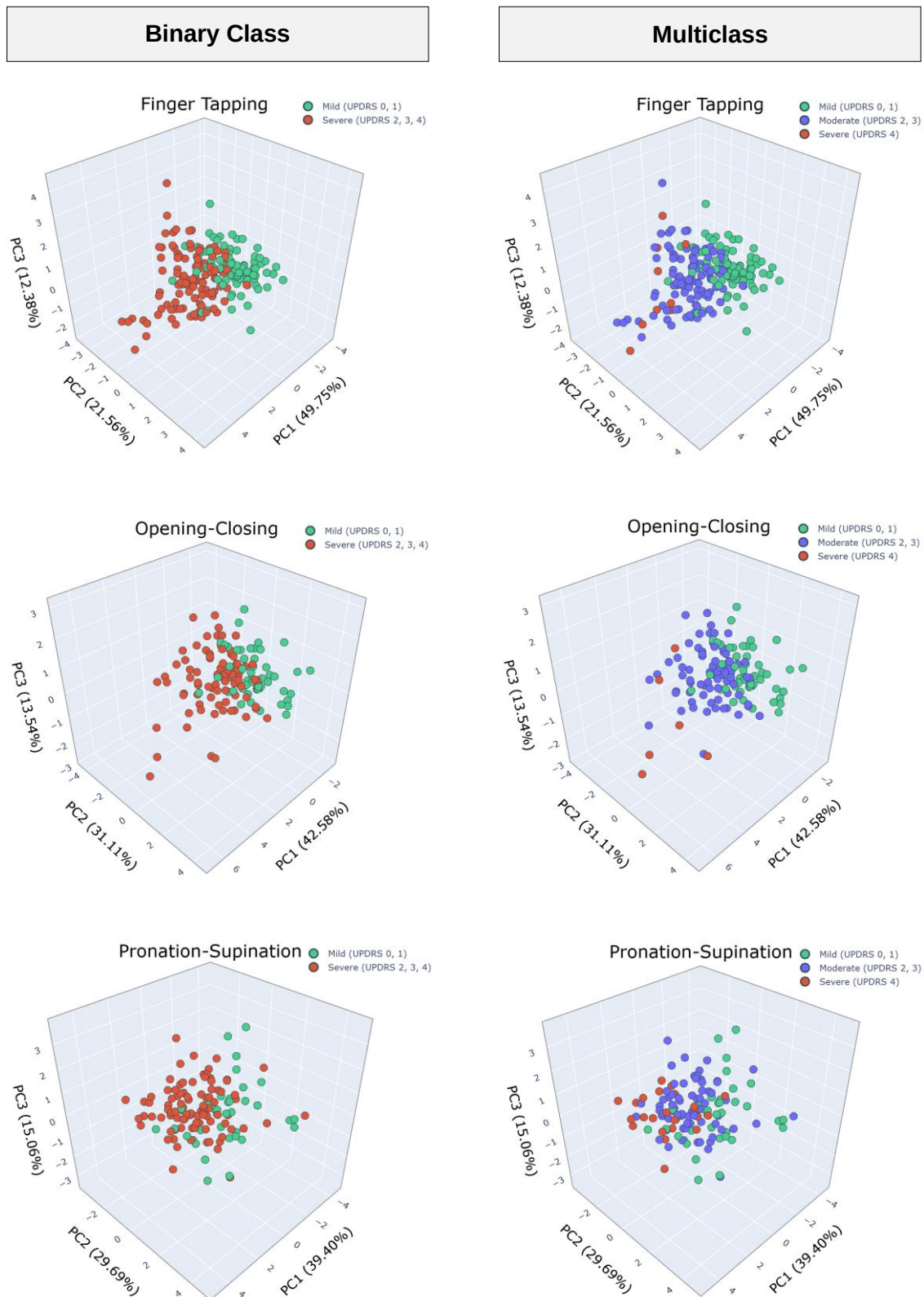


Figure 4.21. Principal component analysis: Finger tapping (row 1), opening-closing (row 2) and pronation-supination (row 3). The 3D plots in the left column label data points into two groups: mild (MDS-UPDRS $0 \leq 1$) and severe (MDS-UPDRS $2 \leq 4$), while the plots in the right column label data into three groups: respectively: mild ($0 \leq 1$), moderate ($2 \leq 3$) and severe (4).

Figure 4.22 shows the final principal component analysis which was performed on the finger tapping dataset. In this evaluation, PCA was used to transform measures of asymmetry into three principal components. For each original feature, the relative asymmetry was determined by normalising the difference between features derived from the right and left hands relative to their mean using **Equation 17**. Right and left hand feature sets were consolidated into a single set of seven asymmetry measures for each participant.

$$\frac{|Feature_{Left} - Feature_{Right}|}{Mean\ of\ Left\ and\ Right\ Feature} \quad (17)$$

The PCA using asymmetry features labelled transformed features by participant group i.e. control or PD. This approach focused on establishing whether asymmetrical movement performance between the right and left upper limbs can be used to distinguish Parkinson's patients from healthy controls. This class labelling strategy was implemented as the resulting asymmetry features for each participant would require a single class label. Moreover, assigning averages of individual hand MDS-UPDRS scores for each participant would eliminate the difference in clinician-assigned symptom severity scores between participants' right and left hands. The analysis was limited to the finger tapping dataset due to the small sample of controls in the opening-closing and pronation-supination datasets. Subsequent analysis used two and three principal components of the asymmetry features in the finger tapping dataset to perform binary classification. Notably, the bar chart in **Figure 4.22** shows the cumulative variance captured by the first three principal components was 94.94%. This indicates that these components capture most of the essential patterns and underlying structure in the data, leaving minimal variance unaccounted for.

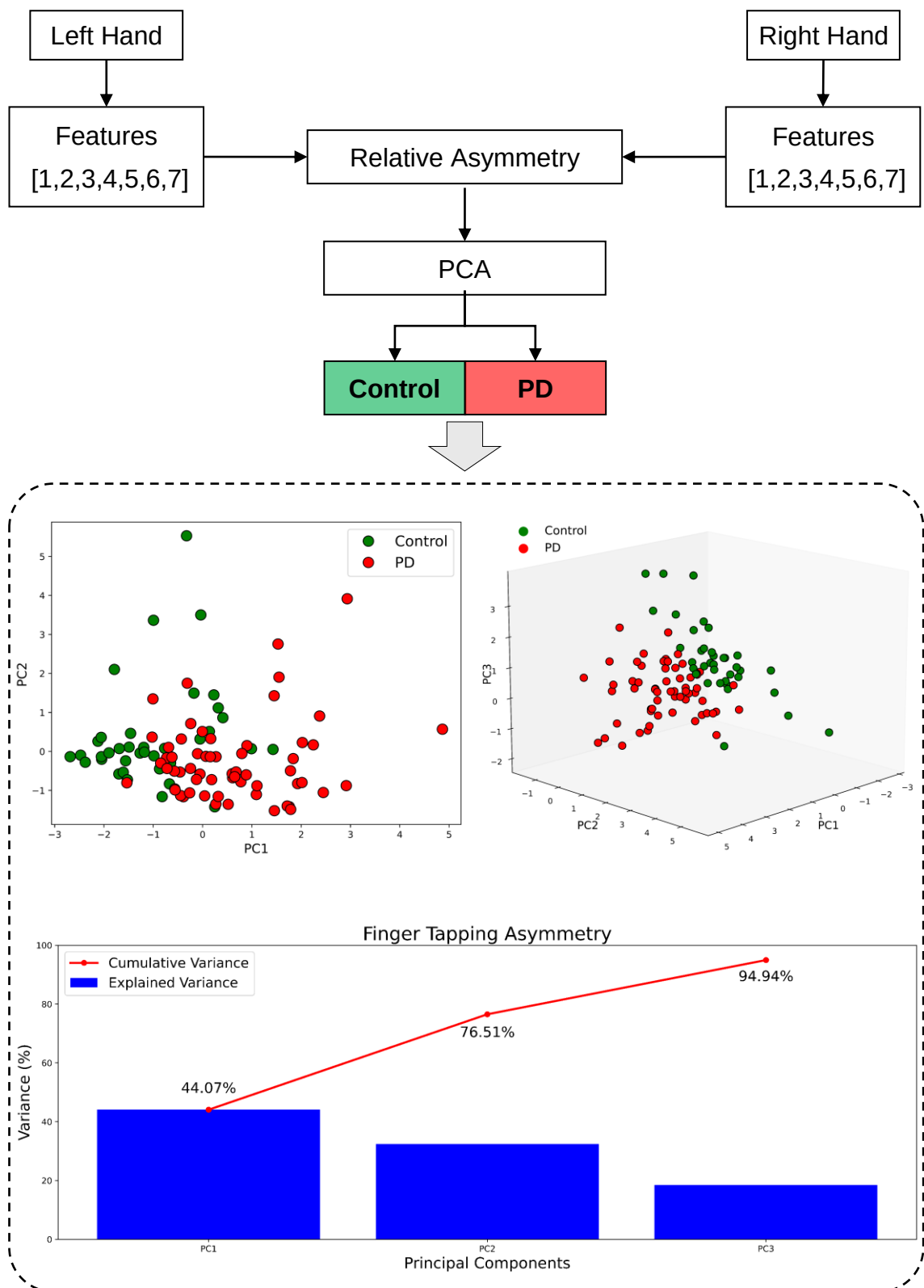


Figure 4.22. Principal component analysis on asymmetry features of the finger tapping test using two and three principal components. The bar chart shows the explained variance (blue bars) of three principal components with a cumulative variance of 94.94% (red line).

4.4 Results

A total of 459 upper limb movement videos were analysed using the four previously outlined configurations. For each analysis configuration, four classification models were employed including a SVM-L (Support Vector Machine with a Linear kernel), SVM-RBF (Support Vector Machine with a Radial Basis Function kernel), SVM-P (Support Vector Machine with a Polynomial kernel) and Logistic Regression to perform symptom severity classification. This evaluation approach enabled a detailed comparison of performance for the finger tapping, opening-closing and pronation-supination hand movement assessments across the classification models using two and three principal components and binary and multiclass classifications.

Each upper limb assessment dataset was divided into training and testing sets with 70% allocated to training and 30% to testing. The Python library Scikit-Learn was used to generate key performance metrics through multiple evaluation functions. To assess each model's performance, the accuracy, sensitivity, specificity and Area Under the Receiving Characteristic Curve (AUC-ROC) performance metrics were used. **Accuracy** indicates the percentage of correct class predictions in the testing data and is computed using the **Equation 18**.

$$\frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100 \quad (18)$$

While accuracy can be directly computed as a percentage of correct predictions across all samples, metrics such as sensitivity, specificity and AUC-ROC are inherently defined for binary classification, where a single positive class is compared against a negative class. Therefore, the performance metrics reported in the results tables of **Chapter 4.4** for the multiclass classification task required

additional calculation to yield single summary values. To achieve this, each symptom severity class (mild, moderate and severe) was first treated as a positive case with the remaining two as a negative, using a One-vs-Rest (OvR) approach. For sensitivity and specificity, the resulting class-wise values were then weighted according to class frequency to obtain a single performance metric for each assessment configuration. Single AUC values were obtained by first computing the area under the ROC curve for each class using predicted probabilities and then calculating a weighted average of these class-wise AUC scores, with the weights determined by the frequency of each consolidated class. This approach was implemented to account for class imbalance across the mild, moderate and severe symptom severity classes.

Sensitivity indicates the model's ability to correctly identify each class. This metric was computed for both binary (mild and Severe) and multiclass (mild, moderate and severe) classifications using **Equation 19**.

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (19)$$

Specificity evaluates how well each model identifies non-target instances within each class and is determined using **Equation 20**.

$$\frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}} \quad (20)$$

Area Under the Receiving Characteristic Curve (AUC-ROC) was determined using the `roc_auc_score` function from Scikit-learn, indicating the model's ability to distinguish between the mild, moderate and severe classes. Higher values of AUC represent more effective discrimination between the symptom severity classes by the model.

Table 13: Classification Performance Metrics for the Finger Tapping Assessment using Four Evaluation Configurations.

Finger Tapping					
Method	Classifier	Accuracy	Sensitivity	Specificity	AUC
2 PC Binary Classification	SVM-L	0.83	0.86	0.79	0.92
	SVM-RBF	0.85	0.92	0.76	0.90
	SVM-P	0.82	0.96	0.61	0.91
	Logistic Regression	0.83	0.86	0.79	0.92
2 PC Multiclass Classification	SVM-L	0.78	0.55	0.86	0.87
	SVM-RBF	0.81	0.56	0.87	0.87
	SVM-P	0.76	0.52	0.84	0.87
	Logistic Regression	0.78	0.54	0.86	0.88
3 PC Binary Classification	SVM-L	0.83	0.78	0.46	0.74
	SVM-RBF	0.81	0.78	0.62	0.76
	SVM-P	0.80	0.89	0.15	0.75
	Logistic Regression	0.85	0.74	0.38	0.75
3 PC Multiclass Classification	SVM-L	0.80	0.55	0.87	0.90
	SVM-RBF	0.78	0.53	0.85	0.84
	SVM-P	0.76	0.51	0.83	0.88
	Logistic Regression	0.78	0.54	0.86	0.88

Table 14: Classification Performance Metrics for the Opening-Closing Hand Movement Assessment using Four Evaluation Configurations.

Opening-Closing					
Method	Classifier	Accuracy	Sensitivity	Specificity	AUC
2 PC Binary Classification	SVM-L	0.85	0.90	0.76	0.90
	SVM-RBF	0.86	0.91	0.75	0.91
	SVM-P	0.80	0.98	0.51	0.90
	Logistic Regression	0.84	0.89	0.76	0.90
2 PC Multiclass Classification	SVM-L	0.80	0.55	0.86	0.84
	SVM-RBF	0.82	0.66	0.87	0.90
	SVM-P	0.78	0.72	0.83	0.87
	Logistic Regression	0.80	0.65	0.87	0.87
3 PC Binary Classification	SVM-L	0.83	0.87	0.76	0.90
	SVM-RBF	0.90	0.96	0.82	0.90
	SVM-P	0.85	1.00	0.65	0.91
	Logistic Regression	0.83	0.87	0.76	0.90
3 PC Multiclass Classification	SVM-L	0.70	0.50	0.82	0.81
	SVM-RBF	0.83	0.59	0.89	0.87
	SVM-P	0.75	0.63	0.85	0.86
	Logistic Regression	0.73	0.62	0.84	0.84

Table 15: Classification Performance Metrics for the Pronation-Supination Hand Movement Assessment using Four Evaluation Configurations.

Pronation-Supination					
Method	Classifier	Accuracy	Sensitivity	Specificity	AUC
2 PC Binary Classification	SVM-L	0.75	0.97	0.79	0.79
	SVM-RBF	0.82	0.94	0.74	0.78
	SVM-P	0.76	0.97	0.69	0.76
	Logistic Regression	0.78	0.95	0.83	0.79
2 PC Multiclass Classification	SVM-L	0.62	0.51	0.74	0.75
	SVM-RBF	0.78	0.60	0.80	0.78
	SVM-P	0.59	0.47	0.73	0.66
	Logistic Regression	0.63	0.53	0.76	0.76
3 PC Binary Classification	SVM-L	0.68	0.78	0.46	0.74
	SVM-RBF	0.73	0.78	0.62	0.76
	SVM-P	0.65	0.89	0.15	0.75
	Logistic Regression	0.63	0.74	0.38	0.75
3 PC Multiclass Classification	SVM-L	0.60	0.63	0.77	0.73
	SVM-RBF	0.69	0.57	0.77	0.73
	SVM-P	0.53	0.40	0.68	0.70
	Logistic Regression	0.58	0.61	0.75	0.73

Table 16: Summary of Model Accuracies for Binary and Multiclass Classifications using Two and Three Principal Components Across the Finger Tapping (FT), Opening-Closing (OC) and Pronation-Supination (PS) Assessments.

Method	Test	2 Principal Components		3 Principal Components	
		Binary	Multiclass	Binary	Multiclass
SVM-L	FT	0.83	0.78	0.83	0.80
	OC	0.85	0.80	0.83	0.70
	PS	0.75	0.62	0.68	0.60
SVM-RBF	FT	0.85	0.81	0.81	0.78
	OC	0.86	0.82	0.90	0.83
	PS	0.82	0.78	0.73	0.69
SVM-P	FT	0.82	0.76	0.80	0.76
	OC	0.80	0.78	0.85	0.75
	PS	0.76	0.59	0.65	0.53
Logistic Regression	FT	0.83	0.78	0.85	0.78
	OC	0.84	0.80	0.83	0.73
	PS	0.78	0.63	0.63	0.58

Table 17: Average Model Accuracies Across the Finger Tapping, Opening-Closing and Pronation-Supination Hand Movement Assessments. The Column highlighted in Green Represents the Highest Average Model Accuracies.

Configuration	SVM-L	SVM-RBF	SVM-P	Logistic Regression
2 PC & Binary Classification	0.81	0.84	0.79	0.82
2 PC & Multiclass Classification	0.73	0.80	0.71	0.74
3 PC & Binary Classification	0.78	0.81	0.77	0.77
3 PC & Multiclass Classification	0.70	0.76	0.68	0.70

Table 18: Classification Performance Metrics for Binary Control vs. PD Classification using Asymmetry Features in the Finger Tapping Dataset.

Method	Classifier	Accuracy	Sensitivity	Specificity	AUC
2 PC Control vs. PD	SVM-L	0.80	0.89	0.78	0.89
	SVM-RBF	0.77	0.92	0.67	0.93
	SVM-P	0.73	0.90	0.62	0.91
	Logistic Regression	0.80	0.87	0.59	0.88
3 PC Control vs. PD	SVM-L	0.93	0.94	0.92	0.94
	SVM-RBF	0.94	0.92	0.94	0.96
	SVM-P	0.80	0.96	0.89	0.97
	Logistic Regression	0.90	0.91	0.83	0.93

4.5 Discussion

The presented assessment methodology evaluated three upper limb movement assessments commonly administered in PD motor examinations. The data acquisition strategy demonstrated the viability of a standardised data capture protocol for three individual PD motor assessments, each involving each involving repetitive alternating hand movements, using standard smartphone video recordings. The comprehensive analysis considered multiple potential sources of error including the reliability of clinician-assigned symptom severity scores across all assessments. Importantly, the scoring complexity associated with both multiple raters and assessments was investigated with the reliability index, measured using interclass correlation coefficient (ICC), demonstrated

moderate to good reliability in the absolute agreement between clinician ratings across finger tapping, opening-closing and pronation-supination (right and left hand) assessments. It must be noted that all three neurologists specialised in movement disorders and scored videos based on their first observation.

Test-specific implementation of MediaPipe Hands enabled clinically relevant features to be extracted from baseline angular displacement time series signals of finger tapping, opening-closing and pronation-supination hand movements. The application of conceptually similar features across each assessment provided a streamlined approach to signal processing while the tailored selection of hand landmarks ensured effective capture of the unique characteristics associated with each movement type. Therefore, variations in classification performance between the upper limb assessments could be attributed to varying movement performance rather than inconsistencies in feature design. The selection of hand landmarks were tailored to capture and appropriately represent each distinct upper limb movement assessment. As such, the proposed method provides an effective assessment framework for PD by evaluating symptom severity through multiple upper limb movements.

Incorporating principal component analysis into the assessment framework allowed clear distinctions in the data derived from each upper limb assessment to emerge through the transformed feature sets. The exploration of four different assessment configurations using PCA provided useful insight into the models' sensitivity to feature complexity. Experimenting with two classification approaches involved the consolidation of the five UPDRS classes into either binary or multiple symptom severity classes, allowing the effects of classification granularity on discriminatory performance to be explored. Symptom severity

classification was explored using multiple classification models including linear SVM, radial basis function SVM, polynomial SVM and logistic regression. Multiple classifiers were explored to determine which algorithmic approach best captures distinct patterns in the datasets for accurate classification. The results demonstrate several patterns in classification performance across the three upper limb assessments when varying the number of principal components and classification approaches (binary vs. multiclass).

The finger tapping assessment consistently demonstrated promising classification performance across the classifiers, with test accuracies ranging from 0.8 using three principal components to perform multiclass classification to 0.85 for binary classification when using both two and three principal components. A potential factor contributing to good class separability is the motor coordination involved in repeated taps of the distal forefinger against thumb. Both the SVM-RBF and logistic regression models achieved slightly higher test accuracies than the other classifiers in binary classification when using two principal components suggesting that the finger tapping test data provides discernible patterns between the mild and severe classes using the more basic two principal component representation. The slight decrease in accuracy observed for multiclass classification indicates reduced discriminatory power across adjacent UPDRS scores. This challenge is expected given that a certain degree of slowness in movement exists in even healthy controls, particularly among older participants. The opening-closing test further demonstrated strong classification performance specifically with SVM-RBF, achieving a peak accuracy of 0.9 for binary classification with three principal components. The stronger predictive performance indicates improved class separation even in the mild-

moderate-severe class setup. This could result from an increased number of hand landmarks being incorporated into the computer vision framework. That being said, the increased demands associated with larger range movements could have resulted in motor symptoms like bradykinesia and rigidity being more pronounced, providing stronger distinctions across symptom severity scores. The pronation-supination assessment dataset generated the lowest overall classification performance, with a highest accuracy of 0.82 achieved with SVM-RBF using two principal components to distinguish participants in the mild class from those in the severe class ($\text{MDS-UPDRS} < 1$). The increase to three principal components did not result in improved classification performance suggesting a lack of distinct distinguishing features in higher-dimensional representations.

It must be noted that the intricate interplay between observational judgments of symptom severity for isolated movement tests and the verified health status of individuals present a significant challenge for selecting clinically meaningful class labels used to train and test the classification models. While allocation of subjects to the mild class (MDS-UPDRS 0, 1) may be indicative of a healthy/ PD-free status given that even healthy elderly participants often exhibit some degree of bradykinesia, it is important to highlight that these scores do not confirm the absence of PD. As such, the final analysis leveraged the unilateral assessment protocol to evaluate intra-limb asymmetries across movement features. The intra-limb asymmetry feature sets in the finger tapping dataset were computed to represent participants using subtle motor variations between the right and left hands, providing an effective approach for distinguishing PD patients from controls. Subsequent classification using three principal components demonstrated strong classification accuracy achieving an accuracy of 0.93 using

a SVM with an RBF kernel. The inclusion of the third principal component increased the cumulative explained variance from 76.51% to 94.94%, enhancing the discriminatory power between healthy controls and those with PD. Furthermore, a good balance between sensitivity (0.92) and specificity (0.94) was obtained suggesting that the SVM-RBF model can reliably distinguish between the participant groups using the transformed asymmetry features, with minimal bias towards either false positives or negatives. In summary, the final classification centred around asymmetrical movement performance demonstrated a clinically meaningful and effective approach to classification, this is likely rooted to the common understanding that PD motor symptoms manifest asymmetrically, with symptoms typically being more pronounced on one side, right from the onset (Barrett et al., 2010). The comprehensive analysis of finger tapping, opening-closing and pronation-supination offer a promising basis for determining the severity of PD motor deficits, offering an effective approach to linking variations across assessments to the unique demands placed on motor function by test-specific movements.

Chapter 5: Accuracy Matters: A Comparative Analysis of Computer Vision and IMU Angular Velocity Measurements in the Pronation-Supination Hand Movement Test

5.1 Introduction

In **Chapter 3**, the group-level analysis demonstrated the discriminatory power of clinically meaningful movement features. The importance of establishing meaningful motor aspects of parkinsonian movement was highlighted by Angel et al. (1970), emphasising that movement poverty can stem from the highest level down to the muscles. This was reaffirmed by the results of the pairwise comparisons which revealed differences even amongst younger and older healthy participants. The correlation analysis between computer features and clinical ratings highlighted two noteworthy findings: 1) movement features hold valuable information but require analysis from multiple viewpoints to pinpoint which features offer clinically meaningful distinctions; 2) the labels assigned to movement data have a notable influence on the patterns identified.

In **Chapter 4**, the addition of the opening-closing and pronation-supination hand movement tests to the finger tapping test demonstrated that different movement types can uncover unique characteristics of movement. Furthermore, the exploration of different class labels highlighted improved separability between broader MDS-UPDRS categories across each assessment, suggesting overlapping data distributions across adjacent scores. In turn, the transition from capturing movements in a 2D plane using YOLO to a 3D plane using MediaPipe Hands facilitated the detection of finer-grained movements. Additionally, the shift to using a smartphone to capture movement data without any efforts to enhance

recording conditions or background noise further underscores the assessment's promise for broader implementation in clinical settings.

In most studies, the classification performance in the concluding stage of the assessment is primarily used to indicate the viability of the proposed methodology. The critical factors that influence the outcomes of objective assessments remain underrepresented, with little mention of the constraints involved in aligning their outputs with clinical ratings. In the study by Williams et al. (2020), it was highlighted that even clinicians may be confounded by comorbidities and factors unrelated to PD. In such cases, agreement between objective assessment methods and benchmark ratings do not always equate to correctness. Instances where both the system and a clinician fail to reflect the ground truth can yield misleading accuracy measures. Therefore, the foundation of these assessments which demand accurate ground truth representations of movements, require deeper investigation. The current chapter presents a novel comparative approach for determining the agreement between peak angular velocities of pronation-supination hand movements in PD patients derived from the VC and IMU methods.

5.2 Methods

5.2.1 Participants

Thirty-five PD patients (recruited in study 2) performed the pronation-supination hand movement test, with both video recordings and IMU used to capture their movements. Tests were conducted unilaterally, hence the corresponding clinician assigned UPDRS scores are summarised for each hand in **Figure 5.1**. One left-hand test was excluded from the analysis, this was due to excessive perspiration preventing the IMU from adhering securely to the skin.

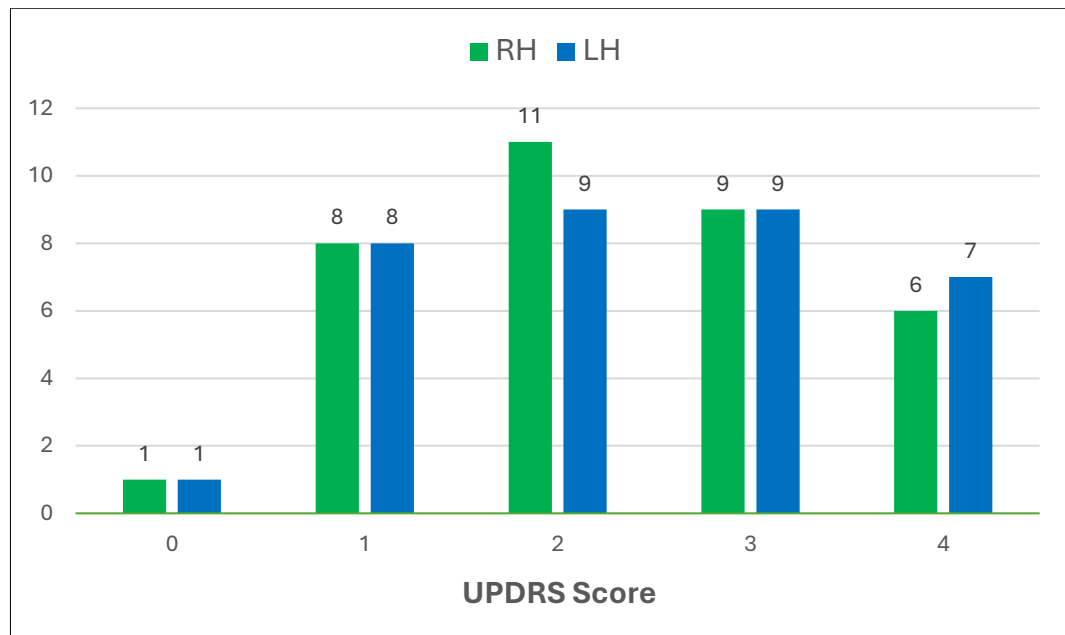


Figure 5.1. Clinician assigned MDS-UPDRS scores for 35 Parkinson's disease patients. Scores are provided individually for the right hand (RH) and the left hand (LH) as the pronation-supination hand movement test is performed unilaterally.

5.2.2 Protocol

The test protocol regarding video recording the pronation-supination hand movements followed that reported in **Chapter 4.2.3**. Hence videos were recorded using an iPhone (60 FPS). An IMU was attached to the dorsal surface of the hand using double-sided tape, aligned along the middle of the hand, to capture angular velocity data- see **Figure 5.2**. In cases where the attachment was insufficient to keep the IMU attached to the hand, an additional layer of tape was applied over the sensor. The Axivity (AX6) IMU with dimension 23 x 32.5 x 8.9 mm and mass 11 g was used, which uses a 6-axis movement sensor that records linear acceleration and angular velocity at high precision. The Open Movement GUI (OMGUI) software was used to setup and configure the AX6 IMU for recording, it was further used to export the captured data. The IMU was set to record data at a sample frequency of 100 Hz and the gyroscope was configured to record within

a range of ± 1000 degrees per second, ensuring all angular velocity movements possible in human motion were captured. The IMU started recording movement data immediately after being disconnected from the OMGUI software. Video recordings started 5 s prior to participants being instructed to begin the test and ended 5 s after completion to ensure the entire test was captured. Subsequent analysis focused on pronation-supination hand movements of the right and left hands using data from both recording modalities.



Figure 5.2. Pronation-supination hand movement tests recorded simultaneously. An IMU placed on the dorsal surface of the hand and video recordings using an iPhone 13 Pro Max, placed on a tripod approximately one meter away from participant hands are used for parallel data capture and analysis.

5.2.3 Data Processing

Although the pronation-supination test was recorded in parallel, the fundamental differences between the VC and IMU systems necessitate processing to align the two datasets before comparing their angular velocity outputs. Both systems

derive motion data differently, with the IMU capturing six-degrees-of-freedom, three for linear acceleration (x, y, z) and three for angular velocity (x, y, z). In contrast, the VC system derives motion data using relative positional coordinate data of hand landmarks over consecutive video frames. Additionally, both measurement methods used different sample frequencies and had asynchronous recording start and end times. **Figure 5.3** provides the analysis framework used to perform direct comparison of peak angular velocity data between VC and IMU across repeated cycles of pronation-supination hand movements of PD patients.

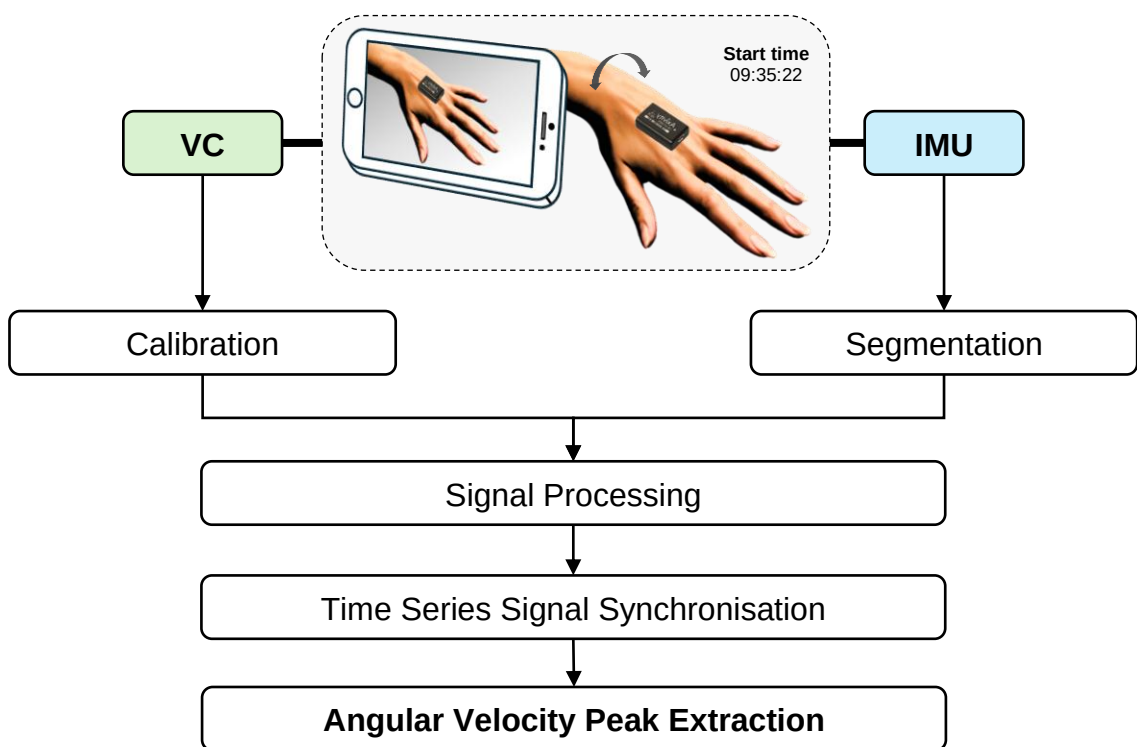


Figure 5.3. Flowchart outlining the steps from parallel data capture using IMU and visual computing (VC), through signal processing and synchronisation, to the direct comparison of angular velocity peaks for identical cycles of pronation-supination hand movements derived from both measurement methods.

The first step in the data processing stage for the IMU was segmentation. Identification of the pronation-supination movement data was performed using Python. The participant ID, sample frequency and recorded test start times were

used to isolate the angular velocity data for each PD participant from noise and other movements. **Table 19** shows the time sheet used to record the test information for each participant.

Table 19: Time Sheets to Record Participant ID, Handedness, and Start and End Times for the Pronation-Supination Hand Movement Test

Subject	Test	Hand	Start Time (hh:mm:ss)	End Time (hh:mm:ss)
PD17	Pronation-Supination	Right Hand	08:20:00	08:20:30
		Left Hand	08:25:30	08:26:00

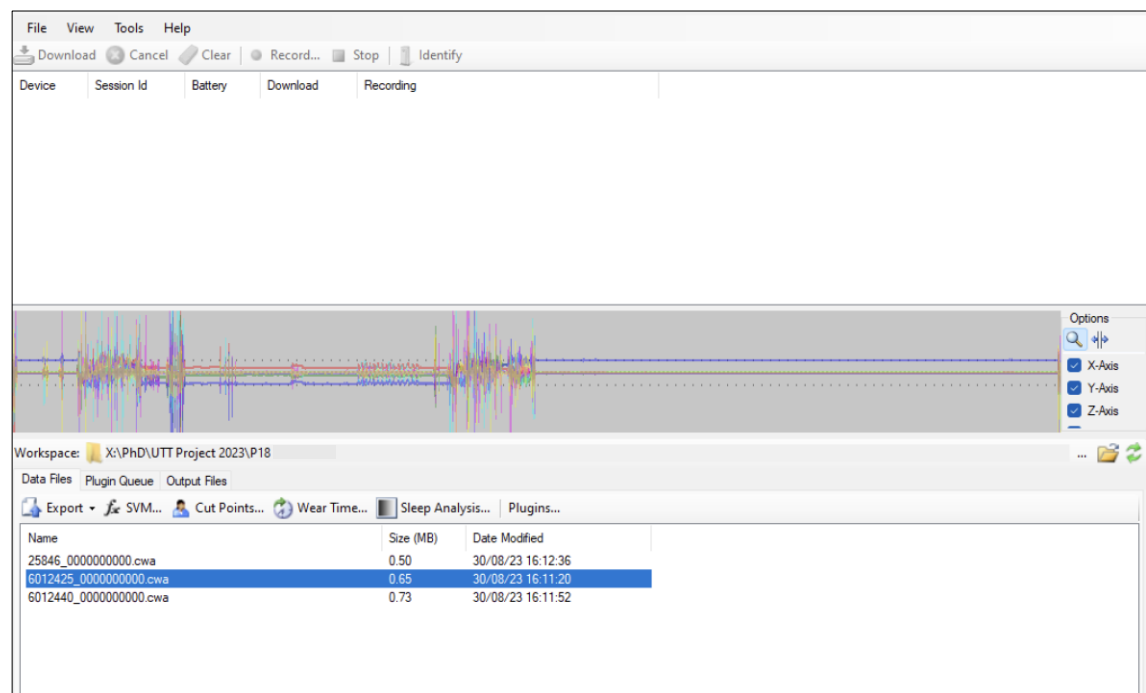


Figure 5.4. Open Movement GUI (OMGUI) software used to configure IMU recording specifications, such as sampling frequency and sensor range, prior to data collection. An example of the signal captured by the Axivity (AX6) IMU after detachment from the computer is shown. Movement data and logged time is then exported to a CSV file for each participant enabling subsequent analysis.

Raw IMU data was exported from the OMGUI software and saved individually for each participant, see **Figure 5.4**. The file naming nomenclature combined the

subject ID with the test handedness i.e. “PD01_R” for the right hand test of Parkinson’s disease participant number 1. The Python code then iterated through each participant’s data file, using the start time to establish the first corresponding z-axis angular velocity data point and the 3000 consecutive data points equivalent to 30s of data at 100 Hz sample rate, **Figure 5.5**. Finally, the code then assigned a 0.01 s time variable to each data point, resulting in the raw IMU angular velocity time series data.

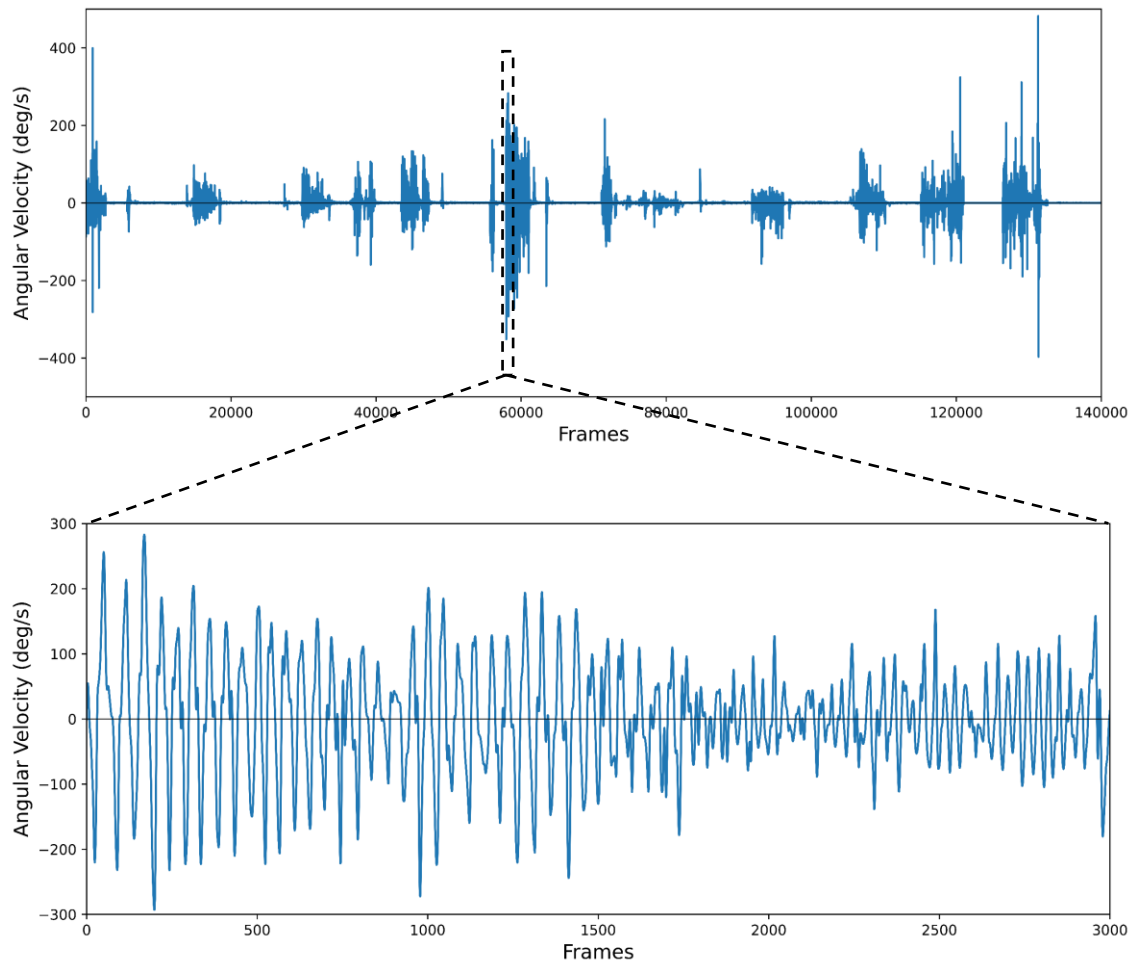


Figure 5.5. Segmentation angular velocity time series signals derived from the IMU data. The pronation-supination test start time is used to establish the first corresponding z-axis angular velocity data point and isolate the 3000 consecutive data points equivalent to 30s of data at 100 Hz sample rate.

For the VC system, the angular velocity time series data derived using MediaPipe Hands was scaled to angular velocity data in units of degrees per second to match the IMU data. This was achieved using a calibration object of known size as illustrated in **Figure 5.6**, allowing for direct comparison of angular velocity time series data between the two systems.

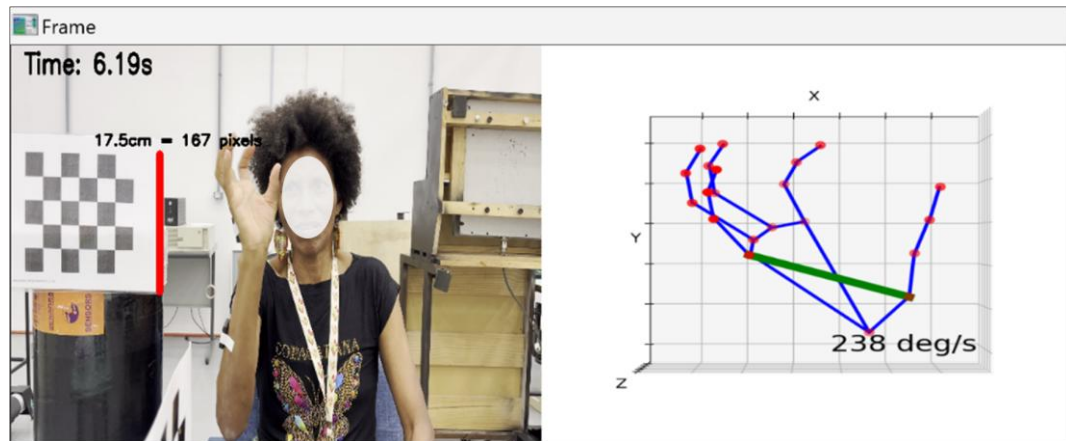


Figure 5.6. Checkerboard position in frame to act as a calibration object. Its known height is manually labelled (red line) in Python and used to scale the measurements, converting the pixel-based movement into degrees per second resulting in angular velocity output data.

5.2.4 Signal Processing

One cycle of pronation-supination is defined as the rotation of the hand from a neutral position to a maximally supinated position (palm facing up), and then pronating back to this neutral position (palm facing down). Each pronation-supination cycle is characterised by five distinct reference points (**Figure 5.7**): an initial positive peak, which reflects the initiation of rapid hand rotation from a neutral position towards maximum supination, this is followed by a rapid decline to zero which corresponds to a maximally supinated hand. Next, a succeeding sharp negative peak in angular velocity marks the point of reversal back to a neutral hand position during pronation. A full repetition is concluded by a second instances of zero angular velocity when the hand reaches the neutral position.

Finally, there is a varying short delay (zero velocity) before the next supination movement is initiated.

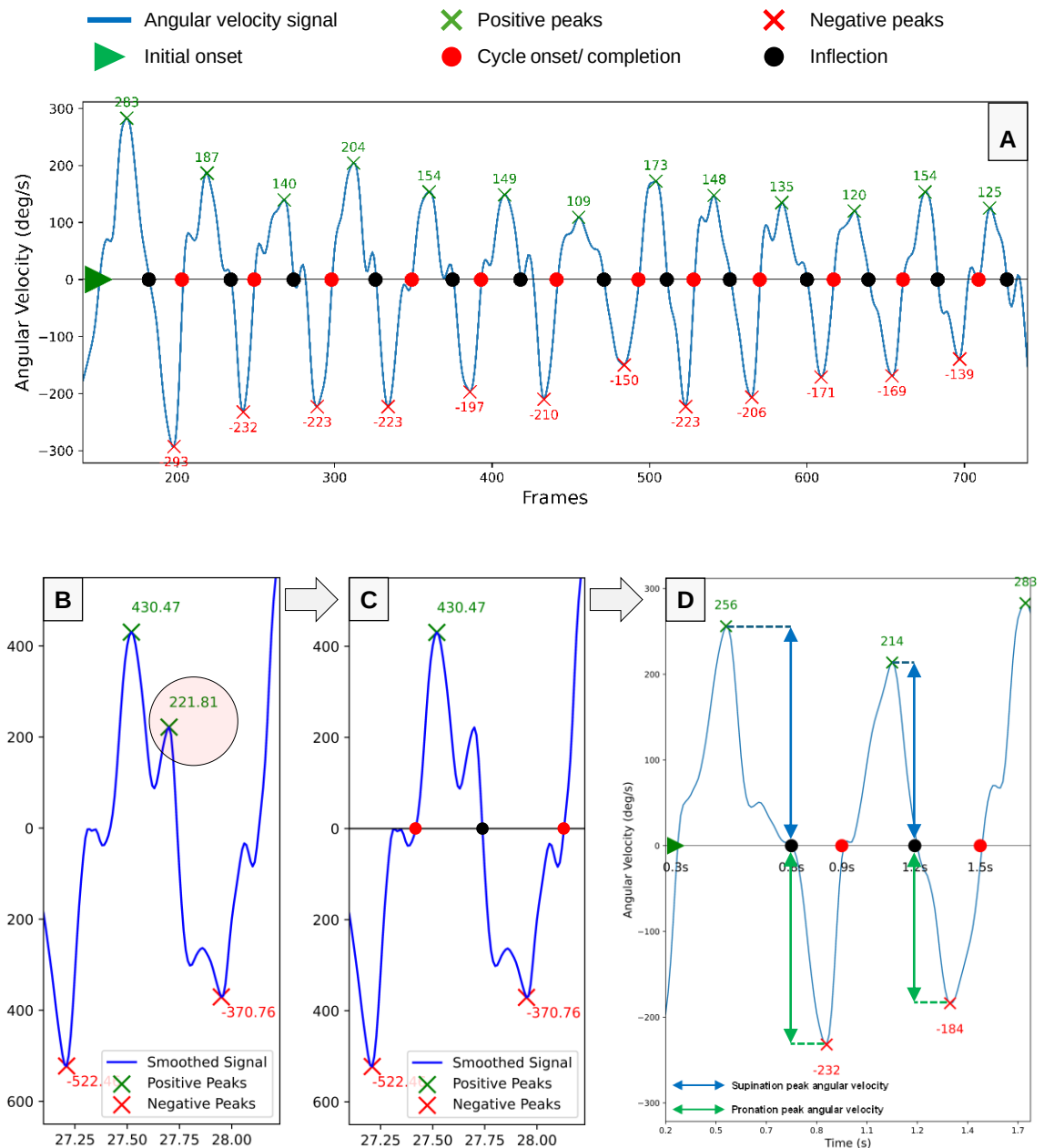


Figure 5.7. Signal processing using x-axis crossovers (where angular velocity is zero) used to identify complete movement cycles (Signal A). Each pronation-supination movement cycle is derived from signals by marking cycle onsets, inflection points and cycle completions. The transition from initial peak extraction to filtered false positive detections of peak angular velocities (Signal B to C). Each cycle contains only one positive peak between the onset and inflection marker and only one negative peak between the inflection and completion markers. The filtered signal is used to extract peak angular velocities (Signal D).

Two signal processing techniques were implemented to allow for direct comparison of both positive and negative peak angular velocities between the VC and IMU modalities, corresponding to supination and pronation angular velocities, respectively. The first step used the 'find peaks' function within the SciPy library in Python to customise the peak extraction process. The prominence, minimum peak height and minimum distance between neighbouring peaks thresholds were used to extract peaks. Prominence is a measure of how much a peak in a time-series signal stands out relative to its neighbouring peaks due to its height and location. A minimum peak height threshold of 10 degrees/second was set, to define the minimum height a data point must exceed to be considered a peak. Additionally, a minimum x-axis distance of 0.1 seconds between two neighbouring peaks was set, to define the minimum separation in time a data point must exceed to be considered a peak. These thresholds further minimised detection of false positives associated with physiologically impossible movements such as angular velocities exceeding 1000 degrees/second or consecutive peaks within less than 0.1 s.

While rapid alternating pronation-supination hand movements generate clean and smooth signals, parkinsonian related movement abnormalities like limb tremor can manifest local maxima and minima values in intermediate movements. The initial peak extract thresholds served as a baseline to detect the kinematic transitions associated with rapid points of reversal during alternating hand rotations. **Figure 5.7** demonstrates the four stages (A-D) that were used to address the erroneous detection of angular velocity peaks. Stage A illustrates the novel approach which incorporated additional signal characteristics to delineate complete cycles of pronation-supination movements by detecting x-axis

crossovers (zero angular velocity). Three critical phases were used to determine complete movement cycles in signals: onset, completion and inter-peak points of inflection. Onset points were marked by the first instance of x-axis crossover preceding a positive peak, the completion as the first x-axis crossover directly after a negative peak and the inflection points as the first x-axis crossover after a positive peak, provided the presence of a subsequent negative peak i.e. two consecutive positive peaks do not constitute an intermediate inflection point marker. The transition from the initial baseline peak extraction to identification and elimination of false positives using the additional signal processing step is shown in stages B and C. Finally, stage D shows the resulting signal used to extract peak angular velocities derived from the VC and IMU time series data.

The agreement between angular velocity peaks from the VC and IMU systems require identical pronation-supination movement cycles to be compared. The IMU began recording data immediately after being disconnected from the computer. While, video recordings were started 5 s prior to the pronation-supination hand movement test leading to an inherent time offset in the time-series signals from the two systems. Consequently, **Figure 5.8** shows the final signal processing stage which synchronised time series signals for each participant in the time domain. This was done by assigning a number to each peak based on their temporal order and aligning the two signals using the first positive angular velocity peak. The time offset between the two signal's first positive peaks was used to shift the signal with the later onset time back along the x-axis. This was done to ensure that the number of peaks were exactly matched between the two signals and that the angular velocity peaks of identical movement cycles were compared.

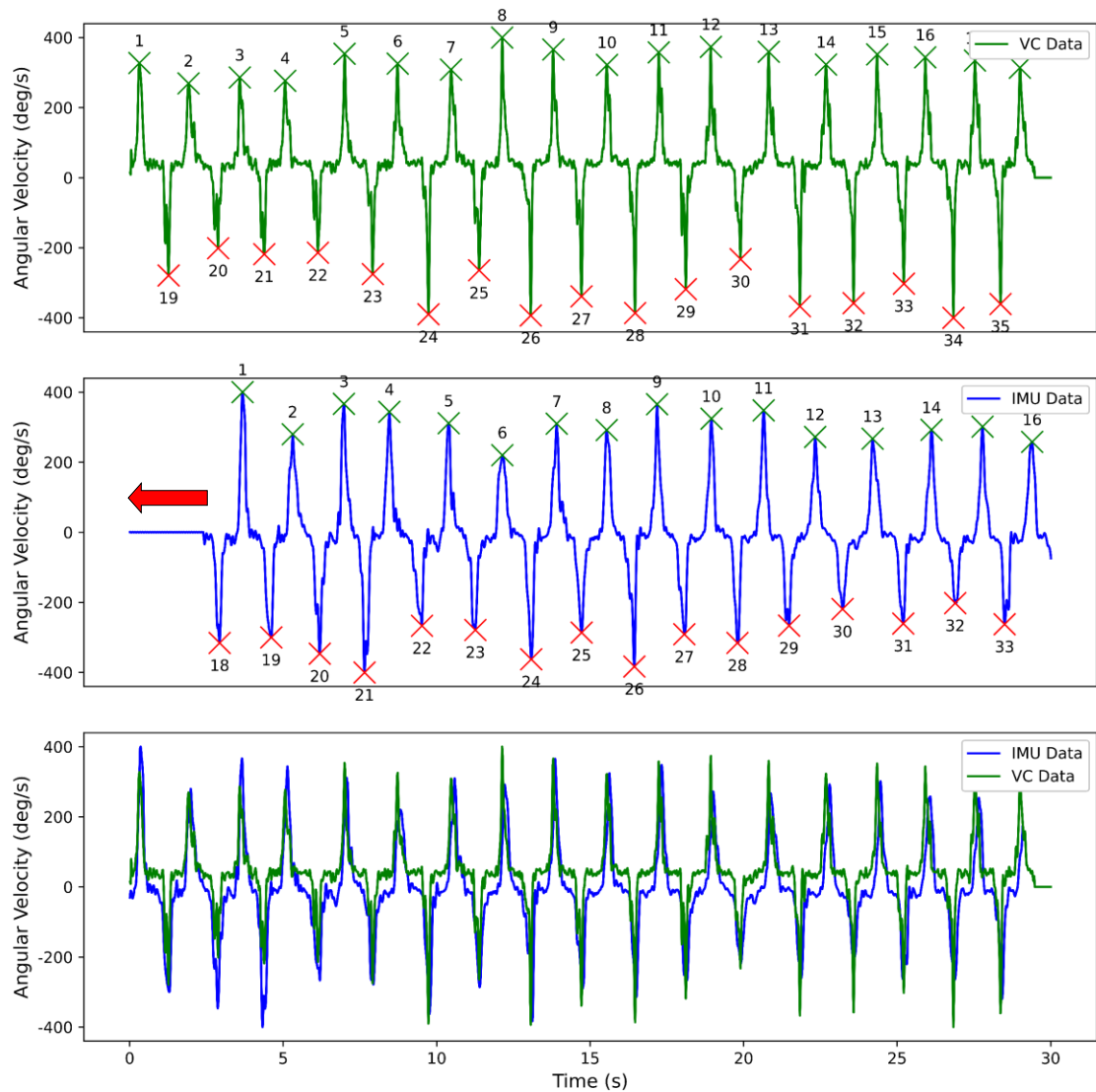


Figure 5.8. Processed VC and IMU signals with angular velocity peaks numbered in temporal order. The signals are then synchronised in the time domain (along the x-axis) using the time index of the first positive angular velocity peak as an anchor point to align the two signals.

5.3 Statistical Analysis

Two analysis approaches were used to evaluate angular velocity data derived from the VC and IMU systems. Regression analysis was used to examine the agreement between the two measurement methods and box plots were used to visualise the distribution of angular velocity peaks across the five UPDRS scores. Two separate linear regression analyses were performed using the VC and IMU

data, one for supination angular velocities (positive peaks) and another for pronation angular velocities (negative peaks). The coefficient of each regression equation (R^2) was calculated to evaluate model fit and visualise the relationships within the scatter plots, **Figure 5.10**. This analysis was implemented to ascertain whether the contactless MediaPipe Hands CV-based approach has capacity for capturing accurate ground truth movement data by evaluating its agreement with the assessment obtained from the IMU system.

The data obtained from 35 PD patients using two measurement methods were grouped by UPDRS classes. Subsequent analysis focused on evaluating the distribution of peak angular velocities of alternating pronation and supination movements across the five clinician assigned UPDRS scores. **Figure 5.11** shows the two separate box plots that were used to visualise the output data obtained from both systems across the MDS-UPDRS scores, one for supination angular velocities (positive peaks) and another for pronation angular velocities (negative peaks). Within each MDS-UPDRS class, summary statistics for the pronation-supination hand movements were computed to provide an overview of the distribution and central tendency of angular velocities across varying symptom severity levels. **Table 20** and **Table 21** summarise the mean, median, standard deviation and interquartile range of angular velocity peaks for supination and pronation movements, respectively. The sum of complete supination-pronation movement cycles are further provided for both VC and IMU derived data.

5.4 Results

All participants completed the pronation-supination hand movement test on their first attempt, resulting in a total of 2672 complete movement cycles. Despite the varying complex angular velocity time series signals obtained across five MDS-

UPDRS classes (**Figure 5.9**), the number of movement cycles detected using the proposed signal processing techniques were exactly matched between the VC and IMU system for all participants within each symptom severity class.

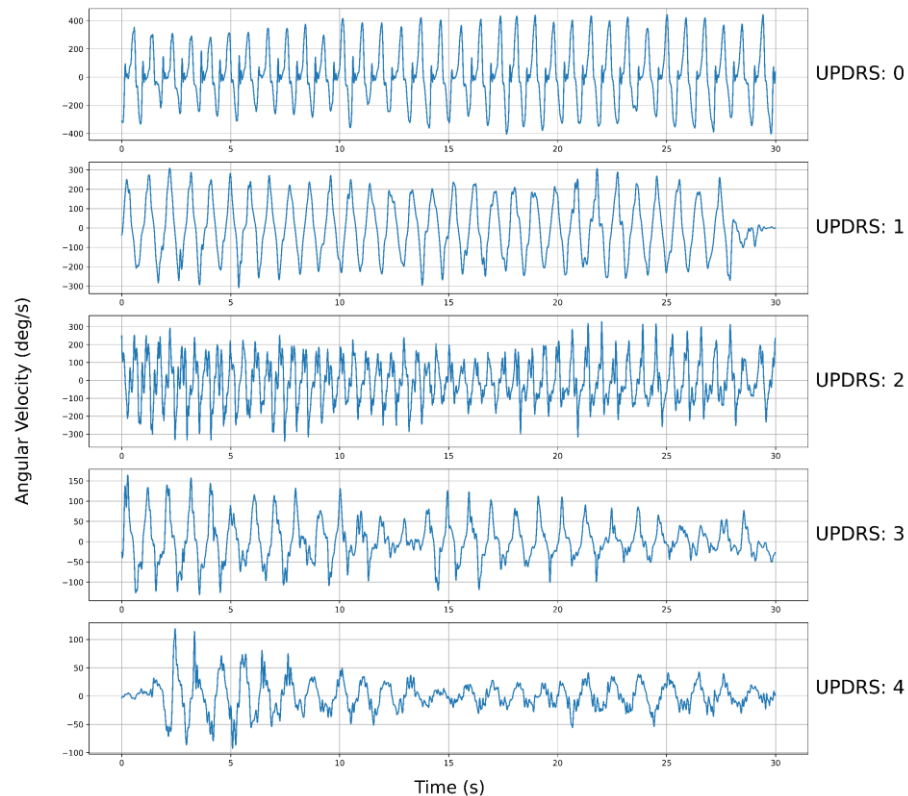


Figure 5.9. Five pronation-supination angular velocity signals progressing from UPDRS 0 to 4. The signals illustrate the decline in movement quality with increasing UPDRS scores.

The regression and box plot analyses highlight high correlation between the VC and IMU systems, with high correlation of determination (R^2) values of 0.98 and 0.99 for supination and pronation peak angular velocities, respectively. These analyses indicates a strong alignment between the two measurement methods, demonstrating the VC system's ability to accurately determine angular velocities during the pronation-supination test similar to the IMU. Furthermore, it confirms the effectiveness of the proposed signal processing method in filtering false positive detections of angular velocity peaks from both the VC-derived signal and the segmented IMU signal.

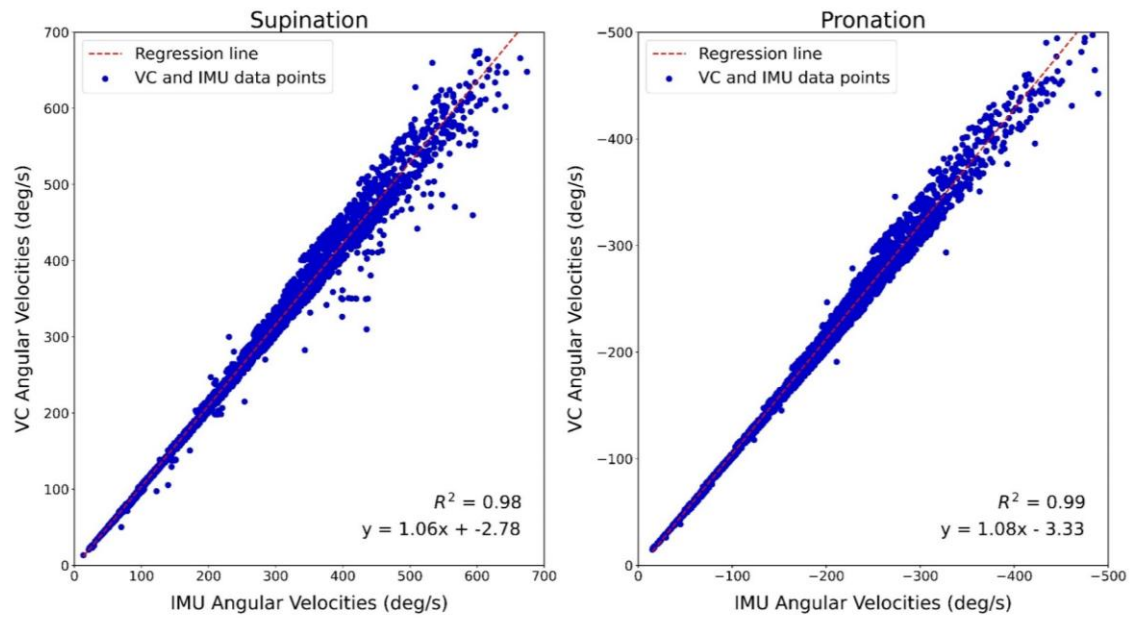


Figure 5.10. Two separate linear regression plots to evaluate model fit using coefficient of regression equation (R^2) and visualise the relationship between peak angular velocity data derived from VC and IMU. The scatter plots represent supination angular velocities (positive peaks) (left plot) and pronation angular velocities (negative peaks) (right plot).

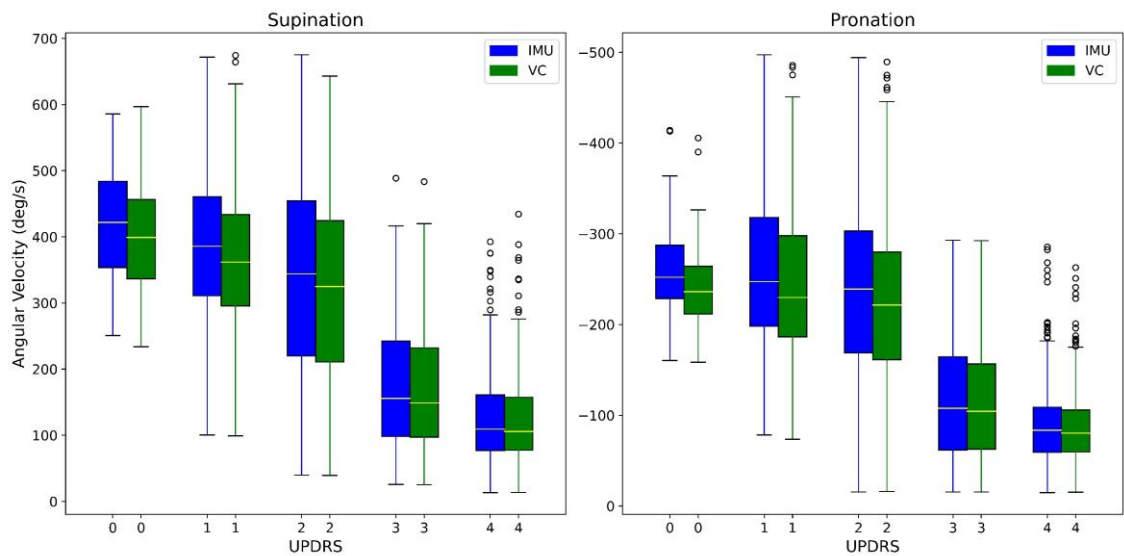


Figure 5.11. Two separate box plots used to visualise the peak angular velocity data across UPDRS scores, derived from the VC and IMU measurement methods. The plots represent supination angular velocities (positive peaks) (left plot) and pronation angular velocities (negative peaks) (right plot).

Table 20: Summary Statistics of Peak Supination Angular Velocities Across UPDRS Scores Derived from the VC and IMU Systems

Summary	UPDRS: 0		UPDRS: 1		UPDRS: 2		UPDRS: 3		UPDRS: 4	
	IMU	VC	IMU	VC	IMU	VC	IMU	VC	IMU	VC
Count	83	83	741	741	857	857	595	595	396	396
Mean	424.32	402.43	387.00	365.21	343.07	324.45	174.84	168.10	124.60	121.81
STD	83.31	81.70	106.07	100.77	146.57	136.38	94.59	88.12	67.50	64.45
Minimum	250.79	234.01	100.52	99.15	39.63	39.21	25.51	25.27	13.43	13.58
25%	353.54	336.75	310.93	295.76	219.90	210.84	98.27	97.37	76.93	77.33
50%	422.06	399.24	385.87	361.88	344.19	324.94	155.65	148.89	109.29	105.71
75%	484.15	456.43	460.78	433.84	454.51	424.46	242.44	232.27	160.99	157.41
Maximum	585.85	596.97	671.84	674.41	675.29	643.16	488.71	483.52	392.60	434.36

Table 21: Summary Statistics of Peak Pronation Angular Velocities Across UPDRS Scores Derived from the VC and IMU Systems

Summary	UPDRS: 0		UPDRS: 1		UPDRS: 2		UPDRS: 3		UPDRS: 4	
	IMU	VC	IMU	VC	IMU	VC	IMU	VC	IMU	VC
Count	83	83	741	741	857	857	595	595	396	396
Mean	-256.97	-240.63	-261.63	-245.49	-237.68	-223.17	-119.72	-115.03	-88.08	-85.46
STD	50.89	46.44	81.13	75.48	93.81	86.73	67.26	62.80	43.45	39.70
Minimum	-414.15	-405.61	-497.33	-485.79	-494.31	-489.47	-292.95	-292.47	-285.54	-262.83
25%	-287.53	-264.19	-318.00	-297.90	-303.03	-279.96	-164.48	-156.58	-108.91	-105.88
50%	-252.09	-236.25	-247.42	-229.80	-239.14	-221.68	-107.73	-104.49	-83.50	-80.34
75%	-228.59	-211.71	-198.11	-186.55	-168.81	-161.24	-61.69	-62.66	-59.09	-59.59
Maximum	-160.36	-158.40	-78.33	-73.54	-15.40	-15.93	-15.40	-15.51	-14.82	-15.07

5.5 Discussion

The proposed assessment framework demonstrates significant promise for establishing agreement between VC and IMU movement data and for further examining the resulting pronation-supination peak angular velocity distributions within each MDS-UPDRS class. Moreover, the exact match in the number of complete movement cycles between the two measurement modalities underscores the effectiveness of the presented novel approach to delineating ‘true’ angular velocity peaks. The implementation of synchronising time series

data in the time domain, prior to comparative analysis, demonstrated strong potential for handling the diverse set of VC and IMU-derived signals associated with a wide spectrum of underlying motor abnormalities. It is noteworthy that in using the highlighted calibration technique for the CV-based assessment, movement velocities could be directly compared to gold standard measurement methods such as the IMU. Additionally, the automation of the segmentation process for signals derived from the IMU made the evaluation of 280 signals (70 right and left hands from two systems) highly robust.

The summary statistical analyses demonstrate a clear decline in absolute magnitudes of both supination and pronation peak angular velocities with increasing UPDRS scores, indicating the reduced movement velocity associated with increased PD motor deterioration. The mean absolute angular velocity peaks derived from both measurement methods were consistently higher for supination movements across all UPDRS scores, likely reflecting the increased strength of the muscles responsible for supination movements. It must be noted that the agreement in the mean angular velocity peaks decreased for higher magnitude velocity values. It is apparent that the values determined from the VC signal consistently have a smaller magnitude than those determined from the IMU signal. This discrepancy is due to the lower sampling frequency used: VC at 60 Hz and IMU at 100 Hz. Although use of a lower sampling rate will always have this underestimation effect, signal with lower velocity values are associated with higher motor symptom severities, resulting in lower magnitude but more complex signal characteristics. This is evidenced by the presence of multiple outliers in the fourth UPDRS class. Therefore, the implementation of appropriate peak extraction techniques is essential prior to subsequent analysis.

The high level of agreement between the VC methodology and the IMU data indicates that any increases or decreases in angular velocity across the pronation-supination repetitions were detected equally as well by both the CV and IMU approaches (despite consistent underestimations of peak angular velocities by the VC system). However, the scatter plots indicate a very close fit to the regression line, particularly for smaller magnitude velocity values (i.e. those <300 deg/s). Another likely factor contributing to differences in angular velocity data is that, unlike the IMU, the VC method is not immune to self-occlusions during rapid orientation changes of the hand. That being said, the IMU may experience signal drift without corrections from accelerometer data. It must be noted that unlike finger tapping, pronation-supination hand movements are not primarily focused in a single up-and-down plane. This poses a greater challenge for discerning rapid alternating hand movements which involve more frequent orientation changes and in turn more self-occlusions. In summary, the assessment methodology demonstrates that the VC-based approach has strong potential for establishing reliable ground truth kinematic data from complex upper limb movement tests across diverse PD motor presentations, offering a promising foundation for accurate clinical movement analysis.

Chapter 6: Key Findings and Conclusion

6.1 Summary of Findings

The findings presented in this research demonstrate excellent promise in making the remarkable leap from raw video recordings of upper limb movements to discerning clinically meaningful distinctions in movement data. Importantly, the critical aspects of the research were guided by a commitment to maintaining realism. Both dataset analyses were conducted using standard video recordings with no extra efforts made to enhance recording conditions. Furthermore, video recordings of upper limb movement tests were intentionally restricted to isolate the hand and forearm. This approach avoided capturing facial features, body posture, or other potentially confounding visual information that could inadvertently influence either the VC-based analysis or clinical evaluations of motor performance. Within this isolated view, an important aspect of the dataset presented in this study was the inclusion of participants with a variety of skin tones introducing variation in the visual clarity, contrast and overall appearance of hands within video recordings. Additionally, clinician ratings were assigned based on initial observations of video recordings, reflecting the realistic patient-clinician interaction in fast-paced clinical settings. It is evident that the visual judgment of PD examination signs are inherently subjective. Assessment outcomes are further complicated by a combination of patient- and clinician-related factors such as the inevitable variability in PD symptom presentations; likewise, clinician experience influences outcomes. Consequently, the extensive interacting factors identified in the research, that impact assessment outcomes, guided the diverse approaches implemented in the research.

In clinical applications, identifying discriminative patterns in data risk driving approaches that focus solely on achieving statistical distinctions over clinically meaningful ones. This challenge can be likened to an observer effectively distinguishing between patients without knowledge of the appropriate course of action. It is for this reason that the assessment frameworks presented in this study evaluate symptom severity in accordance with the MDS-UPDRS scoring criteria. That being said, it is important to highlight that the ordinal nature of scoring PD symptom severity into one of five predefined MDS-UPDRS classes inherently simplifies the representation of individuals. This effect was substantiated in **Chapter 3**, where computer-generated features consistently achieved higher correlation with corresponding amplitude, speed and rhythm components of the MBRS compared to their correlation with UPDRS scores. With this in mind, even the expert eye cannot discern subtle movements accurately, underscoring the significance of extracting detailed motion feature sets through the computer vision methodology. In view of these factors, the assessment frameworks were structured in distinctly progressive analytical stages, with the determination of motion features clarified in the signal processing stage and the classification of summary features made interpretable in the quantitative analysis stage, i.e. principal component analysis in **Chapter 4**.

It is beyond question that the diagnosis of PD is comprehensive and not confined to observations of the isolated movement tests outlined in the motor examination section of the UPDRS. Yet still, the detailed characterisation of core motor examinations remain critical in establishing a baseline for deviations from normal healthy movement as well as for monitoring disease progression. Therefore, the compelling rationale to refine these motor assessments is imminent. Notably, the

shift from standard video recordings using a GoPro Hero 5 camera in **Chapter 3**, to the ubiquitous smartphone (iPhone 13 Pro Max) in **Chapter 4**, highlights the scalability of computer vision-based assessments. The use of two data extraction mechanisms, YOLO in **Chapter 3** and MediaPipe Hands in **Chapter 4**, demonstrate the potential of capturing clinically significant movement features using strategic image annotation for custom trained 2D object detection inference and relevant test-specific landmark extraction for 3D pose estimation. In addition, the ability to quantify these movements accurately was showcased in **Chapter 5**, where the capture of ground truth movement was validated against data obtained from an IMU. In the majority of previous studies, the use of evaluation metrics to indicate model performance was limited to the classification stage, reinforcing the primary focus most studies have on achieving good classification performance. While this is important, classification performance on a specific dataset does not validate a model's capacity to perform well across other datasets, particularly in cases where participant demographics vary significantly. Therefore, the exploration of multiple performance measures in the analytical stages of the assessment framework makes my assessment method highly adaptable for widespread deployment. In **Chapter 5**, the integration of a novel signal processing technique underscores the importance of maintaining accuracy throughout intermediate analysis steps. This focus on step-by-step validation was demonstrated in **Chapter 3**, where the cause of transient jumps in time series signals was identified through straightforward measures like monitoring the number of consecutive frames that include detection of both finger and thumb. Thereby, large displacements in bounding box position linked to abrupt hand movements were distinguished from inconsistencies in detection performance.

It is crucial to recognise one of the most valuable aspects of this research: the direct engagement with individuals affected by Parkinson's disease, together with expert clinicians in the field. Involvement in the data collection process provided an enriching experience, driving a patient-centred approach that is exceptionally challenging to achieve through more remote analyses, such as those conducted on pre-existing datasets collected by others. It is firmly recognised that the intricate factors involved in clinical diagnosis and patient monitoring, particularly with life-altering diseases, extend beyond motion analysis alone. In previous automated assessments of PD, the documentation of these critical design elements lack detail and their implications on practical assessment outcomes are rarely addressed. In PD, the 'on-off' phenomenon relating to dopaminergic treatments like levodopa have been reported to influence both motor symptoms like tremor as well as non-motor symptoms like mood fluctuations including depression, anxiety and elation (Nissenbaum et al., 1987). Consequently, the data acquisition outlined in **Chapter 4** prioritised ethical considerations in the data collection process. This involved the implementation of a booking system to ensure a careful balance between PD participants performing movement assessments in an off-medication state without prolonging this state longer than necessary.

6.2 Clinical Implications

The multiple approaches adopted in this research have highlighted several clinically significant implications for the application of automated assessment techniques to evaluate PD motor impairment. The progressive nature of PD necessitates frequent in-person clinical assessments which ultimately requires patient motivation. Therefore, the potential for remote assessment presents one

of the most compelling rationales for advancing this contactless assessment method. Importantly, this computer vision-based approach offers clear advantages over wearable technologies given its low cost and ease of deployment. Wearable devices often require specialised software and regular calibration to ensure accurate data capture, both of which demand professional setup. Consequently, the low cost and ease of deployment offered by this method make it highly favourable for assessing new and existing PD movement tests.

It is noteworthy to highlight that the presented assessment framework not only provides a contactless alternative to wearable technologies, but facilitates a framework in which continuous improvements can be made through the progressive analytical stages. More specifically, the ability to adapt the data extraction process and refine the selection of extracted motion features allows the continuing advancements in clinical knowledge surrounding PD to be incorporated across the entire framework. In turn, prioritisation of model interpretability makes this automated assessment well-suited to function as a remote clinical decision support tool.

To summarise, the clinical impact of this work extends beyond the proposed solutions, with the review of the critical factors that shape visual judgements of PD being substantiated with findings based on the acquired clinically-rated datasets. The balanced approach of addressing clinician perspectives in PD motor examinations, prior to development of the automated assessment methods, identified areas for improvement right from the outset. It is evident that even isolated upper limb movements allow a good level of distinction to be made between PD patients and controls. Subsequent analysis of intra-limb asymmetry

revealed that differences between the more and less affected sides of individuals can also be used to effectively guide the classification of PD symptom severity.

6.3 Novel Contributions

Data acquisition: In **Chapter 4**, the dataset acquired comprised of three independent upper limb movement assessments, each with three clinician-assigned UPDRS scores. This facilitated insights into movement-specific assessments and their effects on symptom severity classification for PD and control subjects. Importantly, the intraclass-correlation coefficient was computed for the extensive dataset, highlighting the effects of inter-rater reliability in clinician-assigned scores across the three unilateral assessments. Finally, the bimodal data capture presented in **Chapter 5** employed a novel dual-modality approach for capturing unilateral pronation-supination hand movements in PD patients, combining VC and IMU-generated movement data to enable comparative analysis of two measurement methods.

Analysis Approaches: The unique dataset presented in **Chapter 3** provided a clinically-rated benchmark, which enabled finger tapping output features derived from the computer vision-based assessment to be evaluated against both UPDRS and MBRS scores, assigned by 21 neurologists specialising in movement disorders. Furthermore, the allocation of healthy participants into two separate age-specific control groups (younger and older controls), enabled a deeper analysis of PD versus controls by incorporating age as a factor. The analytical approaches adopted in this research provide multiple novel contributions to the automated assessment of parkinsonian movement. Additionally, the custom trained YOLO model demonstrated effective tracking of the distal index finger and thumb. The rolling time window, applied to the

generated finger tapping time series signals, demonstrated an effective approach for computing variations in key signal characteristics over time. In **Chapter 4**, the standardised feature sets derived from three individual upper limb movement tests, using the MediaPipe Hands model, were extracted using test-specific landmarks of the hand. This provided a novel approach to performing a comprehensive analysis using three critical items of the UPDRS motor examination section. In **Chapter 5**, the movements captured using the MediaPipe Hands model were scaled into angular velocity measures using a calibration device, enabling direct comparison with the IMU data. The subsequent analysis showcased a signal processing technique which involved the identification of x-axis crossovers in the signal path to reliably determine complete cycles of pronation-supination hand movements. This was followed by the automated synchronisation of the computer vision and IMU-generated angular velocity signals in the time domain, allowing direct comparison of positive and negative angular velocity peaks between the two measurement systems.

Novel experimentations: In **Chapter 3**, a novel method combining a mouse, acting as a contact sensor, with the computer vision model in parallel was applied to determine the model's ability to capture distinct finger tapping movements accurately. Meanwhile, **Chapter 5** introduced a completely novel comparative analysis framework for establishing the computer vision system's ability to consistently discern ground truth movements, using IMU data as the benchmark.

Real-time capability: The development of a real-time automated assessment framework is detailed in **Chapter 3**. This novel approach demonstrated that appropriate signal processing techniques can be used to identify the onset of finger tapping movements, irrespective of initial index finger and thumb positions.

This establishes groundwork for performing remote assessment for PD movement tests, without the need for manual input.

Novel metrics: The research presented two novel metrics for deriving characteristic movement features from time series signals. In **Chapter 3**, the total tapping distance was computed from finger tapping time series signals, providing an effective motion feature to indicate movement performance. A second novel metric was introduced in **Chapter 4**: the coefficient of variation of inter-cycle path length.

6.4 Conclusion

While numerous published methods have explored wearable technologies to evaluate PD motor abnormalities in hand movements, their reliance on specialised equipment, associated costs and potential to disrupt natural movement restrict their widespread adoption in clinical settings. This thesis demonstrates the feasibility of a contactless, visual computing-based alternative, using standard video recordings alone. It is important to acknowledge that classification performance in symptom severity classification tasks, while informative, is not a definitive indicator of a model's capacity to interpret movement quality. This is because the evaluation process is naturally two-sided: it relies on the model's ability to generate accurate predictions, but also works on the assumption that target class labels are correct. However, as shown in this thesis and supported by previous literature, variability in clinician-assigned clinician ratings introduce a level of uncertainty in these labels. This dual source of variability highlights the need to consider factors beyond classification performance in clinical movement analysis. Moreover, while the quantification of movement removes the subjectivity involved in visual assessments by offering an

objective and repeatable assessment method, the process of transforming multidimensional quantitative data into single ordinal values (UPDRS scores) can itself introduce variation. This decision-making step underscores the complexity of aligning objective computer measures with clinical diagnostic criteria. In light of these challenges, this study's transparent and comprehensive evaluation framework demonstrates promise for identifying these potential sources of error right from the video-based movement capture stage. Consequently, the outcomes of this study support the integration of visual computing and AI techniques into clinical assessments of Parkinson's movement disorders.

Chapter 7: Future Work

7.1 Enhancing the Dataset

The assessment of isolated upper limb movements is useful for establishing objective baseline measures of PD motor symptom severity. However, increasing confidence in treatment response and understanding the impact PD motor impairment has on the practical aspects of daily life requires a holistic evaluation of movement quality. Future work will look to incorporate a broader range of motor examinations into the classification of motor symptom severity. This will allow the assessment method to draw on a more diverse set of movement features, allowing it to construct a more comprehensive representation of PD motor impairment. The goal is to develop a versatile assessment framework capable of adjusting the analysis pipeline to extract test-specific movement features without requiring manual input. Future developments will look to integrate multiple movement analyses and their outputs into a unified decision-making framework. This approach is expected to enhance the classification performance of PD motor symptom severity and better reflect the complex nature of its progression across different areas of the body. As such, four additional lower extremity movement tests will be incorporated into the assessment framework, including: (A) up-on-the-toes stand (UTT-S) followed by 30s of rising up-on-the-toes (UTT-30); (B) seated heel tapping; (C) seated toe tapping; and (D) gait. **Figure B1** illustrates the proposed dual-camera setup designed to capture movement from both sides of the body during the bilateral UTT-30 test, while **Figure B2** shows images of the four lower extremity movement tests being performed with overlaid landmarks using the MediaPipe Pose Landmarker model to track test-specific movements (see **Appendix B**).

7.2 Model Explainability

In clinical practice, variability in the visual interpretation of movement abnormalities is a well-recognised limitation. Identifying the exact cause of differing clinician judgments is challenging, given the inherently subjective nature of visual assessments and the numerous factors that can influence it. Given that the development of a visual computing-based assessment method is intended to support clinicians with an objective and consistent alternative, the interpretability of a model's decision-making is essential. This alignment between objective, data-driven movement analysis and clinical reasoning would encourage the utility and acceptance of visual computing assessment methods in clinical practise. Therefore, future work will focus on enhancing model transparency, particularly during the classification of symptom severity. This involves linking model outputs to clinical reasoning by translating kinematic features into clinically meaningful terms, consistent with the diagnostic criteria for PD. In doing so, classification errors could be linked to misrepresentations of movement performance, grounding errors in clinically recognisable terms.

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Appendices

Appendix A: Supplementary Material for Study 2 (Chapter 4)

 UNIVERSITY of BRADFORD	
Participant Information Sheet	
Computer vision and wearables assessment of performance, movement, and balance problems in older adults during the execution of simple clinical movement tasks.	
This information sheet serves to inform you of what this study is about and what your participation in it involves should you decide to take part. Please take time to read it carefully and then decide whether or not to take part.	
Purpose of Study is to assess: (1) walking ability during two-three minutes of continuous overground walking (2) movement and balance control when rising up on to tiptoes from standing and sitting, (3) movement control when performing simple clinical tasks with your hands, and (4) your confidence about completing everyday physical tasks.	
Why you have been chosen: You have been chosen because you are over the age of 55 years, and are able to walk independently, and are either someone without any diagnosed neurological issues (problems affecting the nervous system) or are someone with Parkinson's and/or dementia.	
Do you have to participate? There is no obligation for you to participate in this research as it is voluntary. You may withdraw without reason at any time during the study should you wish to do so. If you take part, you may withdraw your data from the dataset at a later stage. To do this you will need to contact a member of the research team (contact details below). However, once your data are published it will no-longer be possible to withdraw it.	
What will happen if you take part: You will be asked to visit either Universidade Tecnológica Federal do Paraná – Piscina, or Universidade Federal do Paraná - Departamento de Educação Física - Politécnico Biomecânica Laboratório, where you will be briefed prior to carrying out certain tasks (details below). You will be given the opportunity to ask any questions regarding anything you don't understand. The visit may last up to one and a half hours.	
Your height, weight, leg length and shoe size will be measured. You will be given a questionnaire to complete. The questionnaire will ask about how confident you feel	

Figure A1. Participant information sheet used in Study 2 (Chapter 4). The full document spans four pages.

when completing everyday physical tasks such as getting in and out of a chair, getting dressed and undressed, walking around the house, and so on.

Small sensors will then be attached to your feet, hands, arms, and the middle of your back. You will then be asked to complete the following:

Upper extremity movement tasks:

Task 1: Whilst remaining seated, you will be asked to rest your right elbow on the chair's armrest, and tap your index finger on your thumb as fast as possible and then make the gap between your finger and thumb as wide as possible. This movement is to be repeated as many times as possible within a 30 second period.

This test will be repeated with your left hand.

Task 2: Whilst remaining seated, you will be asked to hold your right hand at face level, and open and close the hand as fast as possible and by spreading the fingers as wide as possible. This movement is to be repeated as many times as possible within a 30 second period.

This test will be repeated with your left hand.

Task 3: Whilst remaining seated, you will be asked to rotate your hand (alternate pronation and supination) as fast as possible. This movement is to be repeated as many times as possible within a 30 second period.

This test will be repeated with your left hand.

Lower extremity movement tasks:

Task 4: You will be asked to walk around a semi-circular path for around three continuous minutes at your normal walking pace.

Task 5: You will be asked to stand in a comfortable position, looking forward and holding a mobile device to your chest throughout. You may place one index finger on the chair in front of you for balance if required. After a short period, you will be instructed to rise on your tiptoes as fast and high as possible, and hold this position for a few seconds. After a short period, you will be instructed to return to normal standing. This task may be repeated in the event you lose your balance.

Figure A1 (continued). Participant information sheet - Page 2 of 4.

Task 6: You will remain standing in a comfortable position, looking forward and holding a mobile device to your chest. You may place one index finger on the chair in front of you for balance if required. You will be asked to rise on your tiptoes as high as possible and as many times as you are comfortably able to do within 30 seconds.

Task 7: You will be seated on a chair, and whilst holding a mobile device resting on your thigh, you will repeatedly rise your right leg on your tiptoes as fast and high as possible within 30 seconds.

This task will be repeated with your left leg.

Task 8: Whilst remaining seated with both feet on the ground, you will be asked to tap your toes of your right leg as fast as possible without lifting your heel off the ground. This movement is to be repeated as many times as possible within a 30 second period.

This test will be repeated with the left leg.

NB. All the above tasks were demonstrated to the participants prior to data collection.

Your movements for each task will also be recorded on video (using smartphone).

The information gained will be used for research purposes. No personal identifiable data will be retained. Data will be stored for up to 5 years after the completion of the study. — contacts details below.

Are there any risks associated with taking part?

Your calf/arm muscles may become tired when performing the repeated rising up on the toes movements and the hand tests. It is possible that you may lose your balance when performing the up on the toes movements. A researcher will be close by to assist you in the event you lose your balance.

There will be a chair placed in front of you to place your fingers/hands on lightly for help in preventing you losing your balance.

You will also be given sufficient breaks between each test to minimize fatigue.

What you need to do in preparation:

Please ensure you wear comfortable flat-soled shoes with low ankle tab support (without any reflective material), comfortable clothing which is not excessively loose, and any corrective vision you usually wear when walking.

Further information and contact details:

Figure A1 (continued). Participant information sheet - Page 3 of 4.

Should any questions arise, or if you wish to discuss any aspect of this research, please communicate through the following contact details:

Sarah Zahid: s.a.zahid@bradford.ac.uk

Ramzi Jaber: r.jaber@bradford.ac.uk

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Prof. Andre Rodacki: rodacki@ufpr.br

Thank you for taking the time to read this Information Sheet.

Figure A1 (continued). Participant information sheet - Page 4 of 4.

Appendix B: Supplementary Material Related to Future Work (Chapter 7)



Figure B1. Simultaneous video recordings of the rising up-on-the-toes (UTT-30) test using two smartphones (iPhone 13 Pro Max).

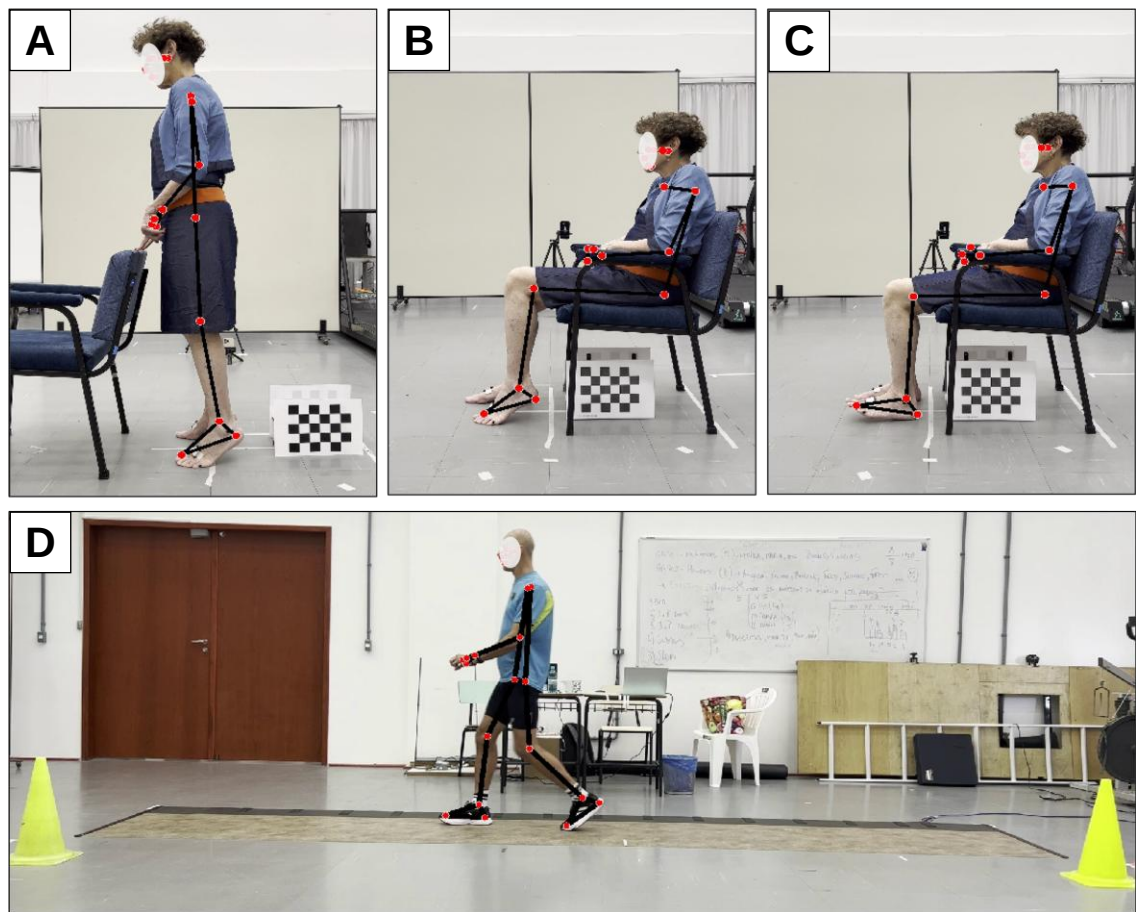


Figure B2. Inference of the MediaPipe Pose model on four lower extremity movement tasks: (A) up-on-the-toes stand (UTT-S) followed by rising up-on-the-toes (UTT-30); (B) seated heel tapping; (C) seated toe tapping; and (D) Gait.